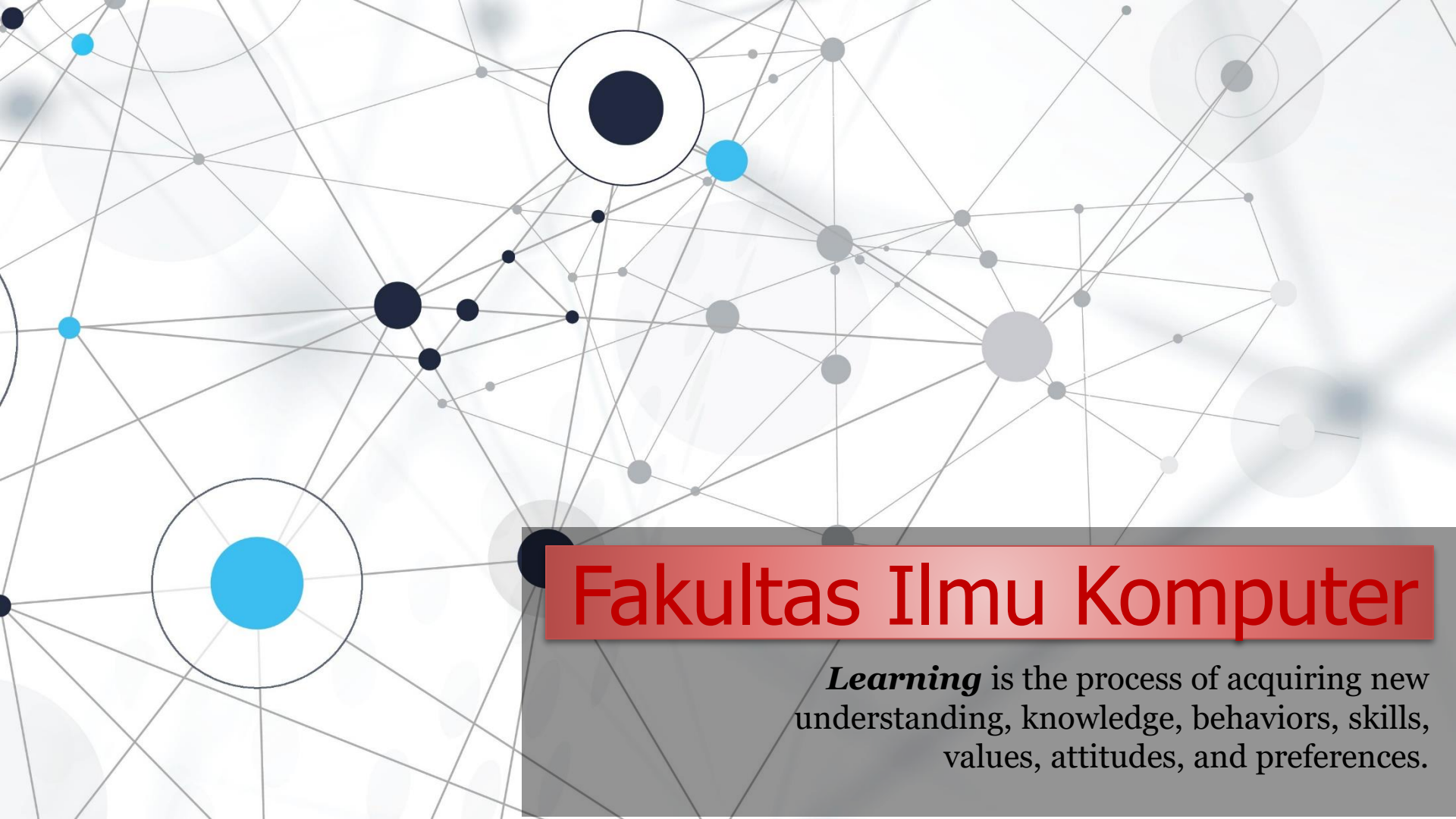


An abstract network diagram with various sized nodes (black, blue, grey) connected by thin grey lines. Some nodes are highlighted with larger circles. The background is white with faint grey circles.

Fakultas Ilmu Komputer

Learning is the process of acquiring new understanding, knowledge, behaviors, skills, values, attitudes, and preferences.

REST API *using* PHP Framework (*in cloud server*)



Welcome to *the* Full Stack Web Development *course*



1 – What is RESTful API ?

(Back-end berkomunikasi dengan client melalui sebuah antarmuka Application Programming Interface (API))

URL, Path, Link, atau Endpoint

(Endpoint adalah URL yang kita request)



API, REST API & RESTful API

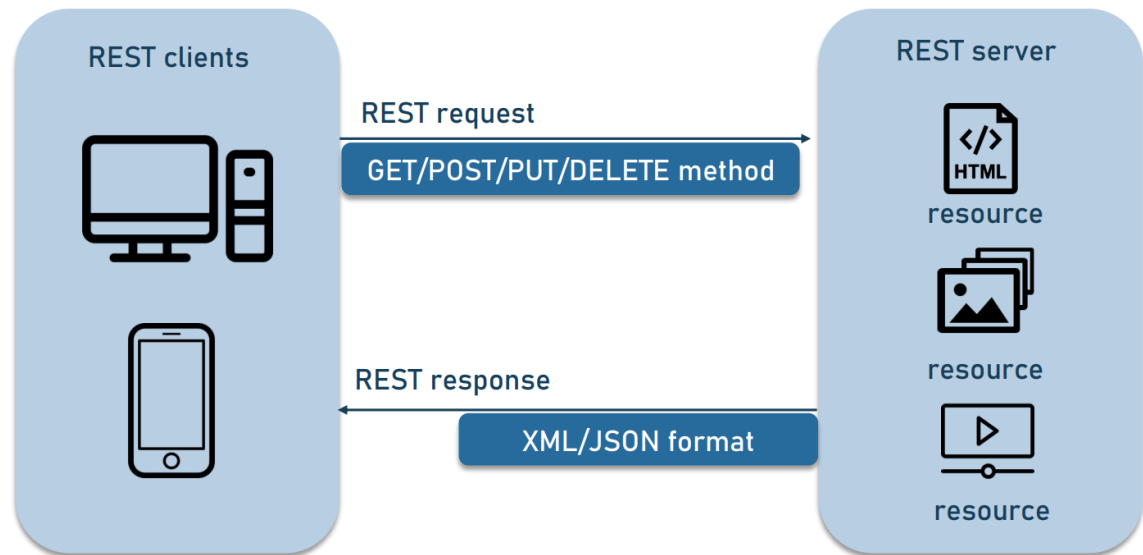
- ❖ **API** adalah sebuah software yang mengintegrasikan antara aplikasi yang dibuat dengan aplikasi yang lain. Tujuan pembuatan API adalah untuk saling berbagi data antar aplikasi yang sudah diintegrasikan tersebut.
- ❖ **REST API** merupakan salah satu dari desain arsitektur yang terdapat di dalam API itu sendiri.
- ❖ **RESTful API** merupakan cara kerja atau proses dari **REST client** (*request*) akan melakukan akses pada data/resource pada **REST server** (*respond*) dimana masing-masing *resource* atau data/resource tersebut akan dibedakan oleh sebuah global ID atau URIs (Universal Resource Identifiers).

REST client on RESTful API is an application or program that uses HTTP to communicate with a server that provides RESTful API. REST clients are usually used by developers or systems that **want to retrieve data or perform certain operations** on resources provided by the server through the API.

REST API Concepts

To get access to a resource, the **client sends an HTTP request**. In return, the server generates an HTTP response with encoded data on the resource. Both types of REST messages are self-descriptive, meaning they contain information on how to interpret and process them.

REST is short for
Representational
State Transfer.

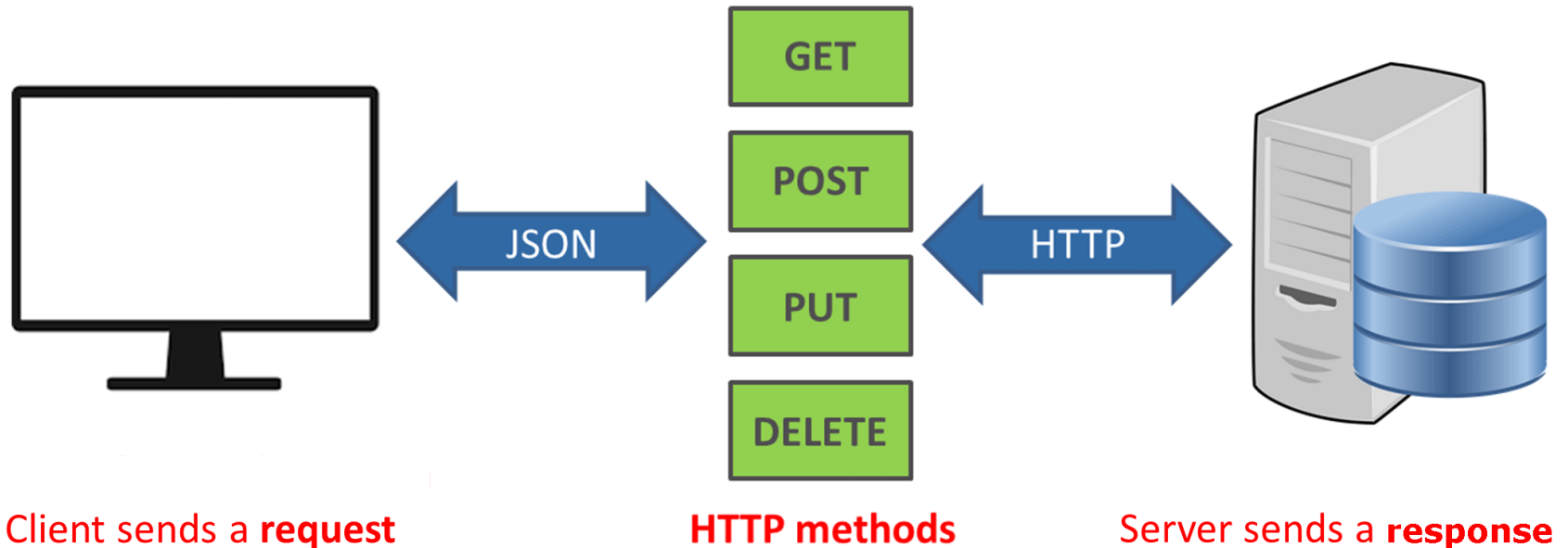


The key elements of the REST API paradigm are:

- 1) a **client** or software that runs on a user's computer or smartphone and initiates communication;
- 2) a **server** that offers an API as a means of access to its data or features; and
- 3) a **resource**, which is any piece of content that the server can provide to the client (for example, a video or a text file).

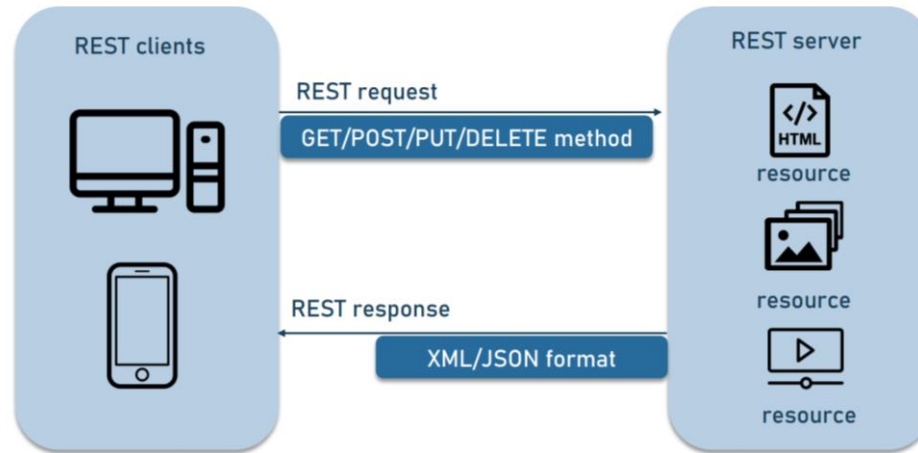
What is RESTful API ?

- 1) RESTful API stands for **RE**presentational **State** Transfer **Application** Programming Interface. It is a *software architectural style* for building *web services* that allows **communication between clients and servers** through the HTTP protocol.
- 2) A RESTful API typically operates by exposing resources or data as unique Uniform Resource Locator (URL). Clients can interact with these resources by making **HTTP requests such as GET, POST, PUT, and DELETE**, which correspond to read, create, update, and delete operations, respectively.
- 3) The server then responds with the requested data in a format such as **JSON, XML, or plain text**.



Dalam konteks RESTful API, state atau statelessness mengacu pada sifat dari interaksi antara client dan server.

API Architecture



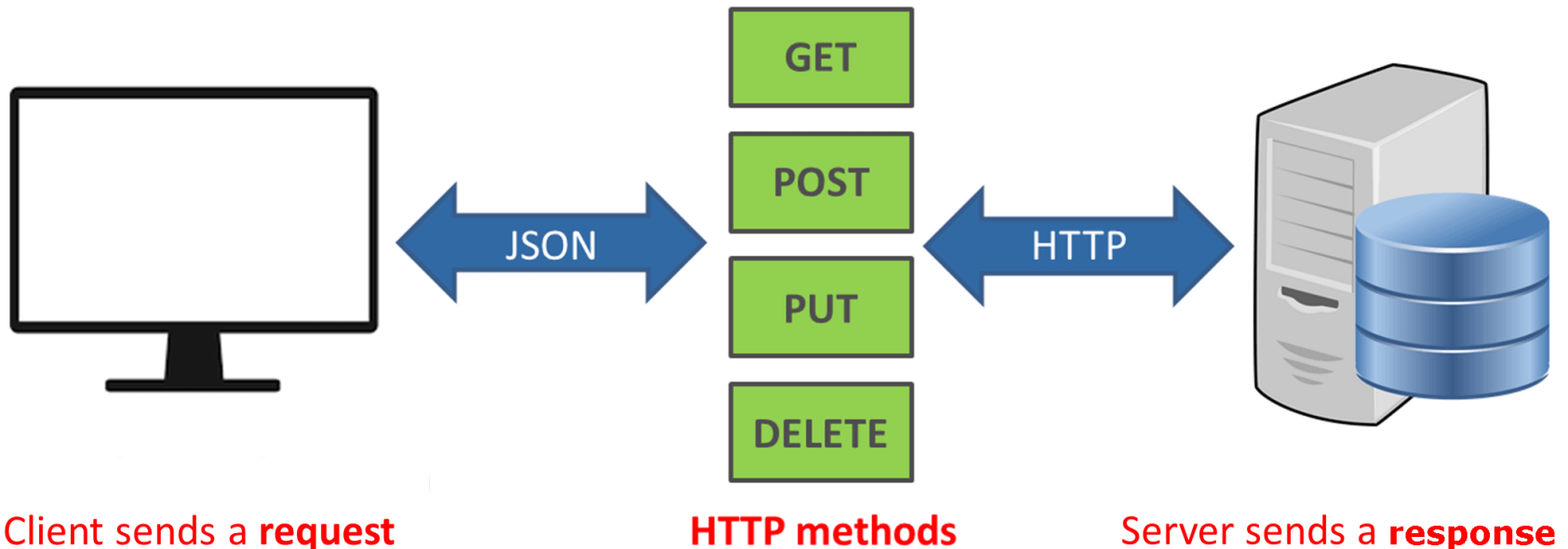
- 1) Dalam konteks REST (Representational State Transfer) API, **sebuah "endpoint"** adalah **URL tertentu** yang diekspos oleh API tersebut untuk berinteraksi dengan sumber daya-sumber dayanya.
- 2) **Endpoint digunakan untuk melakukan berbagai operasi pada sumber daya**, seperti mengambil data, membuat sumber daya baru, memperbarui sumber daya yang sudah ada, atau menghapus sumber daya.
- 3) Setiap **endpoint** biasanya sesuai dengan sumber daya tertentu atau kumpulan sumber daya dalam API.
- 4) An API endpoint is a specific location within an API that **accepts requests and sends back responses**. It's a way for different systems and applications to communicate with each other, by sending and receiving information and instructions via the endpoint.

HTTP Methods

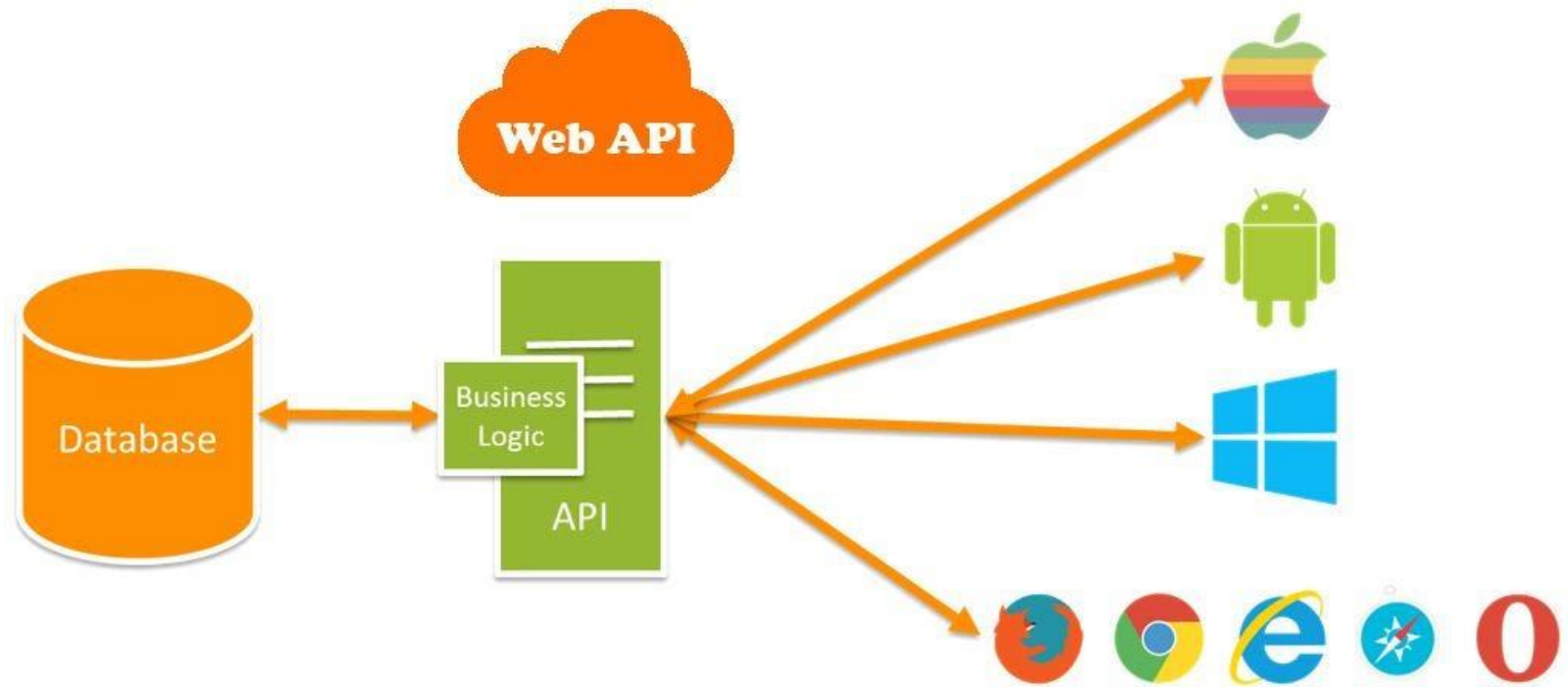
Hypertext Transfer Protocol (HTTP) adalah sebuah jalan atau cara komunikasi dapat terjalin antara client dengan server.

Dalam HTTP Method REST yang sering digunakan antara lain:

- ❖ **GET** : Fungsi ini digunakan untuk perintah menampilkan data.
- ❖ **POST**: Fungsi ini biasanya digunakan untuk perintah dalam menambahkan data (*secure*).
- ❖ **PUT**: Fungsi ini biasanya digunakan untuk perintah edit data.
- ❖ **DELETE**: Fungsi ini biasanya digunakan untuk perintah dalam menghapus data.



What is RESTful API ?



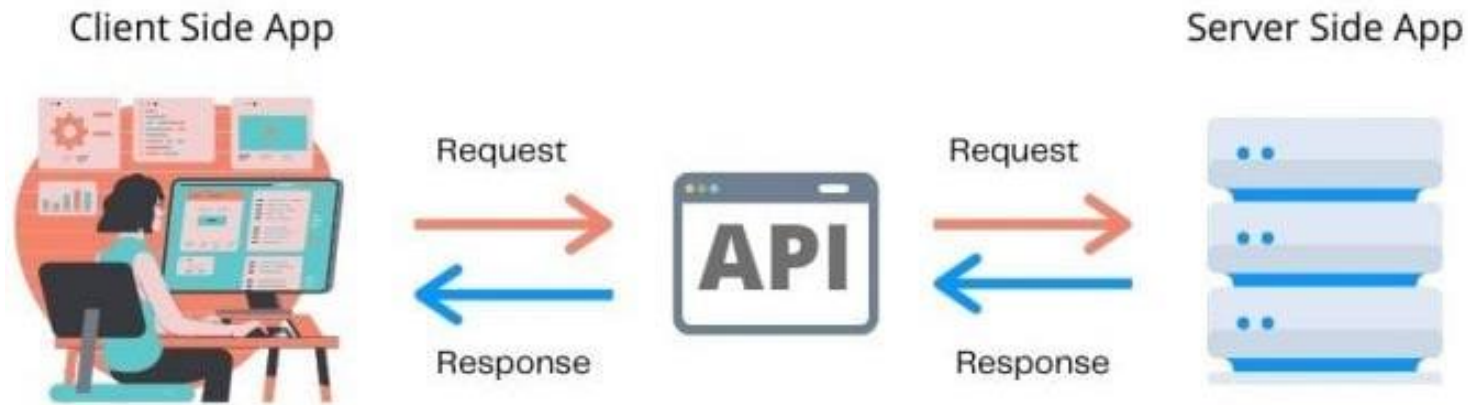
RESTful API is a method for building web applications that allows a **communication system between a client and a server** via the HTTP protocol.

RESTful API can be built using various programming languages, including PHP.

PHP frameworks that support the creation of RESTful APIs such as Laravel, CodeIgniter, and Slim Framework.

Prinsip-prinsip RESTful API

Dalam konteks pembangunan RESTful API menggunakan PHP Framework, developer harus memastikan bahwa API yang dibangun memenuhi prinsip-prinsip RESTful seperti:



- 1) Menggunakan HTTP verbs untuk mengindikasikan tindakan yang dilakukan oleh klien (GET, POST, PUT, DELETE).
- 2) Menggunakan URI untuk mengidentifikasi sumber daya (resource) yang diakses oleh klien.
- 3) Mengirimkan respons dalam format yang sesuai, seperti JSON atau XML.
- 4) Tidak menyimpan status klien pada server, sehingga setiap permintaan (request) harus mencakup semua informasi yang diperlukan.

Web API vs RESTful API

Web API (Application Programming Interface) dan RESTful API (Representational State Transfer API) adalah dua jenis API yang berbeda dalam hal **desain** dan **konsep pengembangan**.

Web API adalah API umum yang dapat digunakan untuk mengakses layanan web melalui **protokol komunikasi tertentu** seperti SOAP (Simple Object Access Protocol) dan XML-RPC (XML Remote Procedure Call).

Web API menggunakan konsep Remote Procedure Call (RPC) untuk memanggil fungsi atau prosedur yang disediakan oleh server, dan biasanya menggunakan format data seperti XML atau JSON untuk mengirimkan dan menerima data.

RESTful API adalah salah satu jenis Web API yang didasarkan pada prinsip-prinsip REST, yaitu suatu gaya arsitektur yang diterapkan pada pembuatan web service.

RESTful API menggunakan **metode HTTP verbs** seperti GET, POST, PUT, dan DELETE untuk memanipulasi sumber daya yang disediakan oleh server.

RESTful API juga **memanfaatkan** representasi sederhana dari sumber daya (*resource*), yaitu **URI (Uniform Resource Identifier)** **untuk mengidentifikasi sumber daya** yang ingin diakses atau dimanipulasi.

Apa Manfaat API RESTful?

Skalabilitas

Sistem yang menerapkan API REST dapat menskalakan secara efisien karena REST mengoptimalkan interaksi klien-server. Statelessness menghapus beban server karena server tidak perlu mempertahankan informasi permintaan klien di masa lalu. Pembuatan cache yang dikelola dengan baik secara parsial atau keseluruhan menghilangkan beberapa interaksi klien-server. Semua fitur ini mendukung skalabilitas tanpa menyebabkan kemacetan komunikasi yang mengurangi performa.

Fleksibilitas

Layanan web RESTful mendukung pemisahan total klien-server. Layanan web RESTful menyederhanakan dan memisahkan berbagai komponen server sehingga masing-masing bagian dapat berkembang secara mandiri. Perubahan platform atau teknologi pada aplikasi server tidak memengaruhi aplikasi klien. Kemampuan untuk melapisi fungsi aplikasi semakin meningkatkan fleksibilitas lebih jauh. Contohnya, developer dapat membuat perubahan pada lapisan basis data tanpa menulis ulang logika aplikasi.

Independensi

API REST independen terhadap teknologi yang digunakan. Anda dapat menulis baik aplikasi klien dan server dalam berbagai bahasa pemrograman tanpa memengaruhi desain API. Anda juga dapat mengubah teknologi mendasar di kedua sisi tanpa memengaruhi komunikasi.

Source: <https://aws.amazon.com/id/what-is/restful-api/>

Format Data API

- ❖ JSON (JavaScript Object Notation) banyak digunakan karena dapat dibaca oleh orang dan mesin dan unggul dalam membantu mempromosikan aksesibilitas web.
- ❖ JSON terutama digunakan untuk mengirim badan permintaan dalam aplikasi web dan aplikasi klien. Ini adalah bentuk yang kompatibel dengan berbagai pengkodean.
- ❖ Format data API selain JSON melibatkan XML , YAML, CSV, HTML, dan teks biasa.



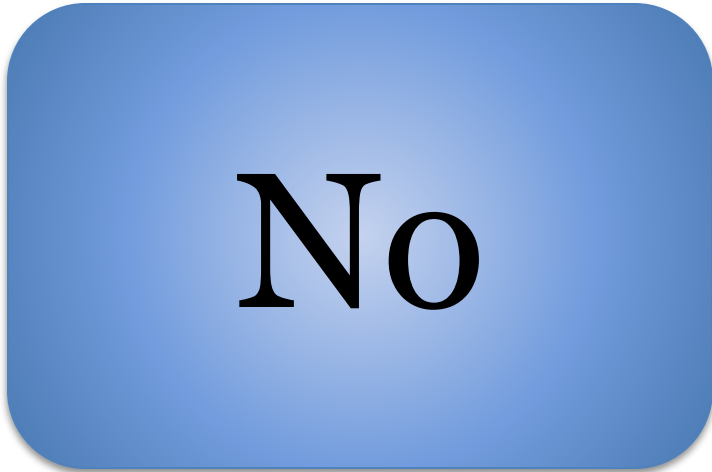
```
1 users : [  
2   {  
3     "userId": 1,  
4     "firstName": "Chris",  
5     "lastName": "Lee",  
6     "phoneNumber": "555-555-5555",  
7     "emailAddress": "cleef@fileinfo.com"  
8   },  
9   {  
10    "userId": 2,  
11    "firstName": "Action",  
12    "lastName": "Jackson",  
13    "phoneNumber": "555-555-5556",  
14    "emailAddress": "ajackson@fileinfo.com"  
15  },  
16  {  
17    "userId": 3,  
18    "firstName": "Ross",  
19    "lastName": "Bing",  
20    "phoneNumber": "555-555-5557",  
21    "emailAddress": "rbing@fileinfo.com"  
22  },  
23  {  
24    "userId": 4,  
25    "firstName": "David",  
26    "lastName": "Reeves",  
27    "phoneNumber": "555-555-5558",  
28    "emailAddress": "dreeves@fileinfo.com"  
29  },  
30  {  
31    "userId": 5,  
32    "firstName": "Josie",  
33    "lastName": "Mac",  
34    "phoneNumber": "555-555-5559",  
35    "emailAddress": "jmac@fileinfo.com"  
36  }  
37 ]
```

Dalam penerapannya, apakah kita harus melakukan proses autentikasi (*metode*) secara langsung saat menerapkan RESTful API?

Proses autentikasi adalah proses untuk memverifikasi identitas atau keaslian pengguna yang ingin mengakses sistem atau layanan. Proses autentikasi bertujuan untuk memastikan bahwa pengguna yang meminta akses benar-benar adalah orang yang berhak untuk mengakses sistem atau layanan tersebut.



Yes



No

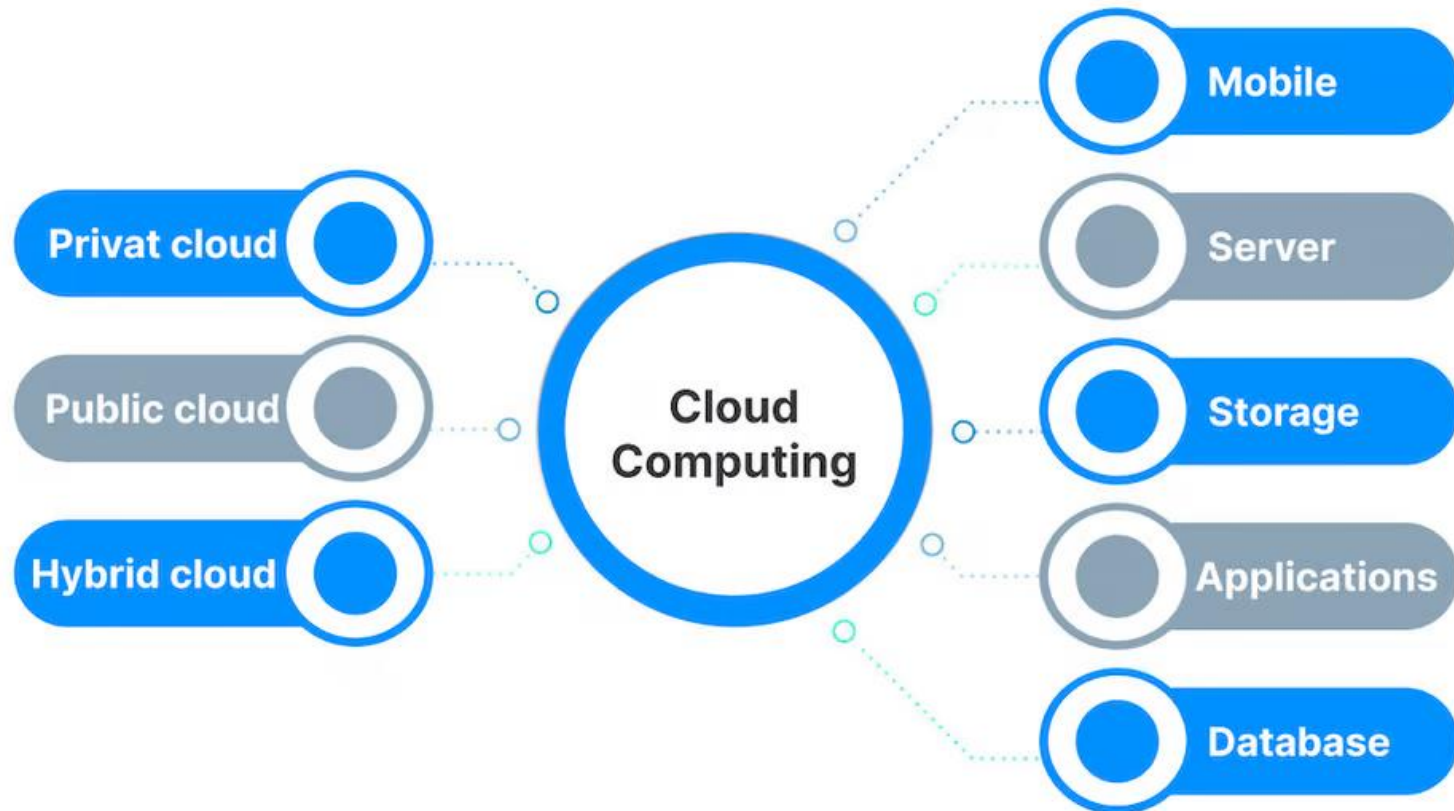
A decorative border of green leaves and branches at the top of the slide.

2 – RESTful API Examples

A decorative border of green leaves and branches at the bottom of the slide.

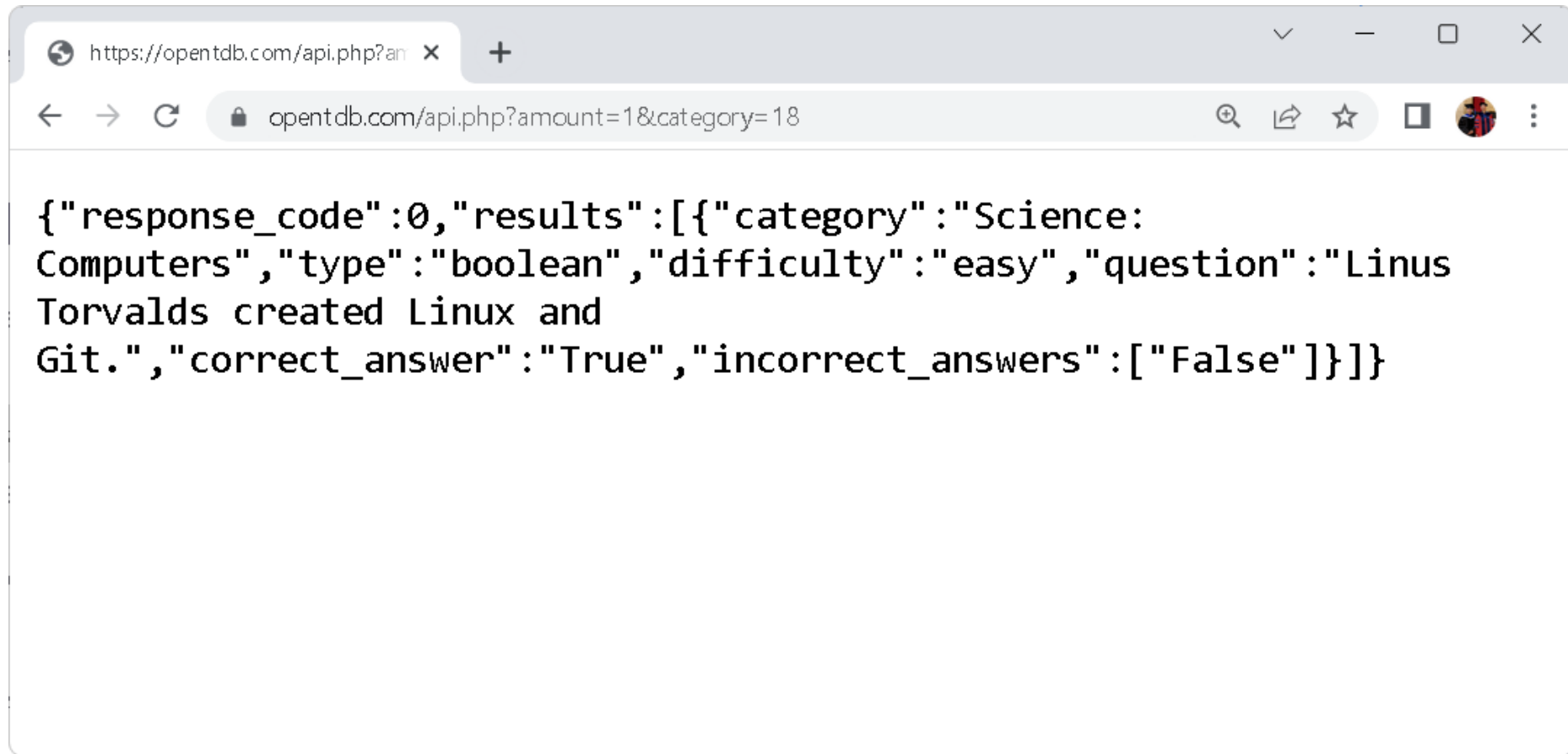
Untuk apa RESTful API digunakan?

RESTful API sangat berguna dalam **pengembangan aplikasi berbasis cloud** karena memungkinkan *users* untuk mengakses dan memanipulasi data dari *server cloud* melalui protokol HTTP dengan cara yang sederhana, aman dan efisien.



RESTful API Web Service

Layanan web RESTful seperti di bawah ini digunakan untuk menyediakan **API web public**.



Keamanan API REST :

- ❖ Manfaatkan HTTPS
- ❖ Gunakan mekanisme otentikasi yang kuat

RESTful API OpenWeatherMap

`api.openweathermap.org/data/2.5/forecast?q={city name}&appid={API key}`

```
{
  "cod": "200",
  "message": 0,
  "cnt": 40,
  "list": [
    {
      "dt": 1611381600,
      "main": {
        "temp": 303.07,
        "feels_like": 305.04,
        "temp_min": 303.07,
        "temp_max": 304.01,
        "pressure": 1009,
        "sea_level": 1009,
        "grnd_level": 993,
        "humidity": 61,
        "temp_kf": -0.94,
        "weather": [
          {
            "id": 500,
            "main": "Rain",
            "description": "light rain",
            "icon": "10d"
          }
        ],
        "clouds": {
          "all": 98
        },
        "wind": {
          "speed": 3.57,
          "deg": 338
        },
        "visibility": 10000,
        "pop": 0.33,
        "rain": {
          "3h": 0.25
        },
        "sys": {
          "pod": "d"
        },
        "dt_txt": "2021-01-23 06:00:00"
      },
      "dt": 1611392400,
      "main": {
        "temp": 301.12,
        "feels_like": 304.48,
        "temp_min": 300.83,
        "temp_max": 301.12,
        "pressure": 1008,
        "sea_level": 1008,
        "grnd_level": 992,
        "humidity": 74,
        "temp_kf": 0.29,
        "weather": [
          {
            "id": 501,
            "main": "Rain",
            "description": "moderate rain",
            "icon": "10d"
          }
        ],
        "clouds": {
          "all": 100
        },
        "wind": {
          "speed": 2.6,
          "deg": 343
        },
        "visibility": 10000,
        "pop": 0.97,
        "rain": {
          "3h": 4.28
        },
        "sys": {
          "pod": "d"
        },
        "dt_txt": "2021-01-23 09:00:00"
      },
      "dt": 1611403200,
      "main": {
        "temp": 298.61,
        "feels_like": 303.68,
        "temp_min": 298.37,
        "temp_max": 298.61,
        "pressure": 1009,
        "sea_level": 1009,
        "grnd_level": 994,
        "humidity": 86,
        "temp_kf": 0.24,
        "weather": [
          {
            "id": 501,
            "main": "Rain",
            "description": "moderate rain",
            "icon": "10n"
          }
        ],
        "clouds": {
          "all": 100
        },
        "wind": {
          "speed": 0.2,
          "deg": 23
        },
        "visibility": 10000,
        "pop": 0.98,
        "rain": {
          "3h": 5.34
        },
        "sys": {
          "pod": "n"
        },
        "dt_txt": "2021-01-23 12:00:00"
      },
      "dt": 1611414000,
      "main": {
        "temp": 297.94,
        "feels_like": 302.78,
        "temp_min": 297.91,
        "temp_max": 297.94,
        "pressure": 1011,
        "sea_level": 1011,
        "grnd_level": 996,
        "humidity": 89,
        "temp_kf": 0.03,
        "weather": [
          {
            "id": 501,
            "main": "Rain",
            "description": "moderate rain",
            "icon": "10n"
          }
        ],
        "clouds": {
          "all": 100
        },
        "wind": {
          "speed": 0.46,
          "deg": 26
        },
        "visibility": 5779,
        "pop": 0.94,
        "rain": {
          "3h": 6.37
        },
        "sys": {
          "pod": "n"
        },
        "dt_txt": "2021-01-23 15:00:00"
      },
      "dt": 1611424800,
      "main": {
        "temp": 297.42,
        "feels_like": 302.23,
        "temp_min": 297.42,
        "temp_max": 297.42,
        "pressure": 1009,
        "sea_level": 1009,
        "grnd_level": 994,
        "humidity": 90,
        "temp_kf": 0,
        "weather": [
          {
            "id": 501,
            "main": "Rain",
            "description": "moderate rain",
            "icon": "10n"
          }
        ],
        "clouds": {
          "all": 100
        },
        "wind": {
          "speed": 0.25,
          "deg": 218
        },
        "visibility": 6163,
        "pop": 0.99,
        "rain": {
          "3h": 3.66
        },
        "sys": {
          "pod": "n"
        },
        "dt_txt": "2021-01-23 18:00:00"
      },
      "dt": 1611435600,
      "main": {
        "temp": 297.21,
        "feels_like": 301.35,
        "temp_min": 297.21,
        "temp_max": 297.21,
        "pressure": 1009,
        "sea_level": 1009,
        "grnd_level": 994,
        "humidity": 87,
        "temp_kf": 0,
        "weather": [
          {
            "id": 500,
            "main": "Rain",
            "description": "light rain",
            "icon": "10n"
          }
        ],
        "clouds": {
          "all": 99
        },
        "wind": {
          "speed": 0.62,
          "deg": 189
        },
        "visibility": 10000,
        "pop": 0.94,
        "rain": {
          "3h": 0.92
        },
        "sys": {
          "pod": "n"
        },
        "dt_txt": "2021-01-23 21:00:00"
      },
      "dt": 1611446400,
      "main": {
        "temp": 298.34,
        "feels_like": 302.84,
        "temp_min": 298.34,
        "temp_max": 298.34,
        "pressure": 1010,
        "sea_level": 1010,
        "grnd_level": 995,
        "humidity": 82,
        "temp_kf": 0,
        "weather": [
          {
            "id": 500,
            "main": "Rain",
            "description": "light rain",
            "icon": "10d"
          }
        ],
        "clouds": {
          "all": 98
        },
        "wind": {
          "speed": 0.2,
          "deg": 355
        },
        "visibility": 10000,
        "pop": 0.9,
        "rain": {
          "3h": 0.78
        },
        "sys": {
          "pod": "d"
        },
        "dt_txt": "2021-01-24 00:00:00"
      },
      "dt": 1611457200,
      "main": {
        "temp": 302.07,
        "feels_like": 305.59,
        "temp_min": 302.07,
        "temp_max": 302.07,
        "pressure": 1011,
        "sea_level": 1011,
        "grnd_level": 996,
        "humidity": 68,
        "temp_kf": 0,
        "weather": [
          {
            "id": 501,
            "main": "Rain",
            "description": "light rain",
            "icon": "10d"
          }
        ],
        "clouds": {
          "all": 47
        },
        "wind": {
          "speed": 1.99,
          "deg": 327
        },
        "visibility": 10000,
        "pop": 0.35,
        "rain": {
          "3h": 0.25
        },
        "sys": {
          "pod": "d"
        },
        "dt_txt": "2021-01-24 03:00:00"
      },
      "dt": 1611468000,
      "main": {
        "temp": 302.85,
        "feels_like": 305.88,
        "temp_min": 302.85,
        "temp_max": 302.85,
        "pressure": 1009,
        "sea_level": 1009,
        "grnd_level": 994,
        "humidity": 67,
        "temp_kf": 0,
        "weather": [
          {
            "id": 500,
            "main": "Rain",
            "description": "light rain",
            "icon": "10d"
          }
        ],
        "clouds": {
          "all": 63
        },
        "wind": {
          "speed": 3.08,
          "deg": 342
        },
        "visibility": 10000,
        "pop": 0.69,
        "rain": {
          "3h": 1.39
        },
        "sys": {
          "pod": "d"
        },
        "dt_txt": "2021-01-24 06:00:00"
      },
      "dt": 1611478800,
      "main": {
        "temp": 298.49,
        "feels_like": 302.89,
        "temp_min": 298.49,
        "temp_max": 298.49,
        "pressure": 1008,
        "sea_level": 1008,
        "grnd_level": 993,
        "humidity": 86,
        "temp_kf": 0,
        "weather": [
          {
            "id": 501,
            "main": "Rain",
            "description": "moderate rain",
            "icon": "10d"
          }
        ],
        "clouds": {
          "all": 75
        },
        "wind": {
          "speed": 1.06,
          "deg": 46
        },
        "visibility": 2949,
        "pop": 1,
        "rain": {
          "3h": 8.43
        },
        "sys": {
          "pod": "d"
        },
        "dt_txt": "2021-01-24 09:00:00"
      },
      "dt": 1611489600,
      "main": {
        "temp": 297.27,
        "feels_like": 301.25,
        "temp_min": 297.27,
        "temp_max": 297.27,
        "pressure": 1010,
        "sea_level": 1010,
        "grnd_level": 995,
        "humidity": 88,
        "temp_kf": 0,
        "weather": [
          {
            "id": 501,
            "main": "Rain",
            "description": "moderate rain",
            "icon": "10n"
          }
        ],
        "clouds": {
          "all": 88
        },
        "wind": {
          "speed": 1.03,
          "deg": 170
        },
        "visibility": 10000,
        "pop": 1,
        "rain": {
          "3h": 10.03
        },
        "sys": {
          "pod": "n"
        },
        "dt_txt": "2021-01-24 12:00:00"
      },
      "dt": 1611500400,
      "main": {
        "temp": 296.48,
        "feels_like": 299.76,
        "temp_min": 296.48,
        "temp_max": 296.48,
        "pressure": 1012,
        "sea_level": 1012,
        "grnd_level": 997,
        "humidity": 86,
        "temp_kf": 0,
        "weather": [
          {
            "id": 501,
            "main": "Rain",
            "description": "moderate rain",
            "icon": "10n"
          }
        ],
        "clouds": {
          "all": 100
        },
        "wind": {
          "speed": 1.18,
          "deg": 207
        },
        "visibility": 10000,
        "pop": 0.96,
        "rain": {
          "3h": 4.08
        },
        "sys": {
          "pod": "n"
        },
        "dt_txt": "2021-01-24 15:00:00"
      },
      "dt": 1611511200,
      "main": {
        "temp": 296.52,
        "feels_like": 300,
        "temp_min": 296.52,
        "temp_max": 296.52,
        "pressure": 1010,
        "sea_level": 1010,
        "grnd_level": 995,
        "humidity": 85,
        "temp_kf": 0,
        "weather": [
          {
            "id": 500,
            "main": "Rain",
            "description": "light rain",
            "icon": "10n"
          }
        ],
        "clouds": {
          "all": 100
        },
        "wind": {
          "speed": 0.8,
          "deg": 159
        },
        "visibility": 10000,
        "pop": 0.93,
        "rain": {
          "3h": 0.68
        },
        "sys": {
          "pod": "n"
        },
        "dt_txt": "2021-01-24 18:00:00"
      },
      "dt": 1611522000,
      "main": {
        "temp": 295.85,
        "feels_like": 298.27,
        "temp_min": 295.85,
        "temp_max": 295.85,
        "pressure": 1009,
        "sea_level": 1009,
        "grnd_level": 994,
        "humidity": 82,
        "temp_kf": 0,
        "weather": [
          {
            "id": 500,
            "main": "Rain",
            "description": "light rain",
            "icon": "10n"
          }
        ],
        "clouds": {
          "all": 100
        },
        "wind": {
          "speed": 1.46,
          "deg": 195
        },
        "visibility": 10000,
        "pop": 0.4,
        "rain": {
          "3h": 0.91
        },
        "sys": {
          "pod": "n"
        },
        "dt_txt": "2021-01-24 21:00:00"
      },
      "dt": 1611532800,
      "main": {

```

Manage Book List Using RESTful API

1. ``GET /buku``: Endpoint ini digunakan untuk mengambil daftar semua buku.
2. ``GET /buku/{id}``: Endpoint ini digunakan untuk mengambil buku tertentu berdasarkan pengidentifikasi uniknya.
3. ``POST /buku``: Endpoint ini digunakan untuk membuat buku baru.
4. ``PUT /buku/{id}``: Endpoint ini digunakan untuk memperbarui buku tertentu berdasarkan pengidentifikasi uniknya.
5. ``DELETE /buku/{id}``: Endpoint ini digunakan untuk menghapus buku tertentu berdasarkan pengidentifikasi uniknya.

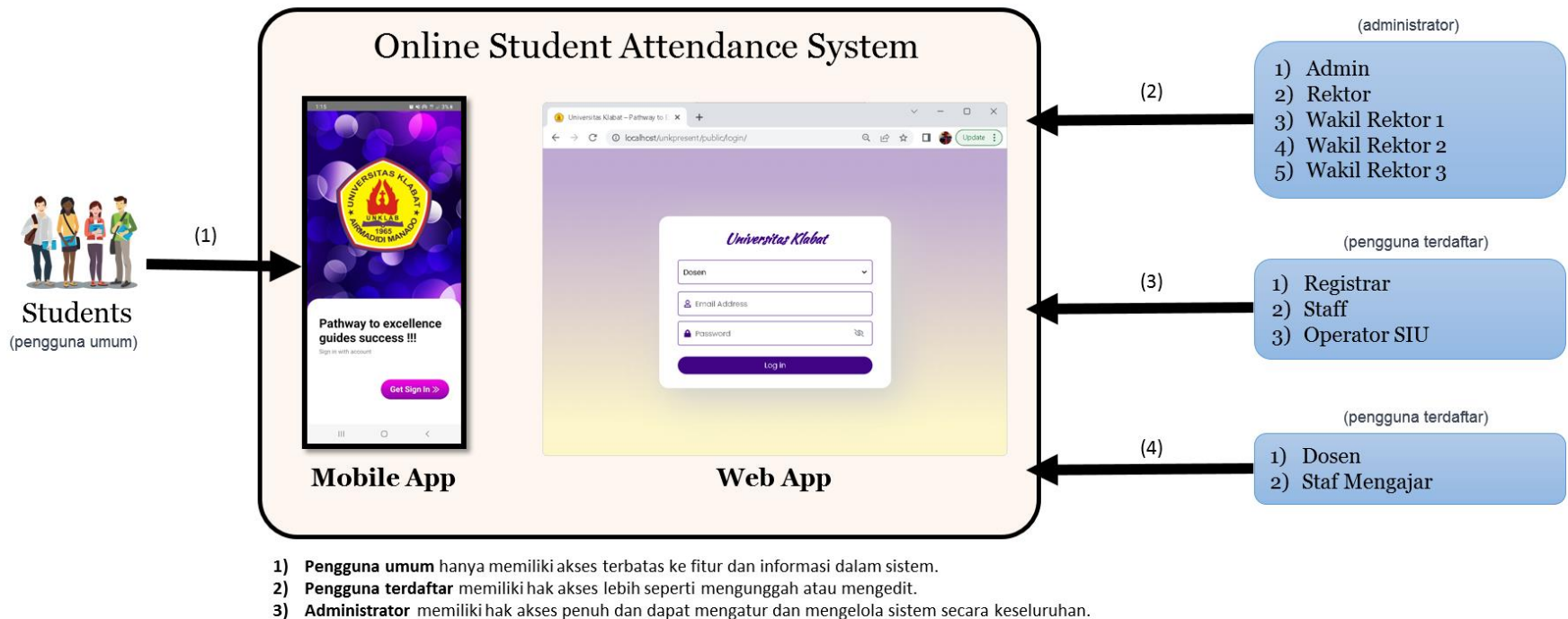
The term "**permanent endpoint**" has no specific meaning in the context of REST APIs. However, in general, **ENDPOINTS in REST APIs** are designed to be relatively stable and not change with high frequency. **Changing endpoints can be disruptive to clients who rely on the API.** Therefore, it is advisable to design and maintain endpoint stability well.

A decorative border of green leaves and branches at the top of the slide.

3 –System Kehadiran Mahasiswa (*fitur QR-code dan Geo-location*)

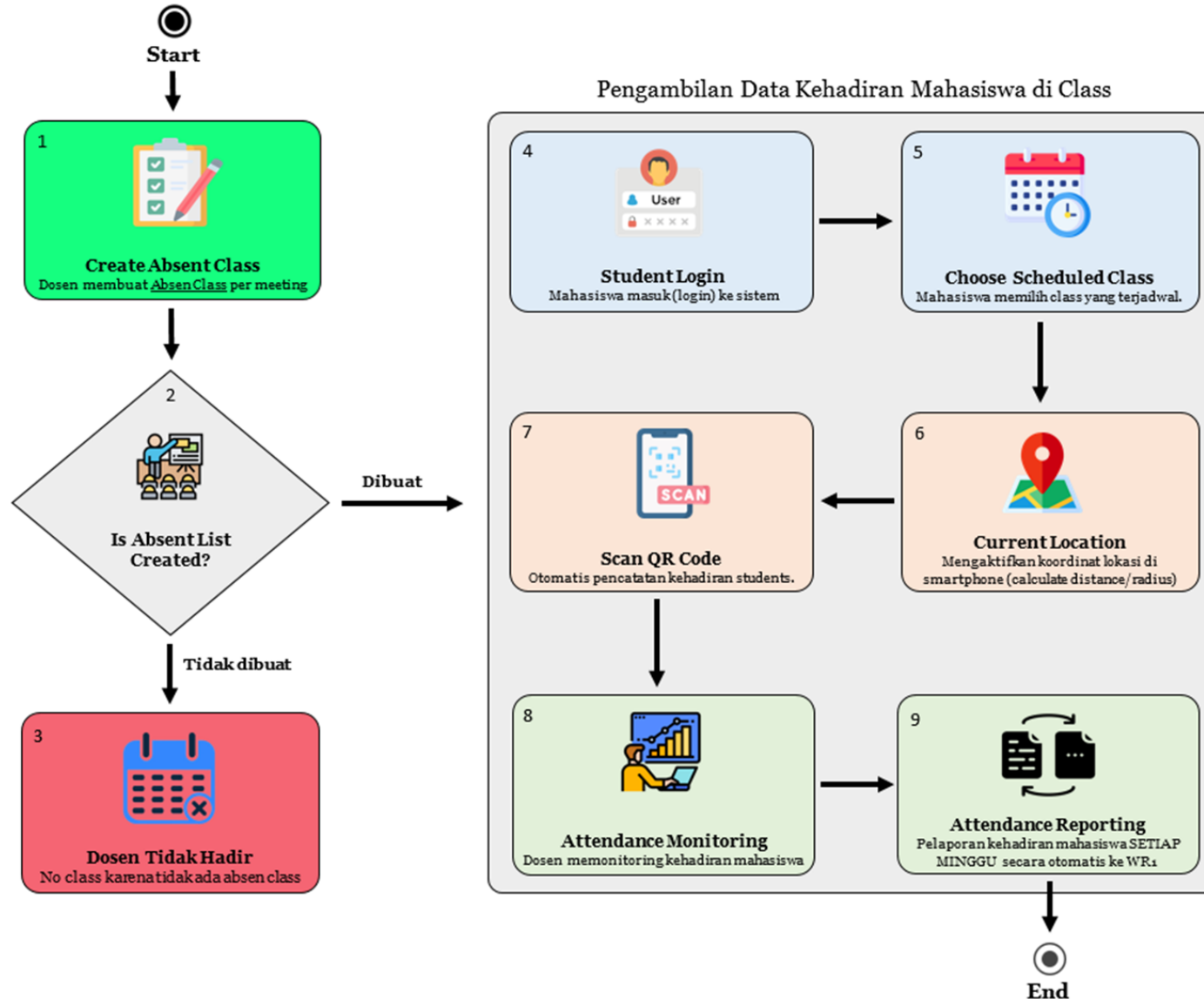
A decorative border of green leaves and branches at the bottom of the slide.

Online Student Attendance System

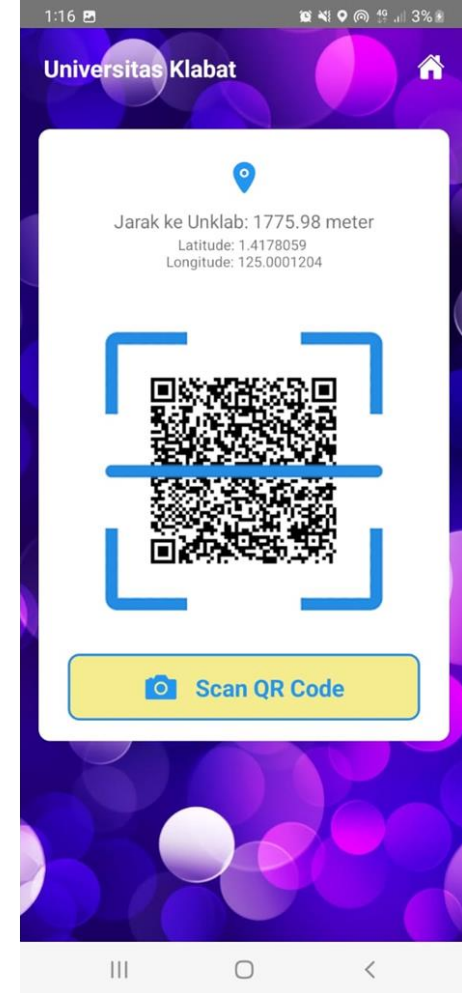
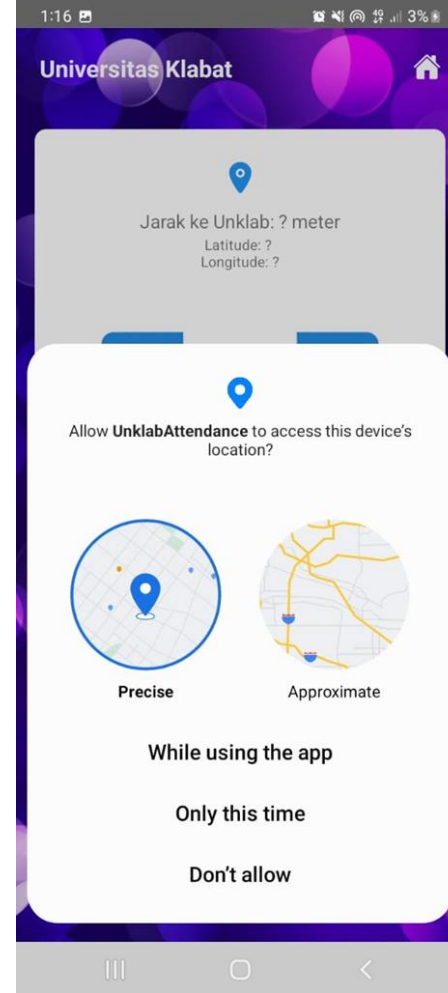
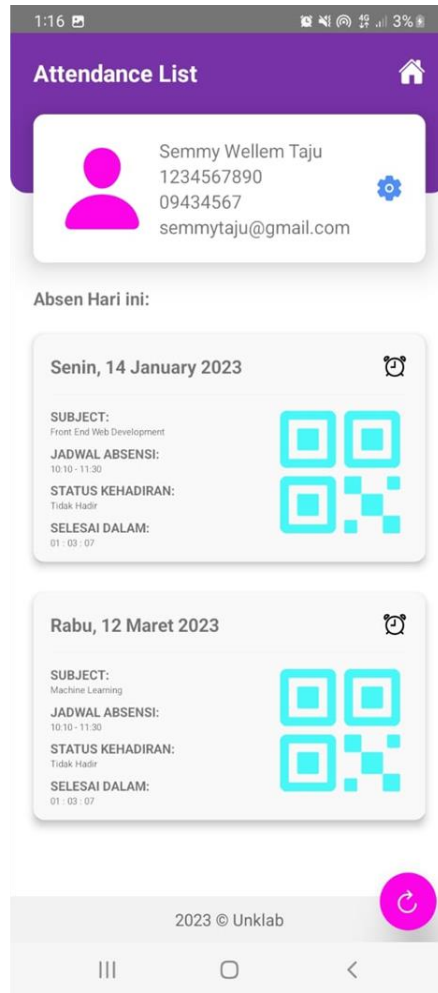
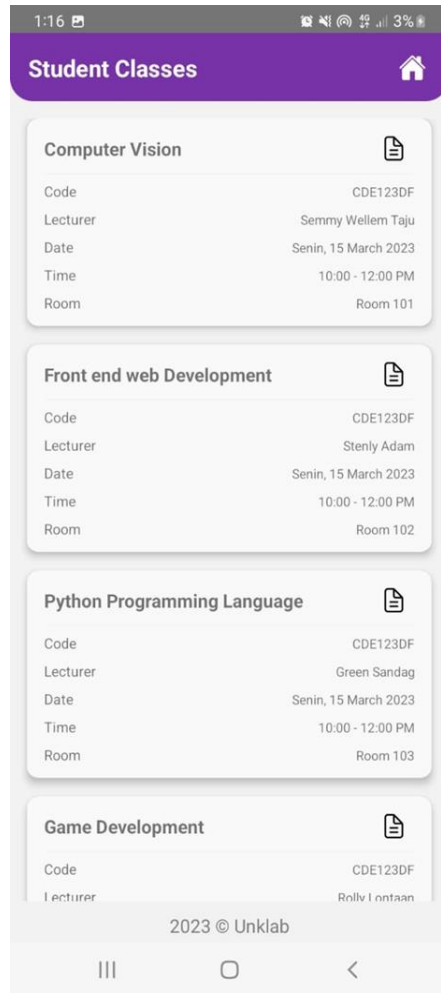


Sistem absensi mahasiswa secara online dengan dua aplikasi (*mobile app* dan *web app*) memiliki tujuan untuk memudahkan *students* dan *staf* dalam melakukan absensi dan pengelolaan kehadiran mahasiswa di kelas.

Alur Pengambil Presensi Class



Mobile Applications



Web Application

Active Class

localhost/unkpresent/public/dosen/active/

Sign out

Universitas Klabat

Search

USER MENU

Home

Profile

Settings

Logout

SHORTCUT LECTURER

Create Class

Active Classes

Archived Classes

Attendance

SHORTCUT ADMIN

Create User

Manage Users

Reports

Active Classes

Add New Class

Back

Course	People/Room	Schedule	Status	Actions
APC496 Fundamental Apache PHP5 Fakultas Ilmu Komputer Semmy Wellem Taju	3 Students GK2-103	Senin ~ Rabu 09.30 - 11.00	active	Options
GAC3678 Sentiment Analysis Fakultas Ilmu Komputer Semmy Wellem Taju	3 Students GK2-103	Selasa ~ Kamis 09.00 - 10.00	active	Options
FED3464 Front-end Web Development Akademi Sekretari Manajemen Indonesia Raissa Maringka	4 Students GA Computer Lab 202/FIK	Senin ~ Rabu 14.30 - 16.00	active	Options
DBA984 Big Data Analysis Fakultas Ilmu Komputer Stelly Adam	7 Students GK3-203	Senin ~ Rabu 17.00 - 18.00	active	Options
MLC4335				

Web Application

Attendance

localhost/unkpresent/public/students/attendance/CLASS6445f8b7ce4de

Universitas Klabat

Search

Sign out

USER MENU

Home

Profile

Settings

Logout

SHORTCUT LECTURER

Create Class

Active Classes

Archived Classes

Attendance

SHORTCUT ADMIN

Create User

Manage Users

Reports

Class Attendance Checklist

Course: [APC496] Fundamental Apache PHP5

New +

Summary

Back →

Attendance	Schedule	Present/Total	Radius	Actions
Attendance #6 Semmy Wellem Taju semmy@unklab.ac.id	Monday 08 May 2023 08.00 - 09.00	2/3 Students	80 Meter	Options
Attendance #5 Semmy Wellem Taju semmy@unklab.ac.id	Thursday 04 May 2023 07.00 - 09.00	0/3 Students	60 Meter	Options
Attendance #4 Semmy Wellem Taju semmy@unklab.ac.id	Friday 05 May 2023 07.00 - 09.00	0/3 Students	60 Meter	Options
Attendance #3 Semmy Wellem Taju semmy@unklab.ac.id	Thursday 04 May 2023 08.00 - 09.00	0/3 Students	70 Meter	Options
Attendance #2 Semmy Wellem Taju semmy@unklab.ac.id	Thursday 27 April 2023 08.00 - 09.00	0/3 Students	150 Meter	Options

Web Application

The screenshot displays a web application interface for managing attendance. A modal titled "Create New Attendance Schedule" is open, allowing users to create a new attendance schedule. The modal contains the following fields:

- Title Schedule:** A text input field with the placeholder text "Judul jadwal presensi ('example: attendance #1')".
- Course:** A dropdown menu currently showing "Fundamental Apache PHP5".
- Lecturer Email:** A text input field containing "semmy@unklab.ac.id".
- Date:** A date picker showing "dd/05/2023".
- Time:** A time picker showing "May 2023" with a calendar view. The calendar shows the month of May 2023, with the 4th of May highlighted.
- Maximum Radius:** A text input field.

At the bottom of the modal, there are two buttons: "Cancel" and "Create Schedule".

The background interface includes a sidebar with a "USER MENU" (Home, Profile, Settings, Logout), "SHORTCUT LECTURER" (Create Class, Active Classes, Archived Classes, Attendance), and "SHORTCUT ADMIN" (Create User, Manage Users, Reports). The top navigation bar shows "Universitas Klabat", a search bar, and a "Sign out" button. The main content area shows a list of attendance schedules with columns for "Title", "Course", "Lecturer", "Date", "Time", "Maximum Radius", and "Actions".

Web Application

Attendance Summary

localhost/unkpresent/public/students/summarize/CLASS643838975a476

Universitas Klabat

Search

Sign out

USER MENU

Home

Profile

Settings

Logout

SHORTCUT LECTURER

Create Class

Active Classes

Archived Classes

Attendance

SHORTCUT ADMIN

Create User

Manage Users

Reports

Student Attendance Summary

[PD23232S] Statistics and Probability

GK2-Tissue Culture Lab

Go to Classes

Back

No.	Students	Number meetings?	Hadir	Late	Izin	Alpa
1	 Radix Basalama s154321@student.unklab.ac.id	5 Meetings	1 times	0 times	1 times	3 times
2	 Unknown moningkatri@gmail.com	5 Meetings	0 times	0 times	0 times	5 times
3	 Unknown brendaabu@yahoo.com	5 Meetings	0 times	0 times	0 times	5 times
4	 Unknown saturundes@gmail.com	5 Meetings	0 times	0 times	0 times	5 times
5	 Unknown eugenie@gmail.com	5 Meetings	0 times	0 times	0 times	5 times
6	 Unknown semmy@yahoo.com	5 Meetings	0 times	0 times	0 times	5 times
7	 Citra Lentari Anada s754321@student.unklab.ac.id	5 Meetings	0 times	1 times	0 times	4 times

Web Application

The screenshot shows a web application development environment with a file explorer on the left, a code editor in the center, and a search bar at the top right. The file explorer shows a project structure with folders like 'controller' and 'model'. The code editor displays PHP code for a 'mobile.php' file. A red box highlights a specific function named 'classes'.

```
36 //}else{
37 //    echo "ERROR";
38 //}
39 }else{
40     echo "ERROR";
41 }
42 }
43
44 // student active class
45 public function classes($email_student = ""){
46     // Cek empty
47     if (!empty($email_student)) {
48
49         // check status login in database
50         $arr_data['classes'] = $this->logic("Mobile_model")->get_all_student_classes($email_student);
51
52         if (!empty($arr_data['classes'])) {
53             //var_dump($arr_data['classes']);
54
55             // to json format
56             $json_data = json_encode($arr_data['classes']);
57             echo $json_data;
58         } else {
59             echo "ERROR";
60         }
61     }else{
62         echo "ERROR";
63     }
64 }
65
66 // student available attendance
67
```

Tebak, REST method yang akan digunakan untuk mengakses fungsi classes diatas?

Web Application

```
localhost/unkpresent/public/mo x +
localhost/unkpresent/public/mobile/classes/semmy@yahoo.com
[{"id_class":"CLASS64383858ec58d","email_lecturer":"semmy@unklab.ac.id","code_class":"AI3456S","name_subject":"Artificial Intelligence","name_lecturer":"Green Sandag","fakultas":"Fakultas Ilmu Komputer","prodi":"Sistem Informasi (Information System)","sks":"3 SKS","building_room":"GK3-205","room_latitude":"1,4178155","room_longitude":"124,9844905","jadwal_class_day_time":"Senin ~ Rabu, 14.30 - 16.00","daftar_email_student":"bobon@yhao.com, jeremy@yahoo.com, s154321@student.unklab.ac.id, s254321@student.unklab.ac.id, s354321@student.unklab.ac.id, semmy@yahoo.com, 04-14 01:14:59","status_class":"active"}, {"id_class":"CLASS643838975a476","email_lecturer":"semmy@unklab.ac.id","code_class":"PD23232S","name_subject":"Statistics and Probability","name_lecturer":"Semmy Wellem Taju","fakultas":"Fakultas Ilmu Komputer","prodi":"Sistem Informasi (Information System)","sks":"3 SKS","building_room":"GK2-Tissue Culture Lab","room_latitude":"1,4178155","room_longitude":"124,9844905","jadwal_class_day_time":"Senin ~ Rabu, 08.00 - 09.00","daftar_email_student":"s154321@student.unklab.ac.id, moningkatri@gmail.com, brendaabu@yahoo.com, saturundes@gmail.com, eugenie@gmail.com, semmy@yahoo.com, s754321@student.unklab.ac.id, 04-14 01:15:47","status_class":"active"}, {"id_class":"CLASS643a940726a3d","email_lecturer":"semmy@unklab.ac.id","code_class":"MLC4335","name_subject":"Machine Learning Algorithms","name_lecturer":"Semmy Wellem Taju","fakultas":"Fakultas Ilmu Komputer","prodi":"Pendidikan Keagamaan Kristen (Religious)","sks":"3 SKS","building_room":"GA Computer Lab 203\\FIK","room_latitude":"1,4178155","room_longitude":"124,9844905","jadwal_class_day_time":"Selasa ~ Kamis, 07.00 - 08.30","daftar_email_student":"semmy@yahoo.com, tajudespies@gmail.com","created_at":"2023-04-15 20:10:28","status_class":"active"}, {"id_class":"CLASS643ac4d56e4a8","email_lecturer":"semmy@unklab.ac.id","code_class":"FED3464","name_subject":"Front-end Web Development","name_lecturer":"Raissa Maringka","fakultas":"Akademi Sekretari Manajemen Indonesia","prodi":"Pendidikan Keagamaan Kristen (Religious)","sks":"3 SKS","building_room":"GA Computer Lab 202\\FIK","room_latitude":"1,4174176","room_longitude":"124,9840658","jadwal_class_day_time":"Senin ~ Rabu, 14.30 - 16.00","daftar_email_student":"budy@gmail.com, s21710268@student.unklab.ac.id, jeremy@unklab.ac.id, semmy@yahoo.com","created_at":"2023-04-15 23:38:52","status_class":"active"}, {"id_class":"CLASS6445f8b7ce4de","email_lecturer":"semmy@unklab.ac.id","code_class":"APC496","name_subject":"Fundamental Apache PHP5","name_lecturer":"Semmy Wellem Taju","fakultas":"Fakultas Ilmu Komputer","prodi":"Pendidikan Keagamaan Kristen (Religious)","sks":"3 SKS","building_room":"GK2-103","room_latitude":"1,4172063","room_longitude":"124,9828819","jadwal_class_day_time":"Senin ~ Rabu, 09.30 - 11.00","daftar_email_student":"semmy@yahoo.com, putra@gmail.com, tampijeni@yahoo.com","created_at":"2023-04-24 11:34:57","status_class":"active"}]
```

A decorative border of green leaves and branches at the top of the slide.

4 –PHP Framework

A decorative border of green leaves and branches at the bottom of the slide.

Framework #1

(File & Directory Structures)

Create Framework Structure

No	Directory/File	Description (<i>optional</i>)
1	<code>app</code>	app directory is the web application's <i>private directory</i> of this new framework.
2	<code>public</code>	<i>public directory</i> is to store all the public static resources like <i>html</i> , <i>css</i> , <i>image</i> and <i>js</i> files.
3	<code>core</code>	core directory is to store all files for routing and it will store the framework's core classes.
4	<code>model</code>	model directory is for all app's Model classes.
5	<code>view</code>	view directory is for all app's View classes.
6	<code>controller</code>	controller directory is for all app's Controller classes.
7	<code>css</code>	css directory is for Cascading Style Sheets (CSS) files.
8	<code>img</code>	Img directory is for images files.
9	<code>js</code>	js directory is for JavaScript files.
10	<code>index.php</code>	index.php file is the 'entry-point' file, the front controller to the users.

Additional Directories:

template directory stores the template files like *header* file and *footer* file.

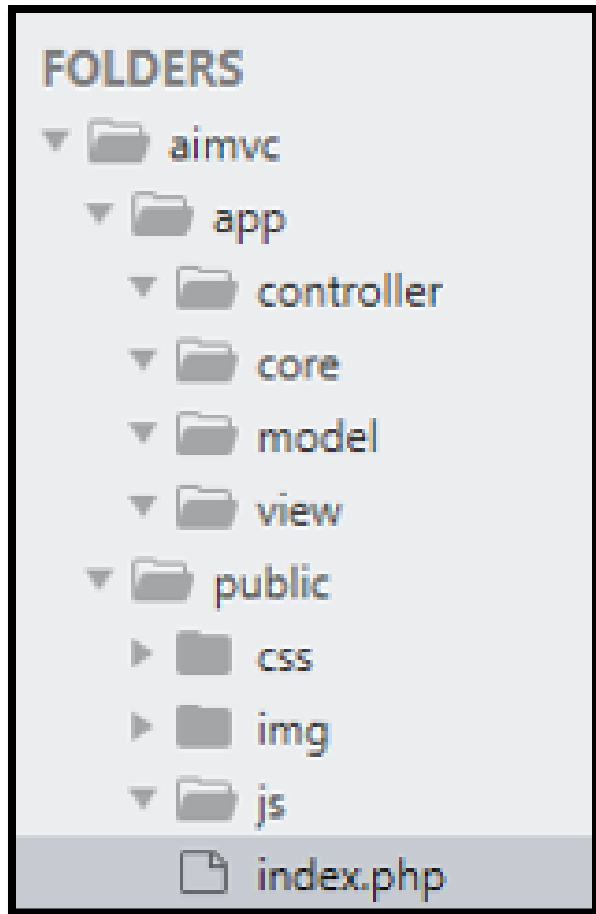
config directory stores the app's configuration files.

upload directory is for uploaded files, such as uploaded images.

db directory is database related classes, such as database driver classes.

helper directory is help/assistant functions.

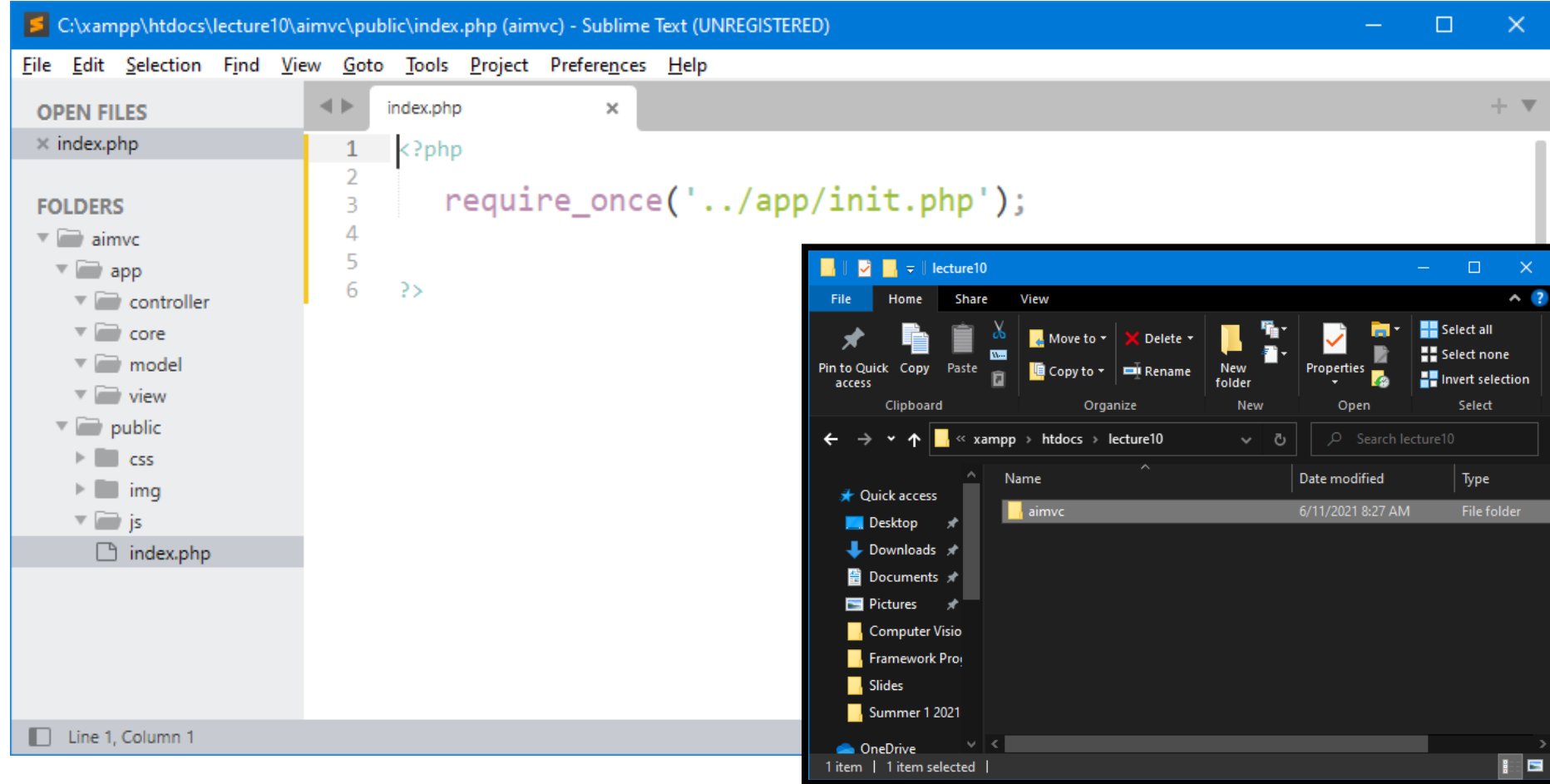
library directory is for class libraries.



As a first step,
Students di minta untuk
mengetahui struktur
folder seperti gambar di
samping.

Basic File & Directory Structures

Directory Path: `C:/xampp/htdocs/lecture10/aimvc/`



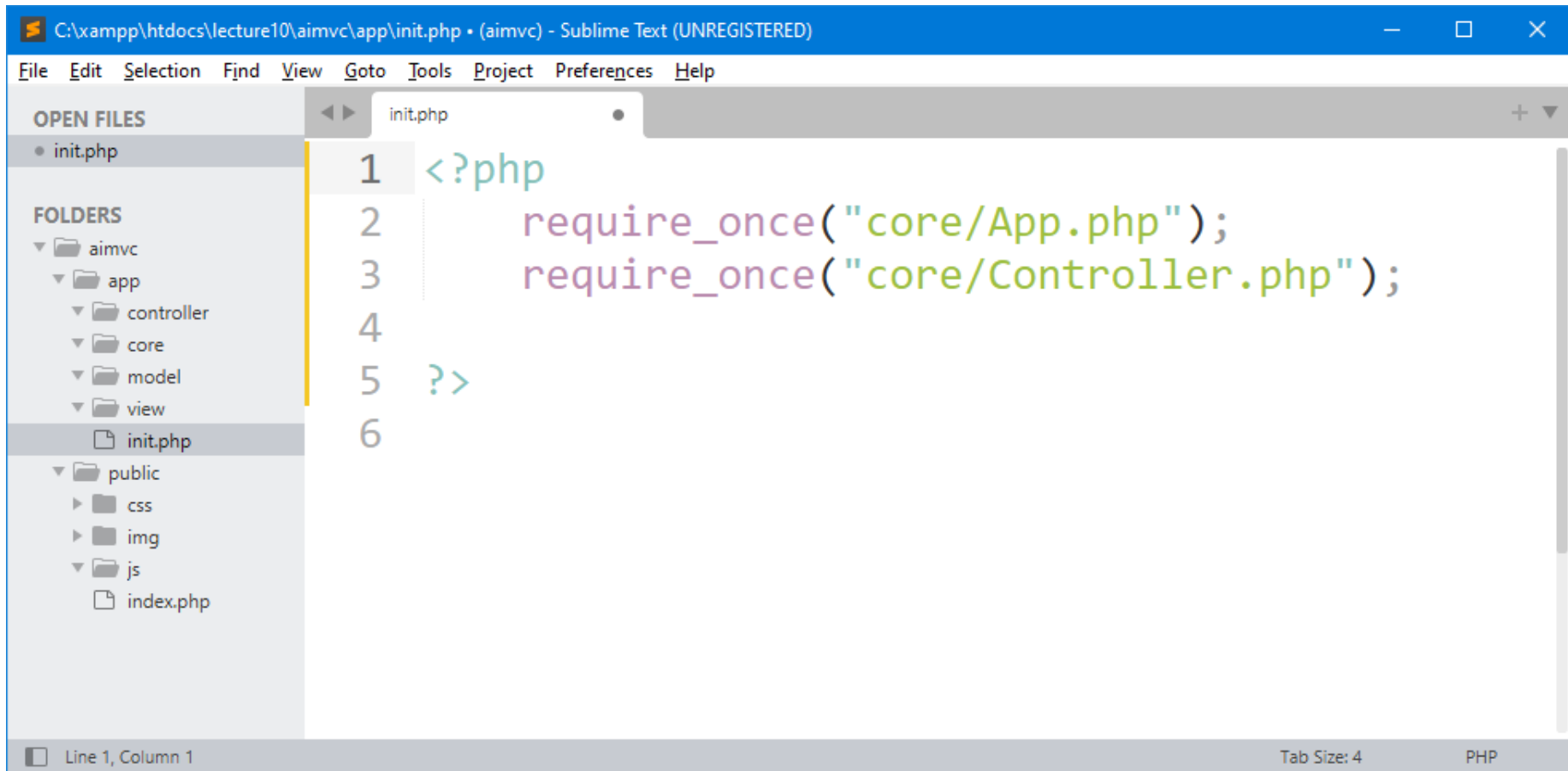
Main directory of our new framework:

aimvc ⇒ Artificial Intelligence Model View Controller ⇒ PHP Framework

Create The file *init.php*

init.php (*initialization*) is an optional file within our Framework file structure and located in *private directory*. This file may contain the initialization of event handlers and connection of additional functions that are common for all websites.

File Path: `C:/xampp/htdocs/lecture10/aimvc/app/` (*private directory*)



```
C:\xampp\htdocs\lecture10\aimvc\app\init.php • (aimvc) - Sublime Text (UNREGISTERED)
File Edit Selection Find View Goto Tools Project Preferences Help

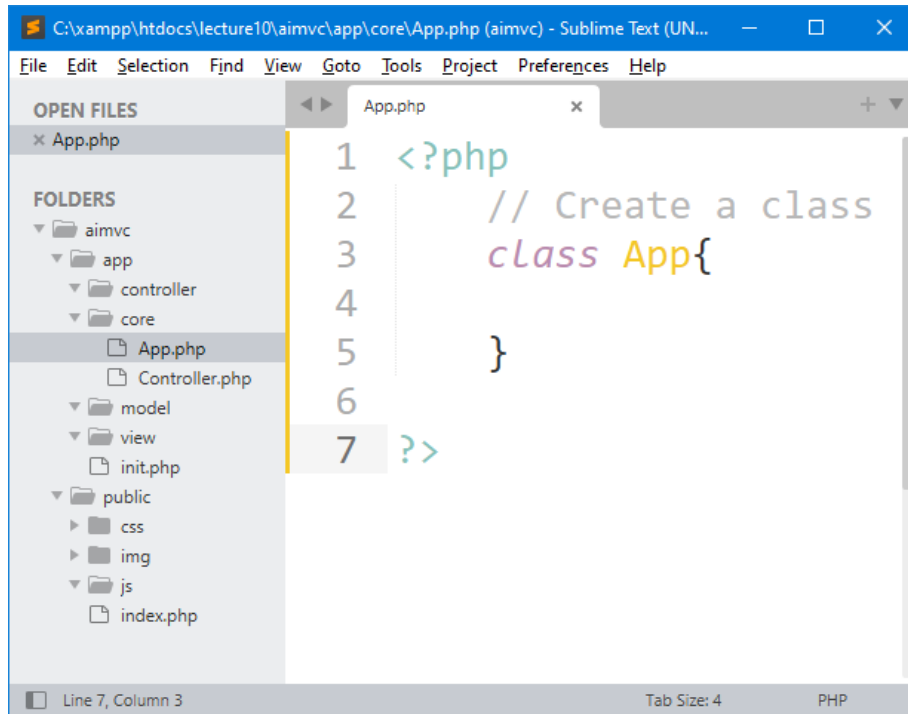
OPEN FILES
• init.php

FOLDERS
▼ aimvc
  ▼ app
    ▼ controller
    ▼ core
    ▼ model
    ▼ view
    init.php
  ▼ public
    ► css
    ► img
    ▼ js
    index.php

1 <?php
2     require_once("core/App.php");
3     require_once("core/Controller.php");
4
5 ?>
6

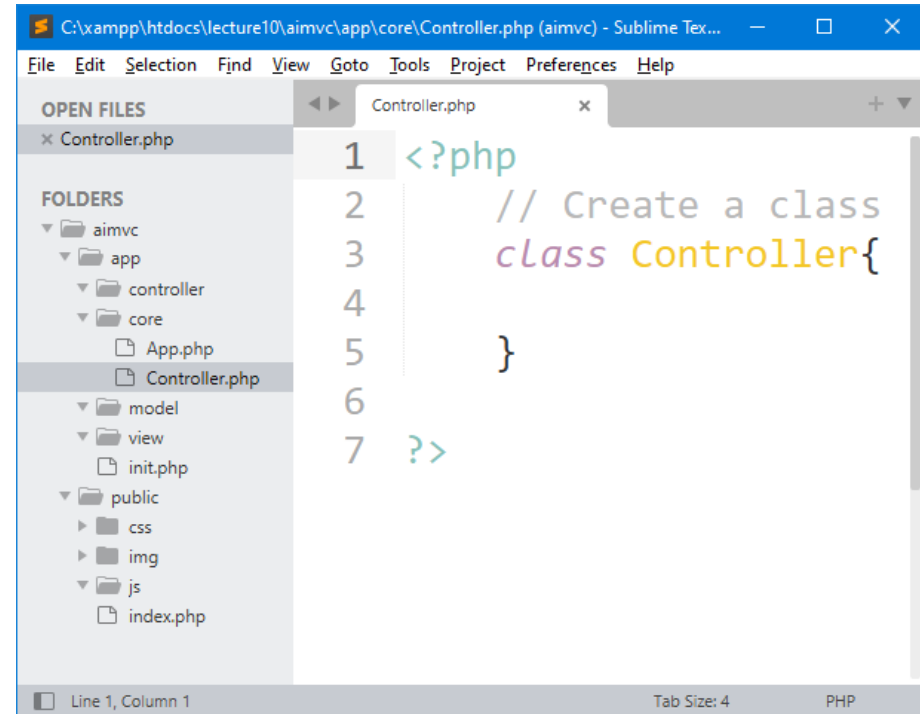
Line 1, Column 1 Tab Size: 4 PHP
```

Create Framework's Core Classes



```
1 <?php
2 // Create a class
3 class App{
4
5 }
6
7 ?>
```

The screenshot shows the Sublime Text editor with the file 'App.php' open. The left sidebar displays the project structure with folders like 'aimvc', 'app', 'controller', 'core', 'model', 'view', 'public', 'css', 'img', 'js', and 'index.php'. The main editor area shows the code for 'App.php'.



```
1 <?php
2 // Create a class
3 class Controller{
4
5 }
6
7 ?>
```

The screenshot shows the Sublime Text editor with the file 'Controller.php' open. The left sidebar displays the project structure with folders like 'aimvc', 'app', 'controller', 'core', 'model', 'view', 'public', 'css', 'img', 'js', and 'index.php'. The main editor area shows the code for 'Controller.php'.

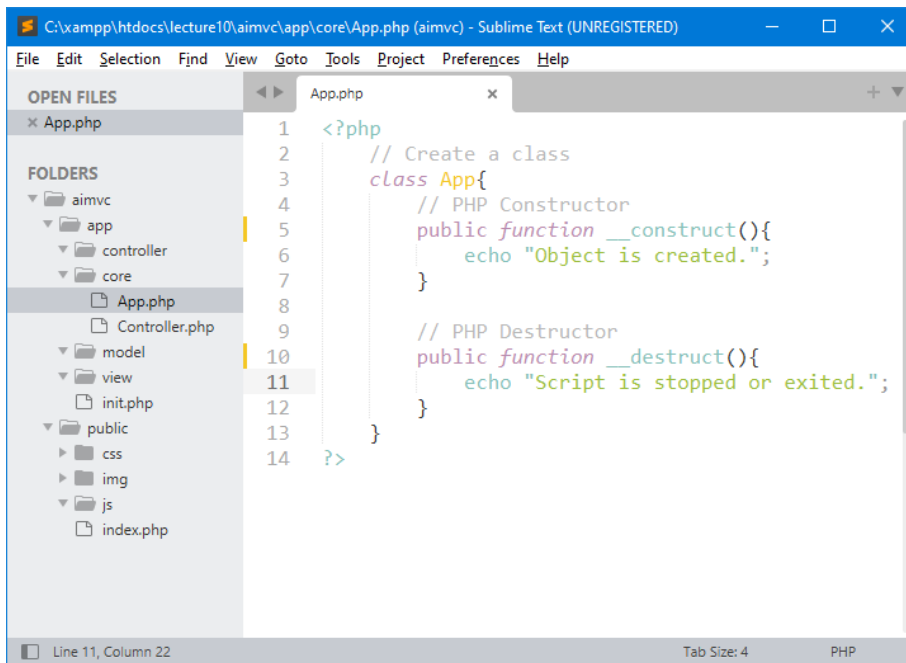
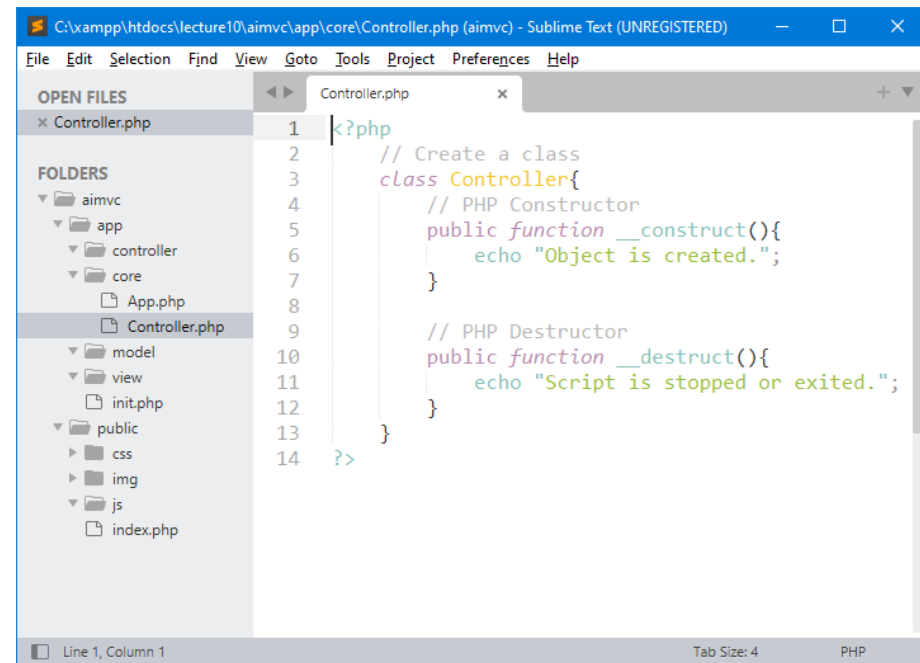
Class App (*App.php*) digunakan sebagai class utama dalam pembentukan aimvc framework. Class Controller (*Controller.php*) digunakan sebagai class yang akan mengatur semua fungsi-fungsi yang akan diletakan di dalam *controller directory*.

Note:

First letter as a capital letter (*uppercase letter*) normally means a php class.

Constructors and Destructors

Tambahkan fungsi **__construct** dan **__destruct** di Class App (**App.php**) dan Class Controller (**Controller.php**), ini berfungsi agar supaya pada saat proses instansiasi atau proses pembuatan instance/objek dari class App (**App.php**) dan class Controller (**Controller.php**) selesai, maka sudah bisa langsung di *display* contoh output-nya.

A screenshot of the Sublime Text editor window titled 'C:\xampp\htdocs\lecture10\aimvc\app\core\App.php (aimvc) - Sublime Text (UNREGISTERED)'. The 'App.php' file is open, showing PHP code for a class. The code includes a class declaration 'class App{', a constructor method 'public function __construct(){ echo "Object is created."; }', and a destructor method 'public function __destruct(){ echo "Script is stopped or exited."; }'. The file explorer on the left shows a project structure with folders like 'aimvc', 'app', 'controller', 'core', 'model', 'view', 'public', 'css', 'img', 'js', and files like 'App.php', 'Controller.php', 'init.php', and 'index.php'. The status bar at the bottom indicates 'Line 11, Column 22' and 'PHP'.A screenshot of the Sublime Text editor window titled 'C:\xampp\htdocs\lecture10\aimvc\app\core\Controller.php (aimvc) - Sublime Text (UNREGISTERED)'. The 'Controller.php' file is open, showing PHP code for a class. The code includes a class declaration 'class Controller{', a constructor method 'public function __construct(){ echo "Object is created."; }', and a destructor method 'public function __destruct(){ echo "Script is stopped or exited."; }'. The file explorer on the left shows the same project structure as the previous screenshot. The status bar at the bottom indicates 'Line 1, Column 1' and 'PHP'.

PHP - The **__construct** Function:

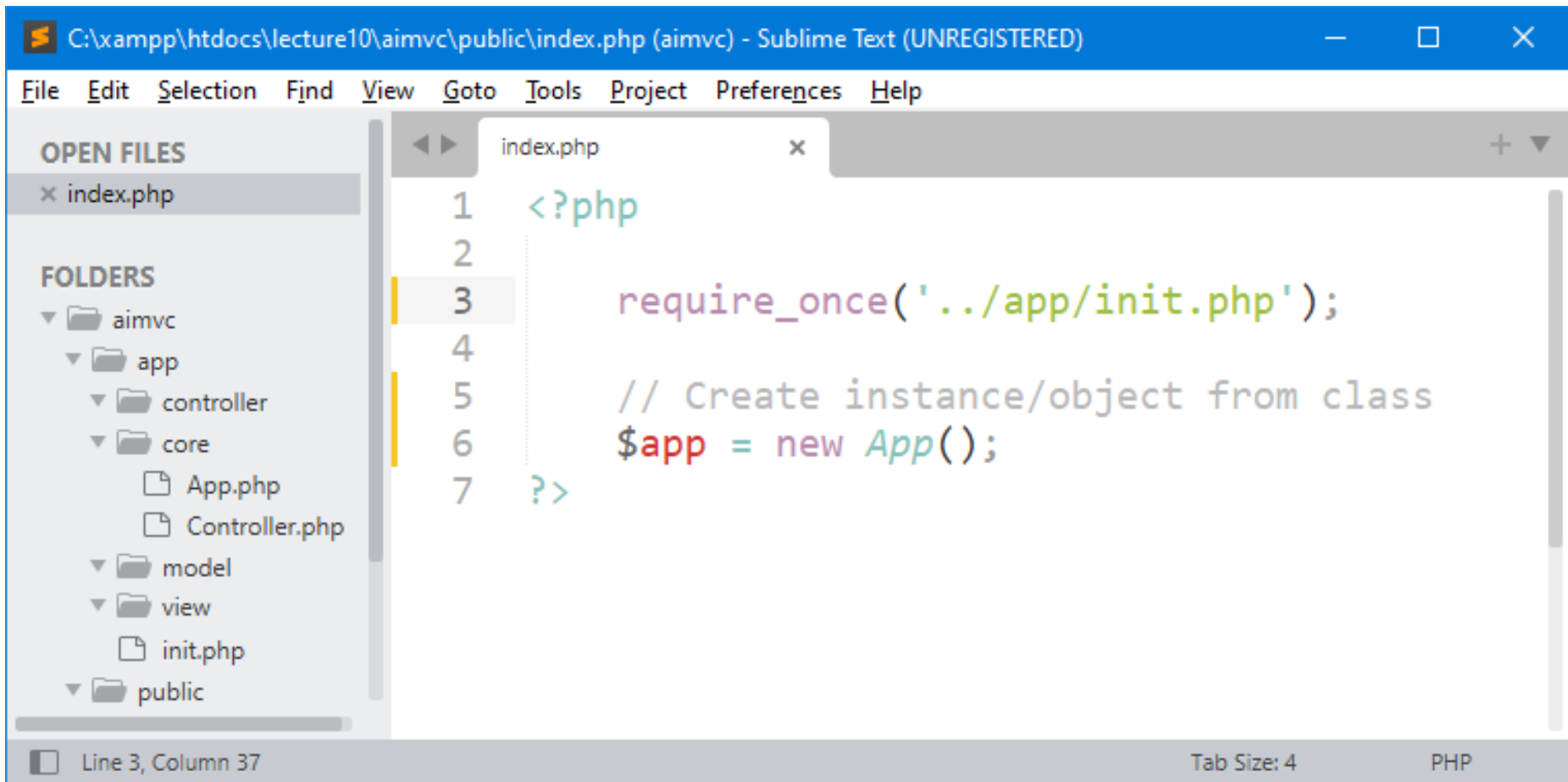
A constructor allows you to initialize an object's properties upon creation of the object.

PHP - The **__destruct** Function:

A destructor is called when the object is destructed or the script is stopped or exited.

Instansiasi Class (Object Class) App

Instansiasi **class App** (*App.php*) dari file **index.php** adalah pembuatan instance/objek dari *class* App, dimana kita akan coba menampilkan/view sesuatu dari *class* tersebut. Perhatikan di dalam *constructor* dari class App sudah terdapat *statement echo* (*which is used to display the output*) dimana akan digunakan untuk menampilkan *output* sederhana.



```
C:\xampp\htdocs\lecture10\aimvc\public\index.php (aimvc) - Sublime Text (UNREGISTERED)

File Edit Selection Find View Goto Tools Project Preferences Help

OPEN FILES
x index.php

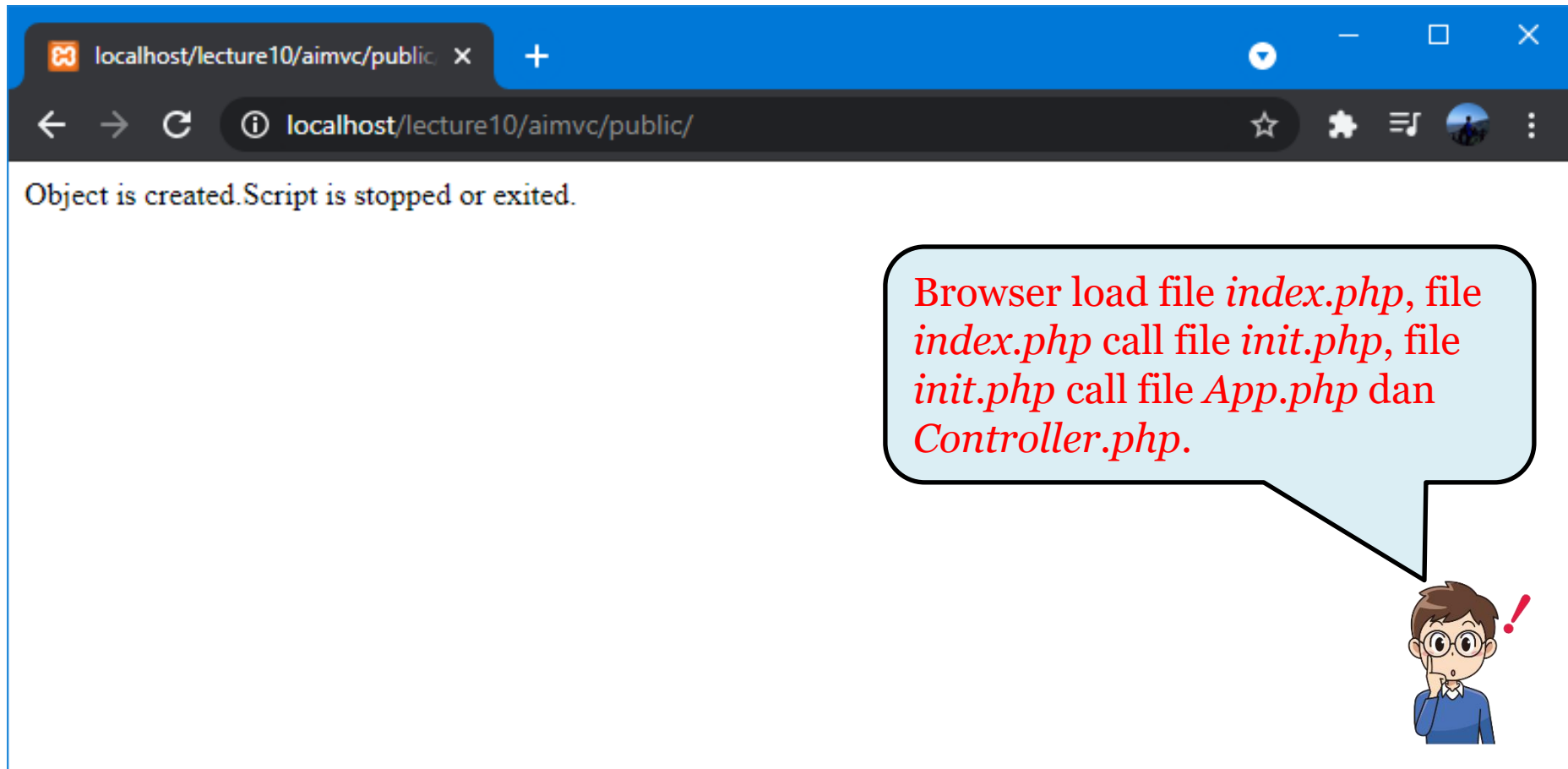
FOLDERS
aimvc
  app
    controller
    core
      App.php
      Controller.php
    model
    view
      init.php
    public

1 <?php
2
3     require_once('../app/init.php');
4
5     // Create instance/object from class
6     $app = new App();
7 ?>
```

Line 3, Column 37 Tab Size: 4 PHP

Test *your* Framework's Structure

Uniform Resource Locator (URL): <http://localhost/lecture10/aimvc/public/>



Pada saat jalankan (*run*) PHP Files di *browser*, maka *output* yang diharapkan (*giving the expected output*) yaitu seperti diatas.

Framework #2

Penerapan Routing (route)

[URL Parsing with Router or URL Routing]

Istilah/Term Routing

- ❑ **Routing** adalah proses dimana suatu *item* dapat sampai ke tujuan dari satu lokasi ke lokasi lain. Dalam hal ini, *item* yang dimaksud adalah halaman aplikasi website. Para *developer* dapat menentukan sendiri halaman yang akan muncul pada saat dikunjungi oleh *users*.
- ❑ **Routing** is the process of selecting a path for traffic in a network or between or across multiple networks. Broadly, routing is performed in many types of networks, including circuit-switched networks, such as the public switched telephone network (PSTN), and computer networks, such as the Internet. - Wikipedia

Contoh penggunaan Routing:

Pada saat *user* mengunjungi halaman *dashboard*, maka *kita* dapat menentukan tampilan apa yang akan muncul, apakah itu hanya berupa tulisan (*text*), berupa halaman *controller*, berupa halaman *view*, maupun halaman *error*.

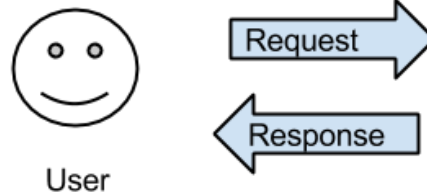
Route dapat handle semua perintah yang telah dideklarasikan atau bahkan **route** bisa dianalogikan sebagai peta petunjuk bagaimana alur navigasi aplikasi yang sedang dibangun atau juga bisa disebut *Router* merupakan suatu rute/jalan/trayek.



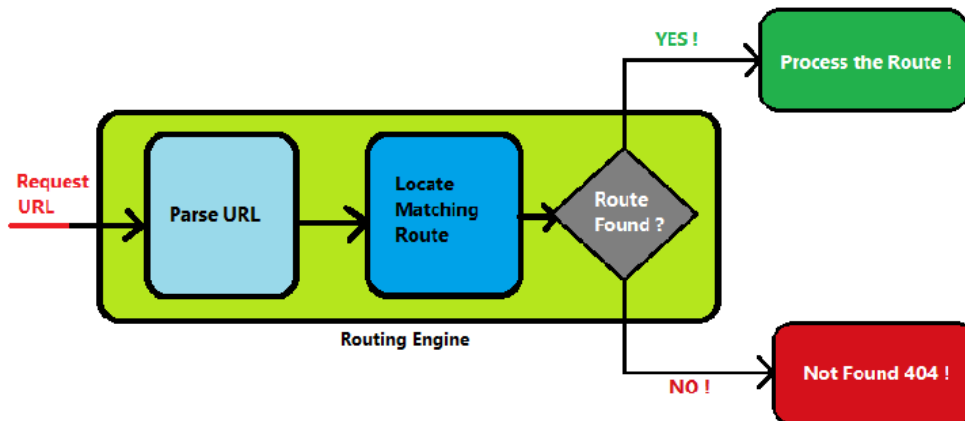
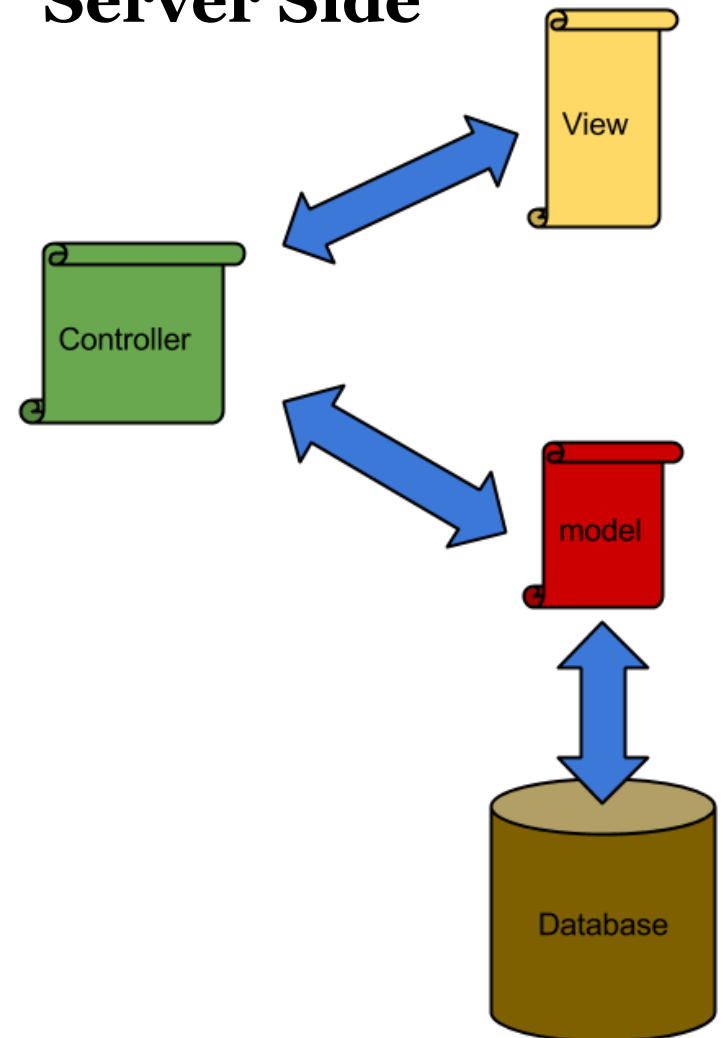
Framework Routing

Pada saat *user* mengunjungi halaman *home*, maka kita dapat menentukan tampilan apa yang akan muncul, apakah itu hanya berupa tulisan (*text*), berupa halaman *controller*, berupa halaman *view*, maupun halaman *error*.

Client Side

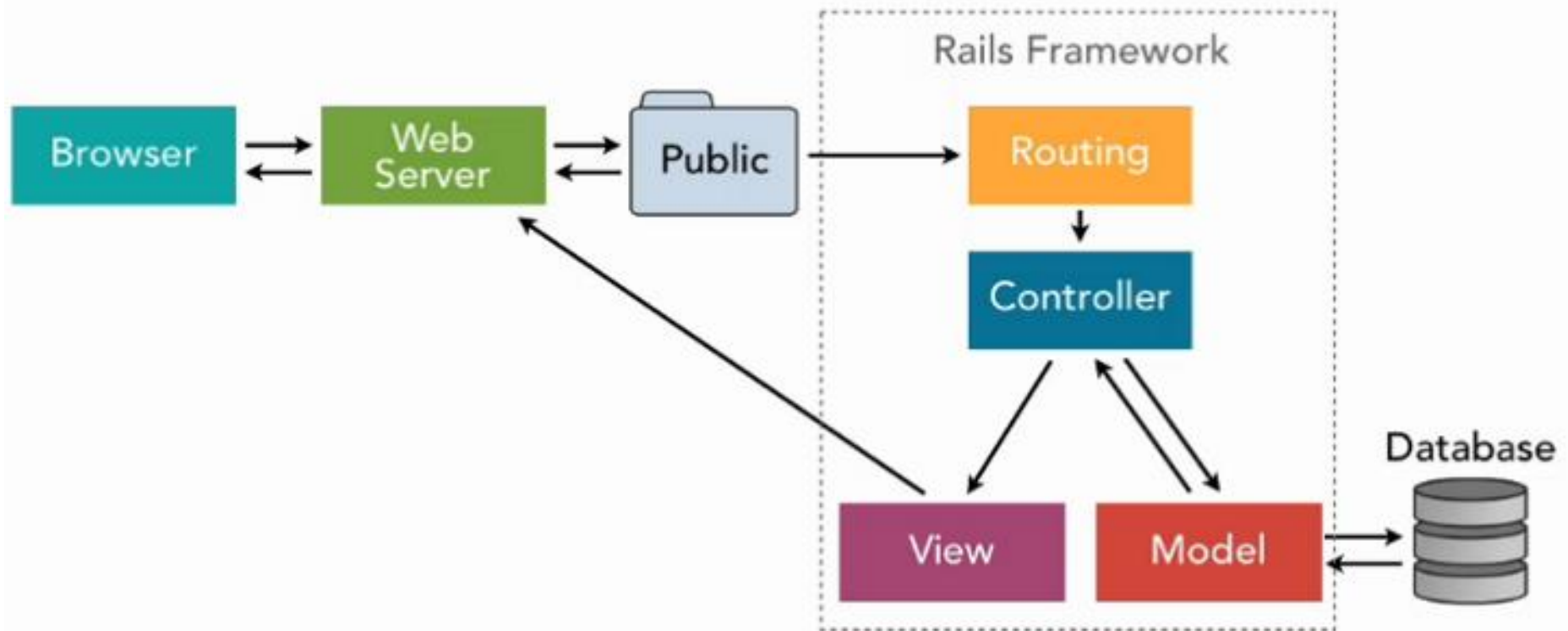


Server Side



Rails Architecture

Ruby on Rails, or **Rails**, is a server-side web application framework written in Ruby under the MIT License. Rails is a ***model–view–controller*** (MVC) framework, providing default structures for a database, a web service, and web pages.



URL Routing

Typically there is a one-to-one relationship (*hubungan satu-ke-satu*) between a URL string and its corresponding controller class/method. The segments in a URL (*bagian dalam URL*) normally follow this pattern:

`http://www.example.com/class/method/parameter1/parameter2/`

What is a Controller?

A Controller is simply a class file that is named in a way that can be associated with a URI.

Class merupakan *controller class* yang sudah dibuat dan biasanya segmen kedua (**method**) dari URL disiapkan untuk nama *metode*.

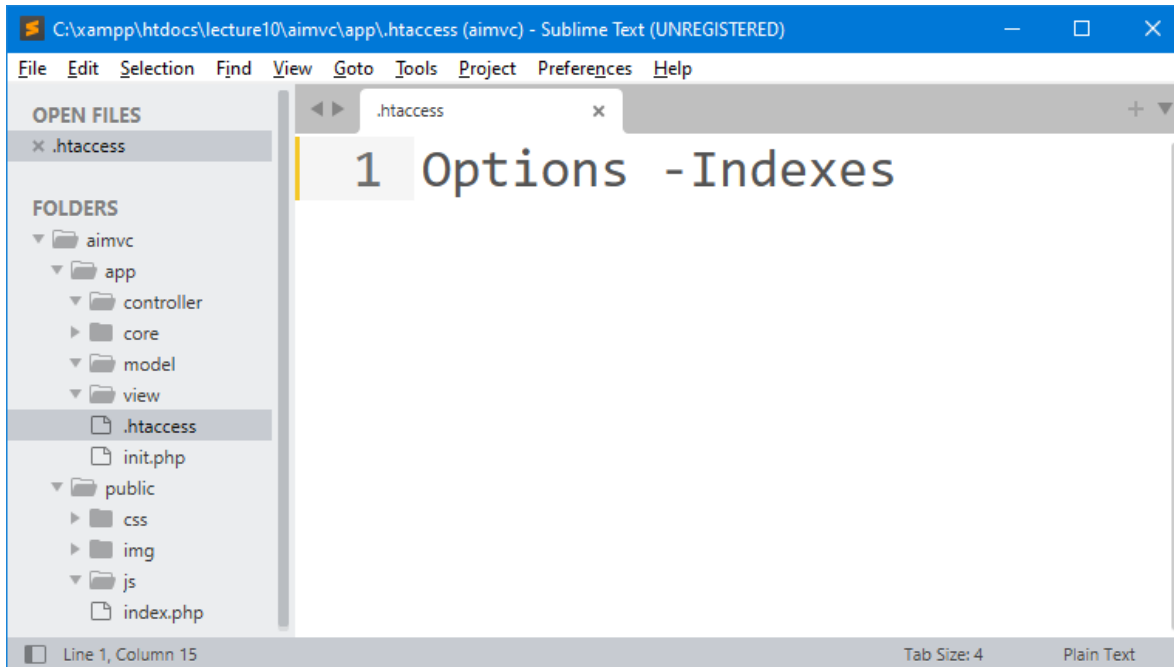
Framework #2 - continue

Penerapan Routing (route)

[Setting your own routing rules]

1 - Restrict Users Access (*private*)

Block/restrict access permission to ***private*** directory (*app directory*):

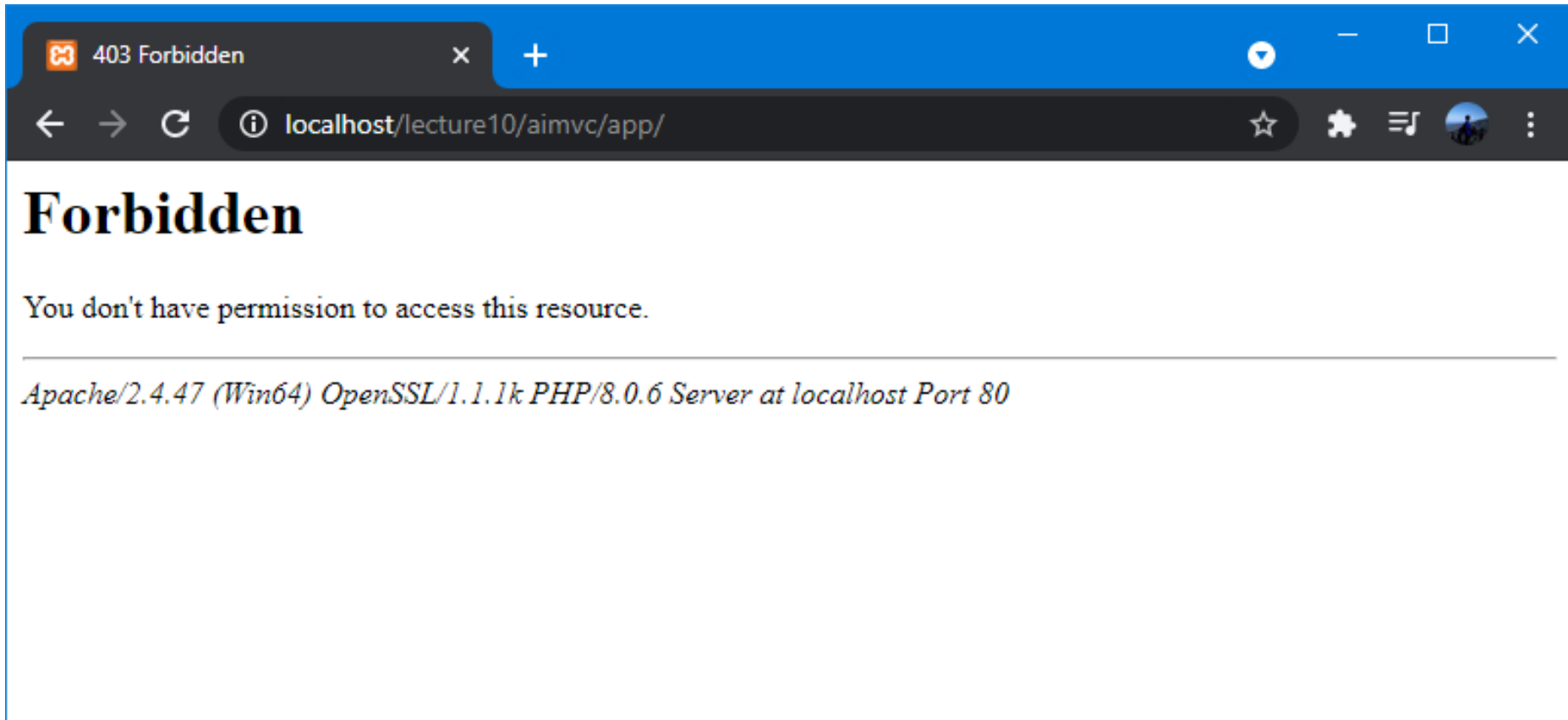


Create file **.htaccess** di dalam folder app. “**Options –Indexes**” artinya jika ada *users* membuka folder app dan folder-folder yang lain di dalamnya maka **jangan tampilkan isi folder-nya**, selama di dalam *folder* tersebut tidak ada file *index*. Dalam arti user tidak dapat diberi izin.

File **.htaccess** adalah cara untuk mengkonfigurasi rincian website kita tanpa perlu mengubah *file konfigurasi server*.

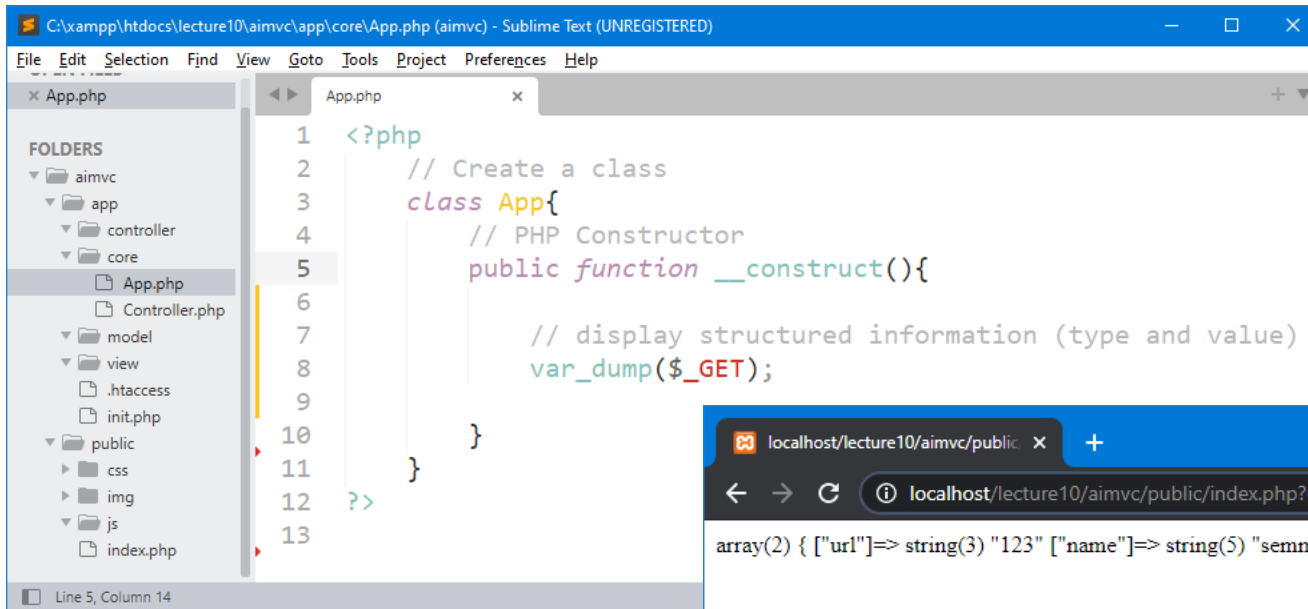
1 - Restrict Users Access (*private*)

Test apakah *users* masih diberi *access* permission ke ***private*** directory (*app directory*) dan folder-folder yang lain di dalamnya? Jika iya, maka dapat mengaktifkan *.htaccess* file pada *httpd.conf* file.



Test dibuka juga directory <http://localhost/lecture10/aimvc/app/view/>

2 - Test Request URL Public Directory

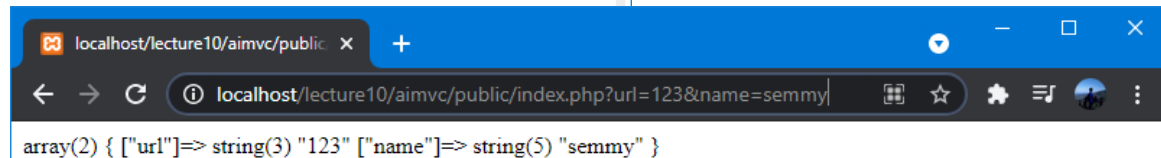


The screenshot shows a Sublime Text editor window titled "C:\xampp\htdocs\lecture10\aimvc\app\core\App.php (aimvc) - Sublime Text (UNREGISTERED)". The left sidebar displays a file explorer with the following structure:

- aimvc
 - app
 - controller
 - core
 - App.php**
 - Controller.php
 - model
 - view
 - .htaccess
 - init.php
 - public
 - css
 - img
 - js
 - index.php

The main editor area shows the content of App.php:

```
1 <?php
2 // Create a class
3 class App{
4     // PHP Constructor
5     public function __construct(){
6
7         // display structured information (type and value)
8         var_dump($_GET);
9
10    }
11 }
12 ?>
13
```



The screenshot shows a web browser window with the address bar displaying "localhost/lecture10/aimvc/public/index.php?url=123&name=semmy". The browser's developer tools or output area shows the following PHP output:

```
array(2) { ["url"]=> string(3) "123" ["name"]=> string(5) "semmy" }
```

Request URL:

<http://localhost/lecture10/aimvc/public/index.php?url=123&name=semmy>

Jika dilihat URL yang di *request* tidak teratur, oleh sebab itu kita akan mencoba Parsing URL (*mengurai URL*) dan membuatnya lebih rapi sesuai dengan *standard framework* yang akan di *design*.

Apply PHP isset() Function in URL

We used *isset function* to check whether a variable (*get url*) is empty and also check whether the variable (*get url*) is set/declared before setting our own routing rules.

Example isset function:

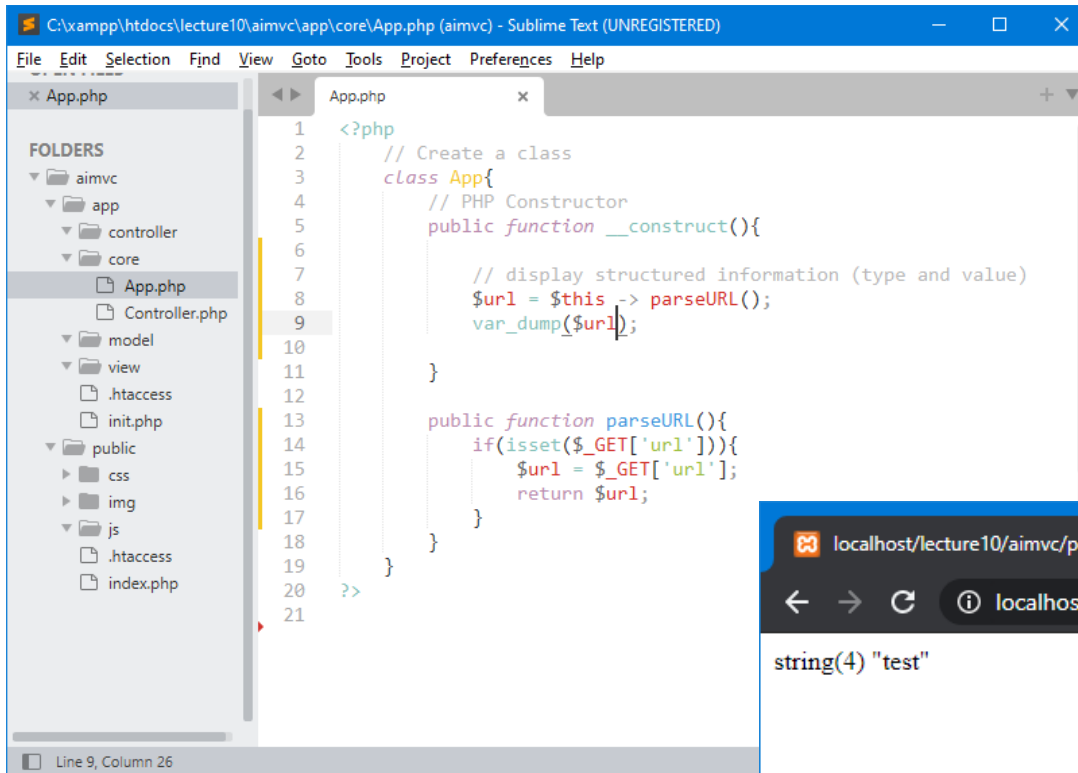
```
<!DOCTYPE html>
<html>
<body>
  <?php
    $a = 0;
    // True because $a is set
    if (isset($a)) {
      echo "Variable 'a' is set.<br>";
    }

    $b = null;
    // False because $b is NULL
    if (isset($b)) {
      echo "Variable 'b' is set.";
    }
  ?>
</body>
</html>
```

Return Value:	TRUE if <i>variable</i> exists and is not NULL, FALSE otherwise
Return Type:	Boolean
PHP Version:	4.0+
PHP Changelog:	PHP 5.4: Non-numeric offsets of strings now returns FALSE

The `isset()` function checks whether a variable is set, which means that it has to be declared and is not NULL. This function returns true if the variable exists and is not NULL, otherwise it returns false.

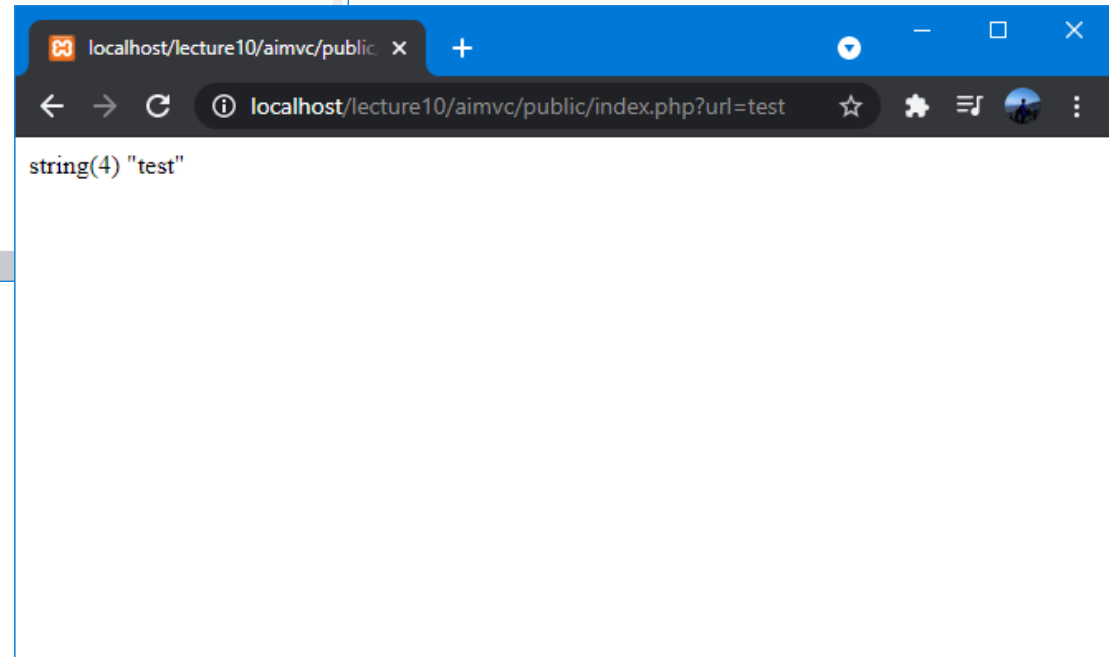
3 – Process Parsing URL



The screenshot shows the Sublime Text editor with the file `C:\xampp\htdocs\lecture10\aimvc\app\core\App.php` open. The left sidebar displays the project structure with folders `aimvc`, `app`, `controller`, `core`, `model`, `view`, `public`, `css`, `img`, and `js`. The `App.php` file is selected in the `core` folder. The main editor window shows the following PHP code:

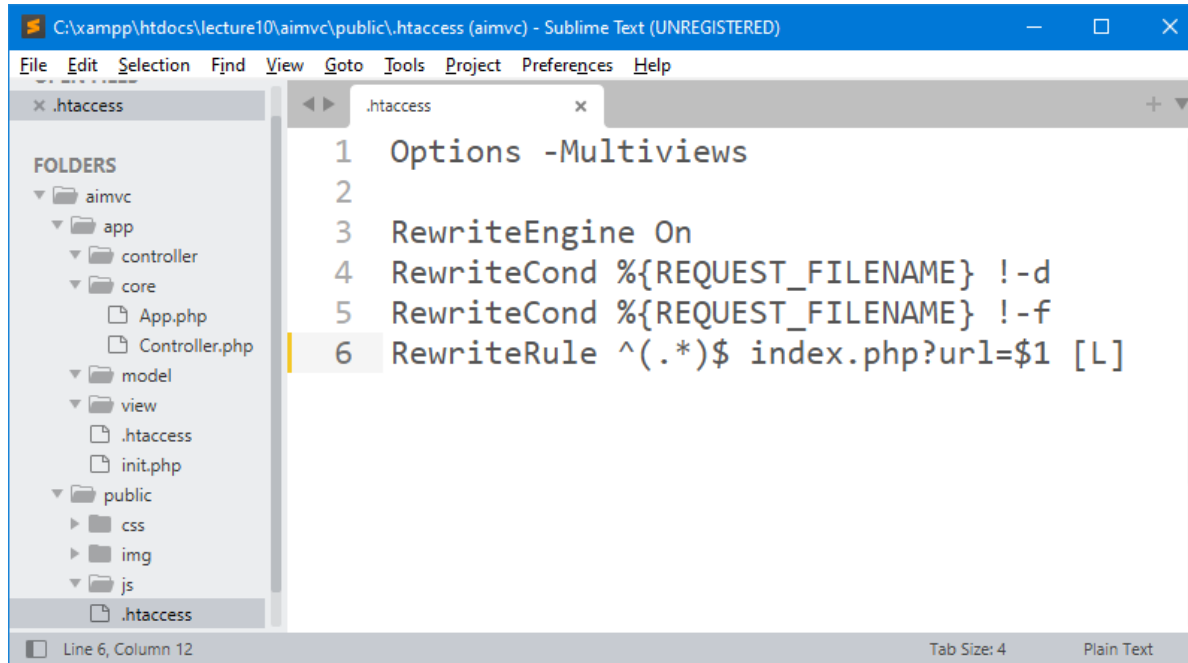
```
1 <?php
2 // Create a class
3 class App{
4     // PHP Constructor
5     public function __construct(){
6
7         // display structured information (type and value)
8         $url = $this->parseURL();
9         var_dump($url);
10
11     }
12
13     public function parseURL(){
14         if(isset($_GET['url'])){
15             $url = $_GET['url'];
16             return $url;
17         }
18     }
19 }
20
21 ?>
```

The status bar at the bottom indicates "Line 9, Column 26".



4 - Avoid Users Access (*public*)

Menghindari *kesalahan access file* atau **public** folder (*public directory*).



The screenshot shows a Sublime Text window titled "C:\xampp\htdocs\lecture10\aimvc\public\.htaccess (aimvc) - Sublime Text (UNREGISTERED)". The left sidebar displays a file explorer with the following structure:

- aimvc
 - app
 - controller
 - core
 - App.php
 - Controller.php
 - model
 - view
 - .htaccess
 - init.php
 - public
 - css
 - img
 - js
 - .htaccess

The main editor area shows the content of the .htaccess file:

```
1 Options -Multiviews
2
3 RewriteEngine On
4 RewriteCond %{REQUEST_FILENAME} !-d
5 RewriteCond %{REQUEST_FILENAME} !-f
6 RewriteRule ^(.*)$ index.php?url=$1 [L]
```

The status bar at the bottom indicates "Line 6, Column 12", "Tab Size: 4", and "Plain Text".

Create file **.htaccess** di dalam folder public. “**Options - Multiviews**” artinya kita akan menghindari kesalahan akses file ataupun directory dari URL (*Uniform Resource Locator*).

Karena *users* akan mengakses langsung *file* class bukan ke *folder* yang ada di **public directory** melalui URL. Maka kesalahan akses ataupun **SQL Injection techniques** (*backend database manipulation to access information*) dapat kita hindari.

4 - Avoid Users Access (*public*)

```
RewriteCond %{REQUEST_FILENAME} !-d  
RewriteCond %{REQUEST_FILENAME} !-f
```

Rewrite condition (**RewriteCond**) tests along with a corresponding rule (**RewriteRule**) if the prior conditions pass.

Jika URL yang di *request* adalah *directory/folder* (*d*) dan *file* (*f*) maka akan diabaikan (*ignored*). Dalam arti, *request* yang diinginkan hanyalah berupa *method*.

4 - Avoid Users Access (*public*)

RewriteRule **^(.*)\$ index.php?url=\$1 [L]**

RewriteRule defines a particular rule.

Fungsi Regex **^(.*)\$** artinya :

1. Symbol caret (^) artinya yang ditulis mulai dari awal url (*match the beginning of a line*) setelah penulisan url “/aimvc/public/”.
2. in regular expressions, symbol dot (.) is a special character used to match any one character. Jadi, akan ambil semua character sesudah “aimvc/public”.
3. A regular expression followed by an asterisk (*) matches zero or more occurrences of the regular expression. Jadi ambil character satu persatu dari url.
4. If a **dollar sign** (\$) is at the end of the entire **regular expression**, it matches the end of a line. Jadi akan dicek sampai semua character (*text*) yang ada dalam url selesai.

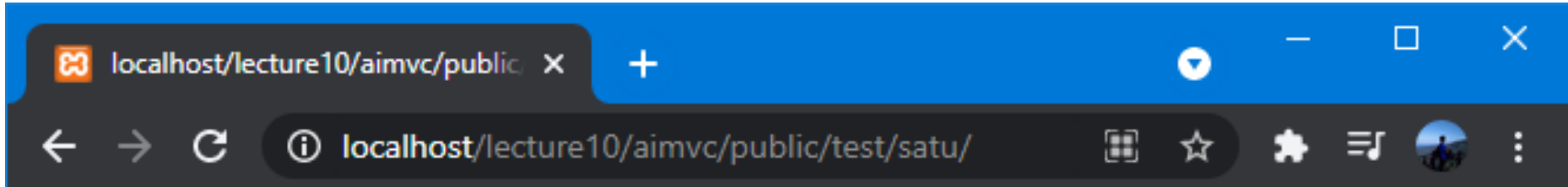
Fungsi Regex **index.php?url=\$1** artinya:

Arahkan ke *file* index.php yang mengirimkan url dan disimpan di \$1 (*placeholder*).

Flag **[L]** artinya:

- ☐ The [L] flag causes *mod_rewrite* to stop processing the rule set.
- ☐ In most contexts, this means that if the rule matches, *no further rules will be processed*.

Request URL without Format



string(10) "test/satu/"

Langkah awal untuk *Request URL, parsing URL* dan dapatkan controller class & method dengan memanfaatkan *.htaccess file* success !!!

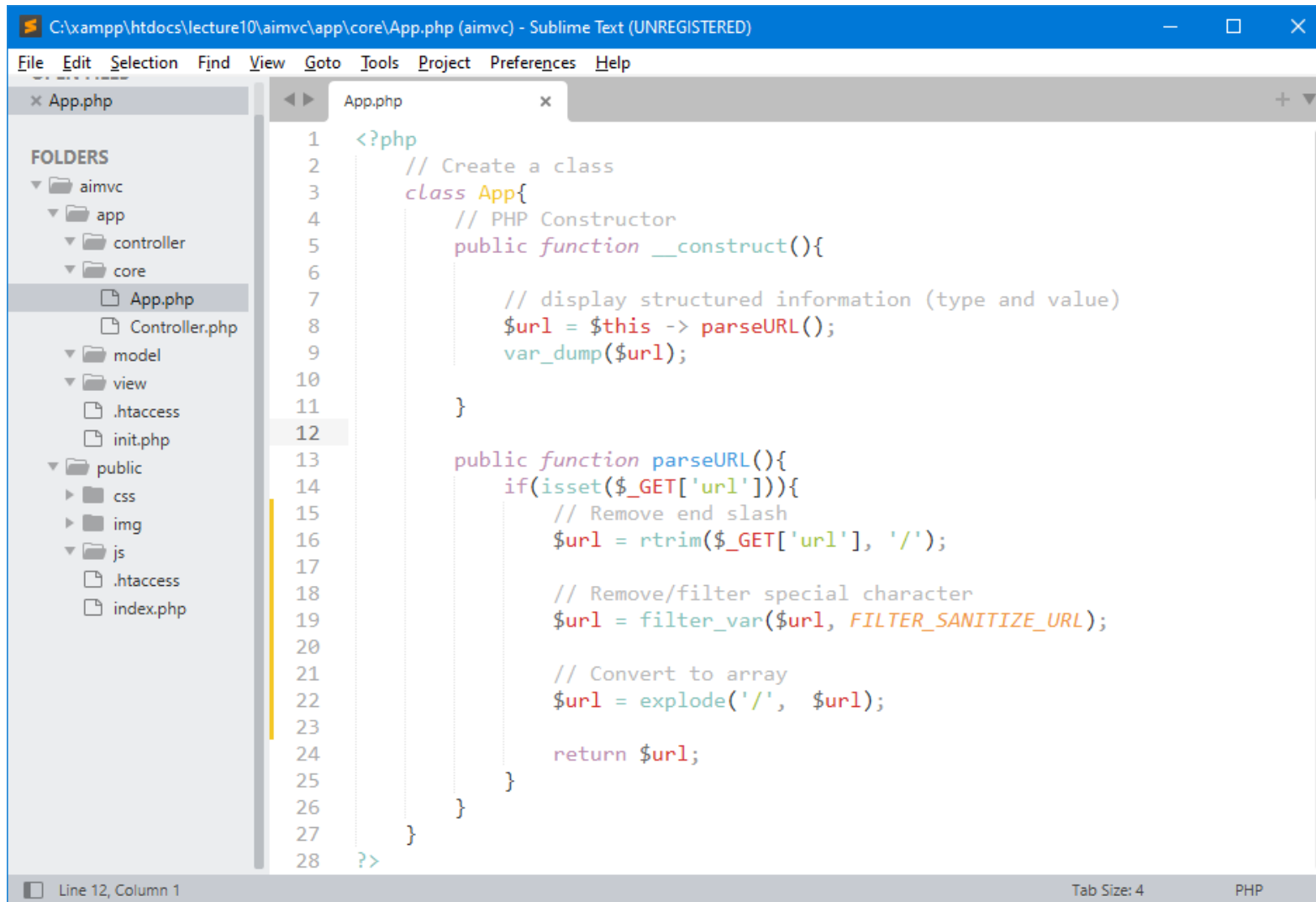
Framework #3

Remove end “/” in Url, Special
Character and Covert to Array.

Setting your own routing rules

Handling Request URL

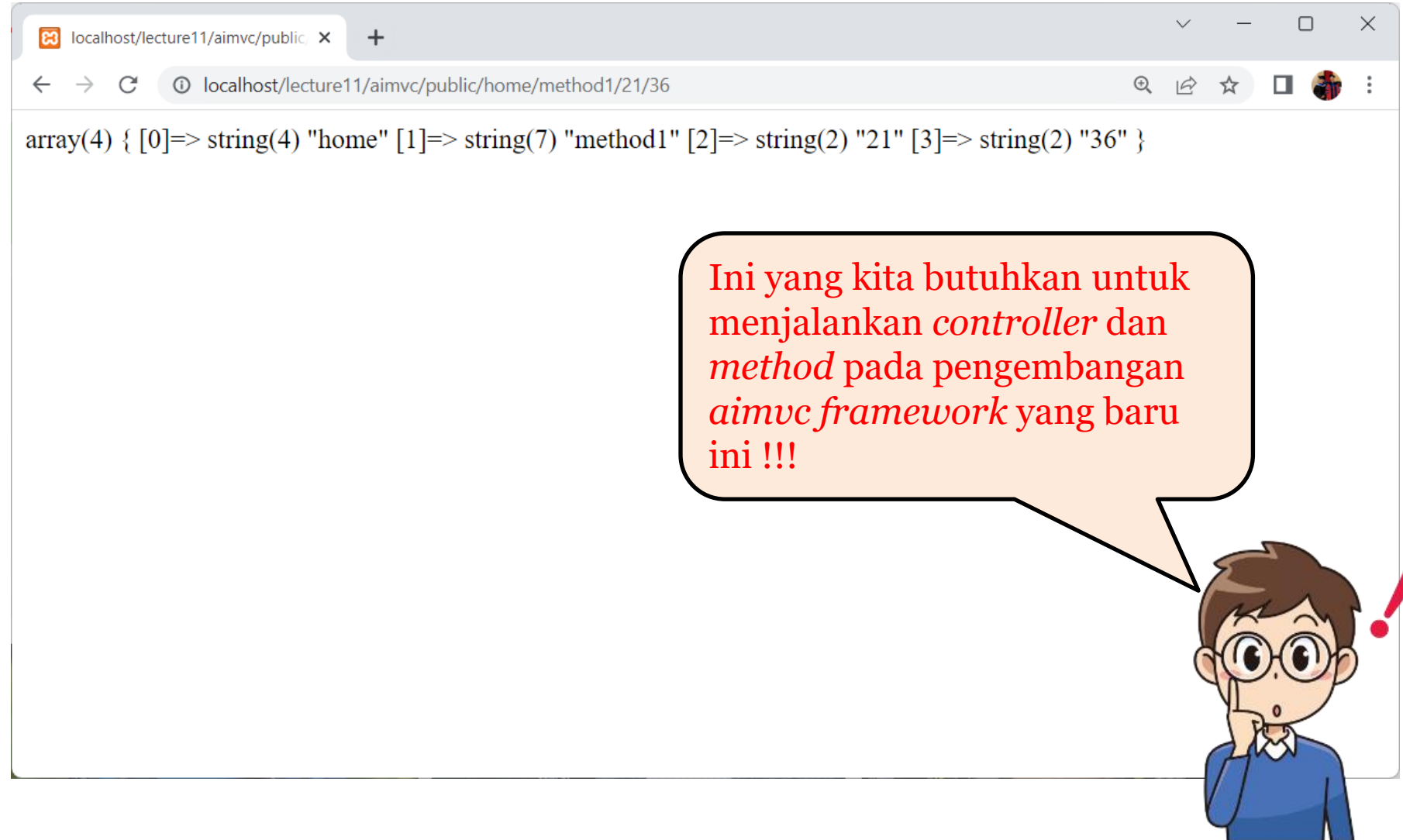
Handle more *routing rules* when *users* try to request through URL.



```
1 <?php
2 // Create a class
3 class App{
4 // PHP Constructor
5 public function __construct(){
6
7 // display structured information (type and value)
8 $url = $this -> parseURL();
9 var_dump($url);
10
11 }
12
13 public function parseURL(){
14 if(isset($_GET['url'])){
15 // Remove end slash
16 $url = rtrim($_GET['url'], '/');
17
18 // Remove/filter special character
19 $url = filter_var($url, FILTER_SANITIZE_URL);
20
21 // Convert to array
22 $url = explode('/', $url);
23
24 return $url;
25 }
26 }
27 }
28 ?>
```

Expected Output

Testing the router by passing the Request URL.



The screenshot shows a web browser window with the address bar displaying `localhost/lecture11/aimvc/public/home/method1/21/36`. The main content area displays the following PHP array output:

```
array(4) { [0]=> string(4) "home" [1]=> string(7) "method1" [2]=> string(2) "21" [3]=> string(2) "36" }
```

A speech bubble from a cartoon character points to the output, stating:

Ini yang kita butuhkan untuk menjalankan *controller* dan *method* pada pengembangan *aimvc framework* yang baru ini !!!

REST API (Representational State Transfer Application Programming Interface) output adalah data yang dikirimkan oleh server API sebagai *respons* terhadap permintaan yang diberikan oleh klien.

Output REST API biasanya berupa data dalam format tertentu seperti JSON (JavaScript Object Notation) atau XML (eXtensible Markup Language), meskipun format lain seperti CSV (Comma Separated Values) dan plain text juga dapat digunakan.





Exercise *for* Students

Exercise #1

Create database in MySQL server

Pada tahap awal silahkan masuk ke MySQL server:

- sudo mysql
- mysql -u root -p

```
mysql> CREATE DATABASE db_unklab;
mysql> CREATE DATABASE db_unklab;
Query OK, 1 row affected (0.78 sec)

mysql> |
```

mysql> SHOW DATABASES;

```
mysql> SHOW DATABASES;
+-----+
| Database |
+-----+
| db_attendance |
| db_unklab |
| information_schema |
| mysql |
| performance_schema |
| sys |
| testadmin |
+-----+
7 rows in set (0.00 sec)
```

mysql> use db_unklab;

```
mysql> use db_unklab;
Database changed
mysql> |
```

Show struktur table:

mysql> DESCRIBE tbl_customer;

```
CREATE TABLE nama_tabel (
    id INT AUTO_INCREMENT PRIMARY KEY,
    nama VARCHAR(50),
    umur INT,
    alamat VARCHAR(255)
);
```

SETELAH ITU BUAT TABEL DAN ISI DATA PADA TABEL YANG TELAH DIBUAT.

mysql> CREATE TABLE tbl_students (reg_number VARCHAR(255), nim_number VARCHAR(255), email VARCHAR(255), fullname VARCHAR(255), password VARCHAR(255), created_at VARCHAR(255), updated_at VARCHAR(255));

mysql> INSERT INTO tbl_students (reg_number, nim_number, email, fullname, password, created_at, updated_at) VALUES

-> ('1001', '1234567890', 'john@example.com', 'John Doe', 'password123', NOW(), NOW()),

-> ('1002', '0987654321', 'jane@example.com', 'Jane Smith', 'password456', NOW(), NOW()),

-> ('1003', '2468013579', 'bob@example.com', 'Bob Johnson', 'password789', NOW(), NOW()),

-> ('1004', '1357902468', 'alice@example.com', 'Alice Brown', 'passwordabc', NOW(), NOW()),

-> ('1005', '3141592653', 'jim@example.com', 'Jim Black', 'passworddef', NOW(), NOW());

```
mysql> CREATE TABLE tbl_students (reg_number VARCHAR(255), nim_number VARCHAR(255), email VARCHAR(255), fullname VARCHAR(255), password VARCHAR(255), created_at VARCHAR(255), updated_at VARCHAR(255));
Query OK, 0 rows affected (2.46 sec)
```

```
mysql> INSERT INTO tbl_students (reg_number, nim_number, email, fullname, password, created_at, updated_at) VALUES
-> ('1001', '1234567890', 'john@example.com', 'John Doe', 'password123', NOW(), NOW()),
-> ('1002', '0987654321', 'jane@example.com', 'Jane Smith', 'password456', NOW(), NOW()),
-> ('1003', '2468013579', 'bob@example.com', 'Bob Johnson', 'password789', NOW(), NOW()),
-> ('1004', '1357902468', 'alice@example.com', 'Alice Brown', 'passwordabc', NOW(), NOW()),
-> ('1005', '3141592653', 'jim@example.com', 'Jim Black', 'passworddef', NOW(), NOW());
```

```
Query OK, 5 rows affected (0.28 sec)
Records: 5 Duplicates: 0 Warnings: 0
```

```
mysql> |
```

Insert satu baris data ke dalam tabel

```
INSERT INTO nama_tabel (nama_kolom1, nama_kolom2, nama_kolom3)  
VALUES ('nilai1', 'nilai2', 'nilai3');
```

Contoh insert table

```
INSERT INTO tbl_mahasiswa (nim, nama, jurusan) VALUES ('123456', 'John  
Doe', 'Teknik Informatika');
```

```
MySQL 8.0 Command Line Client  
mysql> CREATE TABLE Student (  
->   Stud_ID int AUTO_INCREMENT PRIMARY KEY,  
->   Name varchar(45) DEFAULT NULL,  
->   Email varchar(45) NOT NULL UNIQUE,  
->   City varchar(25) DEFAULT NULL  
-> );  
Query OK, 0 rows affected (0.88 sec)  
  
mysql> INSERT INTO Student(Stud_ID, Name, Email, City)  
-> VALUES (1,'Stephen', 'stephen@javatpoint.com', 'Texax'),  
-> (2, 'Joseph', 'Joseph@javatpoint.com', 'Alaska'),  
-> (3, 'Peter', 'Peter@javatpoint.com', 'california');  
Query OK, 3 rows affected (0.18 sec)  
Records: 3  Duplicates: 0  Warnings: 0  
  
mysql> SELECT * FROM Student;  
+-----+-----+-----+-----+  
| Stud_ID | Name   | Email                               | City      |  
+-----+-----+-----+-----+  
|      1 | Stephen | stephen@javatpoint.com | Texax    |  
|      2 | Joseph  | Joseph@javatpoint.com  | Alaska   |  
|      3 | Peter   | Peter@javatpoint.com   | california |  
+-----+-----+-----+-----+
```

Ikuti steps di atas untuk membuat database baru dengan nama
“**db_attendance**” di Cloud Server

tbl_operator

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1 nip	varchar(150)	utf8mb4_0900_ai_ci		No	None			Change Drop More
☐	2 email	varchar(50)	utf8mb4_0900_ai_ci		No	None			Change Drop More
☐	3 fullname	varchar(100)	utf8mb4_0900_ai_ci		No	None			Change Drop More
☐	4 password	varchar(100)	utf8mb4_0900_ai_ci		No	None			Change Drop More
☐	5 phone_number	varchar(50)	utf8mb4_0900_ai_ci		No	None			Change Drop More
☐	6 role	varchar(100)	utf8mb4_0900_ai_ci		No	None			Change Drop More
☐	7 created_at	varchar(150)	utf8mb4_0900_ai_ci		No	None			Change Drop More
☐	8 updated_at	varchar(150)	utf8mb4_0900_ai_ci		No	None			Change Drop More

tbl_students

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1 reg_number	varchar(100)	utf8mb4_0900_ai_ci		No	None			Change Drop More
☐	2 nim_number	varchar(80)	utf8mb4_0900_ai_ci		No	None			Change Drop More
☐	3 email	varchar(100)	utf8mb4_0900_ai_ci		No	None			Change Drop More
☐	4 fullname	varchar(200)	utf8mb4_0900_ai_ci		No	None			Change Drop More
☐	5 password	varchar(50)	utf8mb4_0900_ai_ci		No	None			Change Drop More
☐	6 created_at	varchar(100)	utf8mb4_0900_ai_ci		No	None			Change Drop More
☐	7 updated_at	varchar(100)	utf8mb4_0900_ai_ci		No	None			Change Drop More

Contoh cara insert data via MySQL Query

SQL query/queries on table:

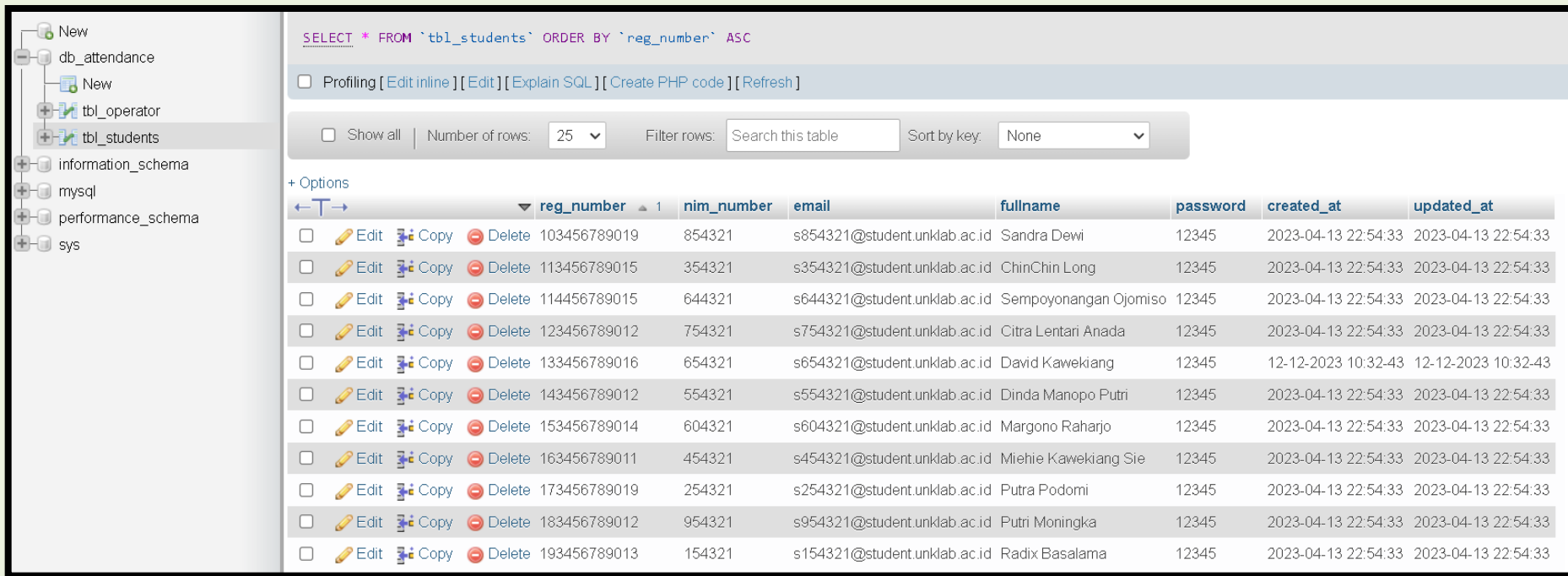
```
INSERT INTO tbl_students (reg_number, nim_number, email, fullname, password, created_at, updated_at)
VALUES
('1001', '1234567890', 'john@example.com', 'John Doe', 'password123', NOW(), NOW()),
('1002', '0987654321', 'jane@example.com', 'Jane Smith', 'password456', NOW(), NOW()),
('1003', '2468013579', 'bob@example.com', 'Bob Johnson', 'password789', NOW(), NOW()),
('1004', '1357902468', 'alice@example.com', 'Alice Brown', 'passwordabc', NOW(), NOW()),
('1005', '3141592653', 'jim@example.com', 'Jim Black', 'passworddef', NOW(), NOW());
```

SQL query/queries on table:

```
INSERT INTO tbl_operator (nip, email, fullname, password, phone_number, role, created_at, updated_at)
VALUES
('1234567890', 'john@example.com', 'John Doe', 'password123', '081234567890', 'admin', NOW(), NOW()),
('0987654321', 'jane@example.com', 'Jane Smith', 'password456', '087654321098', 'staff', NOW(), NOW()),
('2468013579', 'bob@example.com', 'Bob Johnson', 'password789', '082468013579', 'staff', NOW(), NOW()),
('1357902468', 'alice@example.com', 'Alice Brown', 'passwordabc', '081357902468', 'staff', NOW(), NOW()),
('3141592653', 'jim@example.com', 'Jim Black', 'passworddef', '083141592653', 'admin', NOW(), NOW());
```

Row Data in Table Students

Ubah data pada “tbl_students” dan buat sama seperti data dibawah ini. Terdapat paling kurang 11 mahasiswa dengan email yang berbeda.



The screenshot shows a database management interface. On the left is a sidebar with a tree view containing: New, db_attendance, New, tbl_operator, tbl_students, information_schema, mysql, performance_schema, and sys. The main area displays the SQL query: `SELECT * FROM `tbl_students` ORDER BY `reg_number` ASC`. Below the query are controls for Profiling, Show all, Number of rows (set to 25), Filter rows (Search this table), and Sort by key (set to None). The table data is as follows:

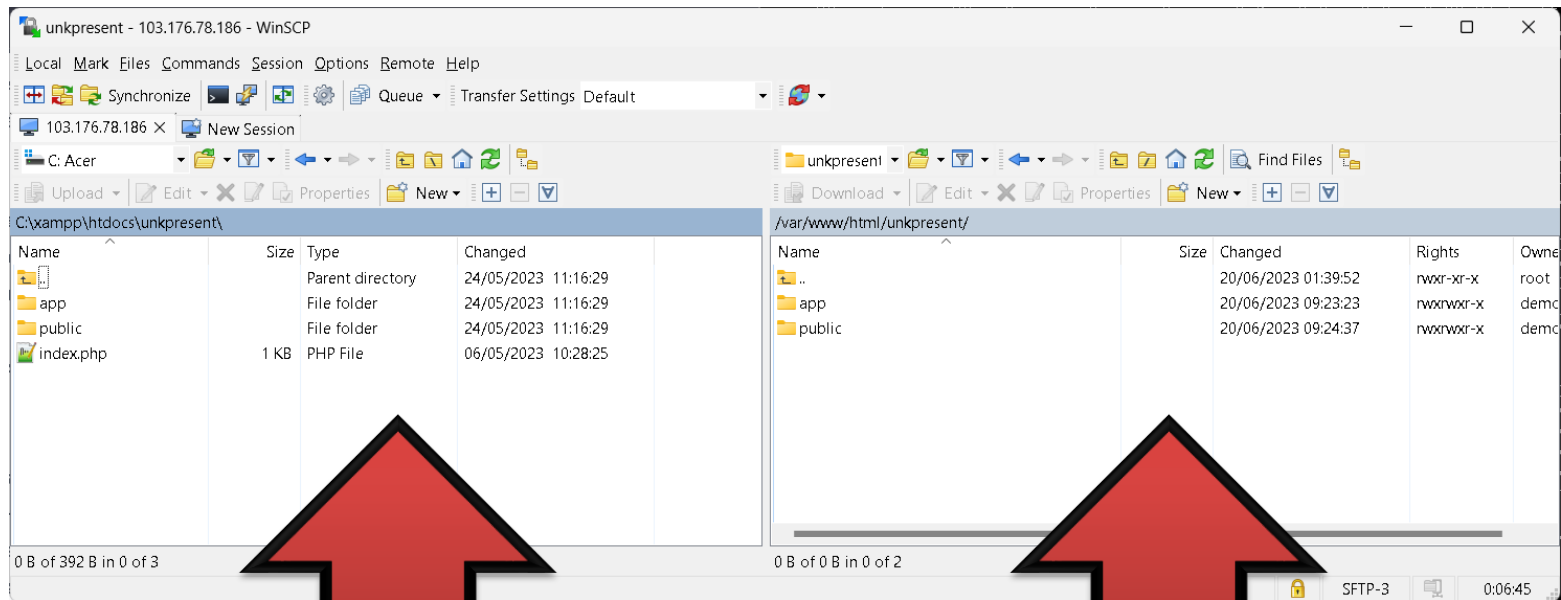
				reg_number	nim_number	email	fullname	password	created_at	updated_at
☐	Edit	Copy	Delete	103456789019	854321	s854321@student.unklab.ac.id	Sandra Dewi	12345	2023-04-13 22:54:33	2023-04-13 22:54:33
☐	Edit	Copy	Delete	113456789015	354321	s354321@student.unklab.ac.id	ChinChin Long	12345	2023-04-13 22:54:33	2023-04-13 22:54:33
☐	Edit	Copy	Delete	114456789015	644321	s644321@student.unklab.ac.id	Sempoyonangan Ojomiso	12345	2023-04-13 22:54:33	2023-04-13 22:54:33
☐	Edit	Copy	Delete	123456789012	754321	s754321@student.unklab.ac.id	Citra Lentari Anada	12345	2023-04-13 22:54:33	2023-04-13 22:54:33
☐	Edit	Copy	Delete	133456789016	654321	s654321@student.unklab.ac.id	David Kawekiang	12345	12-12-2023 10:32:43	12-12-2023 10:32:43
☐	Edit	Copy	Delete	143456789012	554321	s554321@student.unklab.ac.id	Dinda Manopo Putri	12345	2023-04-13 22:54:33	2023-04-13 22:54:33
☐	Edit	Copy	Delete	153456789014	604321	s604321@student.unklab.ac.id	Margono Raharjo	12345	2023-04-13 22:54:33	2023-04-13 22:54:33
☐	Edit	Copy	Delete	163456789011	454321	s454321@student.unklab.ac.id	Miehie Kawekiang Sie	12345	2023-04-13 22:54:33	2023-04-13 22:54:33
☐	Edit	Copy	Delete	173456789019	254321	s254321@student.unklab.ac.id	Putra Podomi	12345	2023-04-13 22:54:33	2023-04-13 22:54:33
☐	Edit	Copy	Delete	183456789012	954321	s954321@student.unklab.ac.id	Putri Moningka	12345	2023-04-13 22:54:33	2023-04-13 22:54:33
☐	Edit	Copy	Delete	193456789013	154321	s154321@student.unklab.ac.id	Radix Basalama	12345	2023-04-13 22:54:33	2023-04-13 22:54:33

Exercise #2

Buat login form berikut ini di cloud server
(file project akan disiapkan)

WinSCP (Windows Secure Copy)

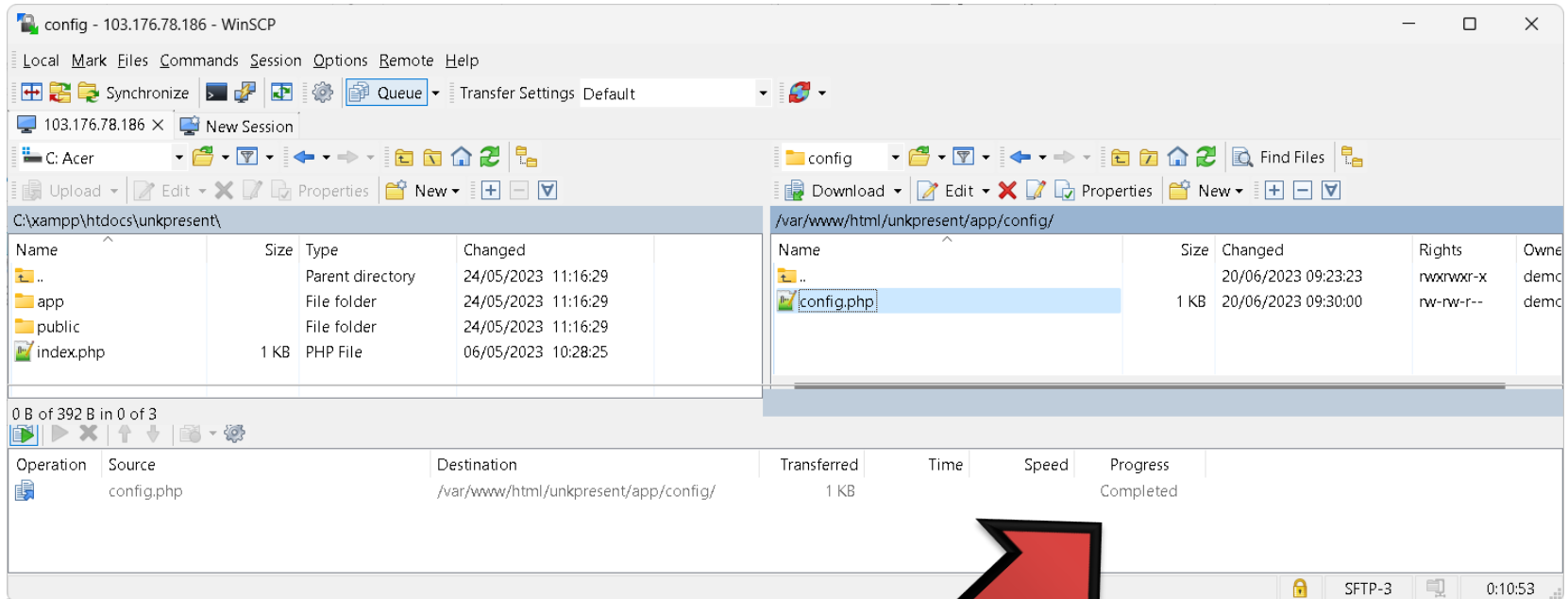
WinSCP (Windows Secure Copy) menyediakan interface grafis untuk mentransfer file antara komputer lokal dan *cloud server* melalui protokol Secure Copy (SCP) atau File Transfer Protocol Secure (SFTP).



Local Computer

Cloud Server

Check Transfer File Progress



Mengecek transfer file progress di WinSCP sangat penting karena memberikan informasi tentang status dan kemajuan proses transfer file yang sedang berlangsung antara komputer lokal dan cloud server.

Create Project Directory

[1] Masuk sebagai *administrator* atau *root*:

➤ ***sudo su***

[2] Buat project folder baru dengan perintah (Parent directory):

➤ ***sudo mkdir -p /var/www/html/unkpresent***

[3] Konfigurasi permisi hak akses dan pemilik dari folder:

➤ ***sudo chmod -R 755 /var/www/html/unkpresent***

➤ ***sudo chown -R semmy:semmy /var/www/html/unkpresent***

[4] Restart Apache Web Server:

➤ ***sudo systemctl restart apache2***

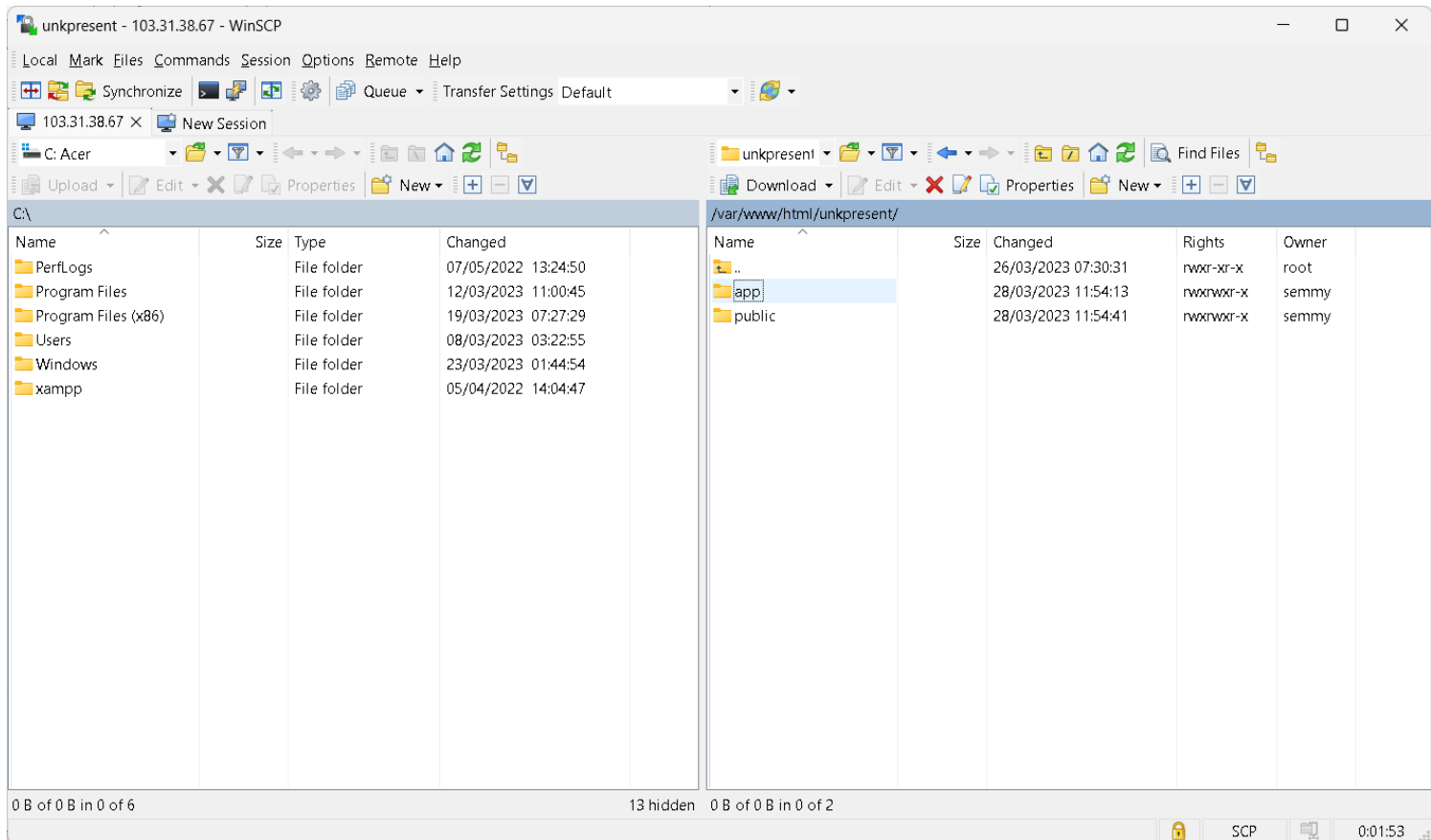
Note:

Perhatikan di point yang ke-3, terdapat nama *user* ***semmy*** yang sudah diberi tanda merah. Nama *user* tersebut harus disesuaikan dengan nama *user server* kalian.

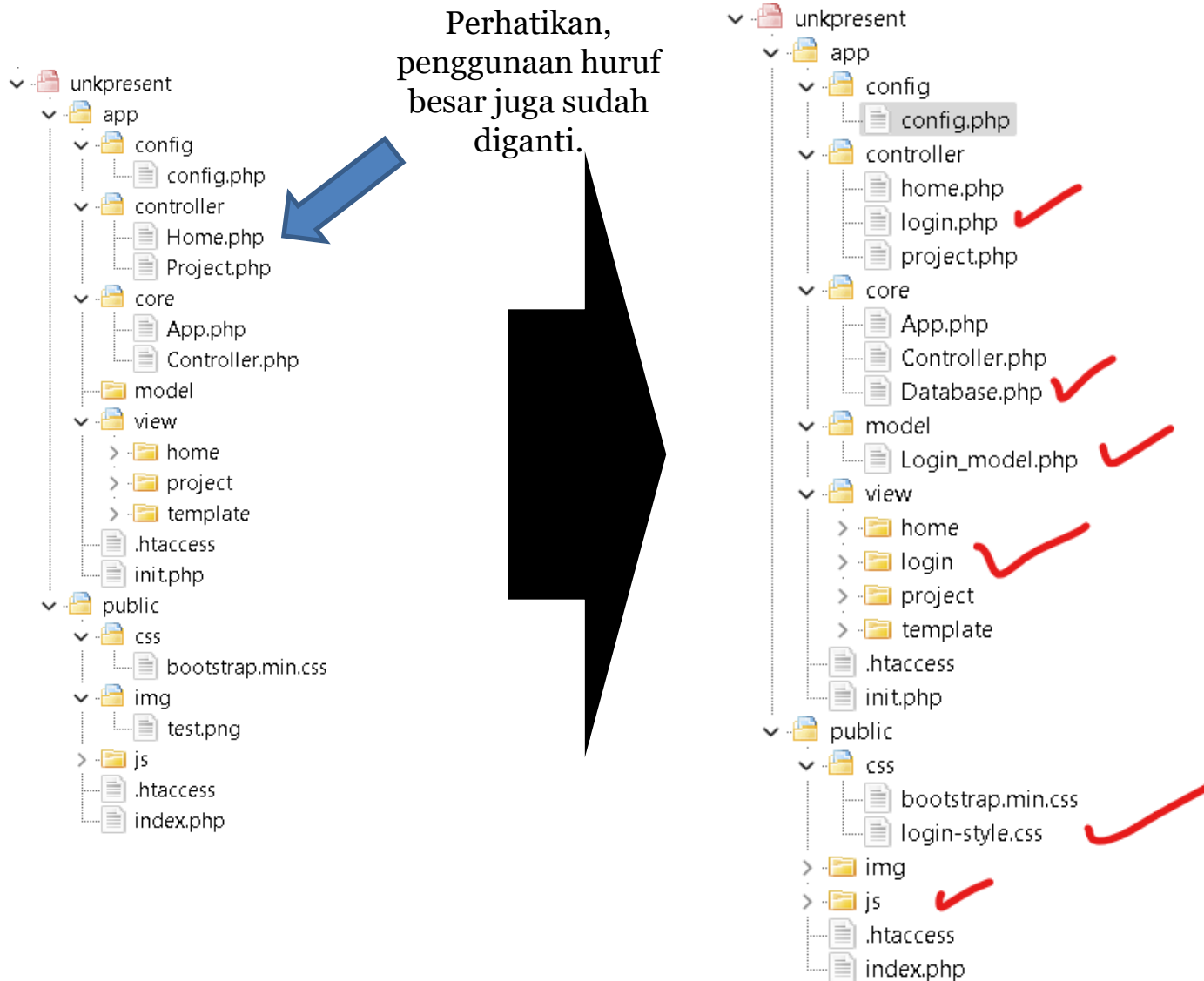
```
root@unklab: /var/www/html
root@unklab:/home/semmy# sudo mkdir -p /var/www/html/unkpresent
root@unklab:/home/semmy# sudo chmod -R 755 /var/www/html/unkpresent
root@unklab:/home/semmy# cd ..
root@unklab:/home# cd ..
root@unklab:/# ls
bin  dev  home  lib32  libx32  media  opt  root  sbin  srv  tmp  var
boot  etc  lib  lib64  lost+found  mnt  proc  run  snap  sys  usr
root@unklab:/# cd var
root@unklab:/var# ls
backups  crash  local  log  opt  snap  tmp
cache  lib  lock  mail  run  spool  www
root@unklab:/var# cd www/
root@unklab:/var/www# ls
html
root@unklab:/var/www# cd html/
root@unklab:/var/www/html# ls
index.html  test.php  unkpresent
root@unklab:/var/www/html# ls -l
total 20
-rw-r--r-- 1 root root 10671 Mar 26 07:06 index.html
-rw-r--r-- 1 root root 20 Mar 26 07:29 test.php
drwxr-xr-x 2 root root 4096 Mar 26 07:30 unkpresent
root@unklab:/var/www/html# sudo chown -R semmy:semmy /var/www/html/unkpresent
root@unklab:/var/www/html# sudo systemctl restart apache2
root@unklab:/var/www/html# ls -l
total 20
-rw-r--r-- 1 root root 10671 Mar 26 07:06 index.html
-rw-r--r-- 1 root root 20 Mar 26 07:29 test.php
drwxr-xr-x 2 semmy semmy 4096 Mar 26 07:30 unkpresent
root@unklab:/var/www/html#
```

Open WinSCP WorkSpace

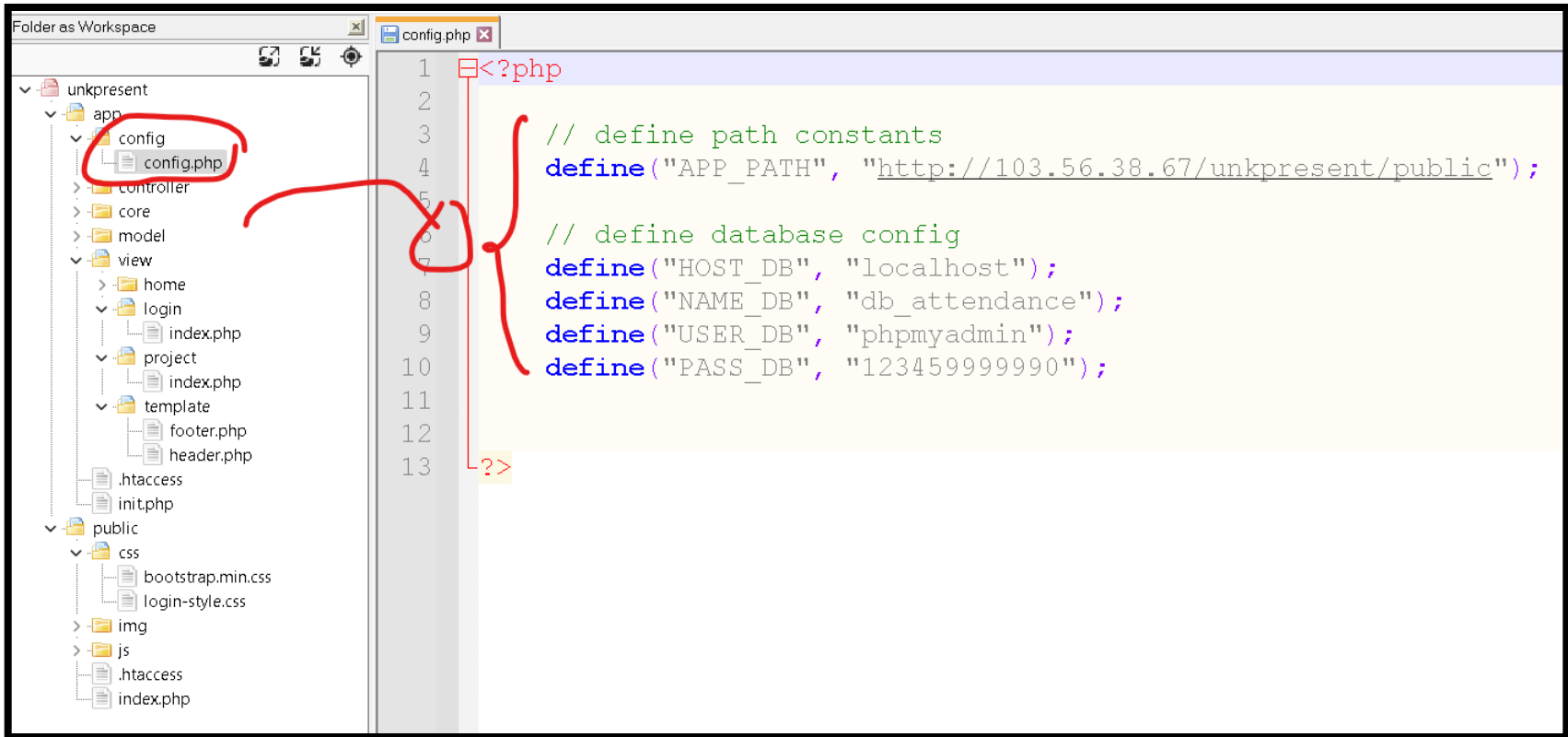
Root directory:
/var/www/html/unkpresent



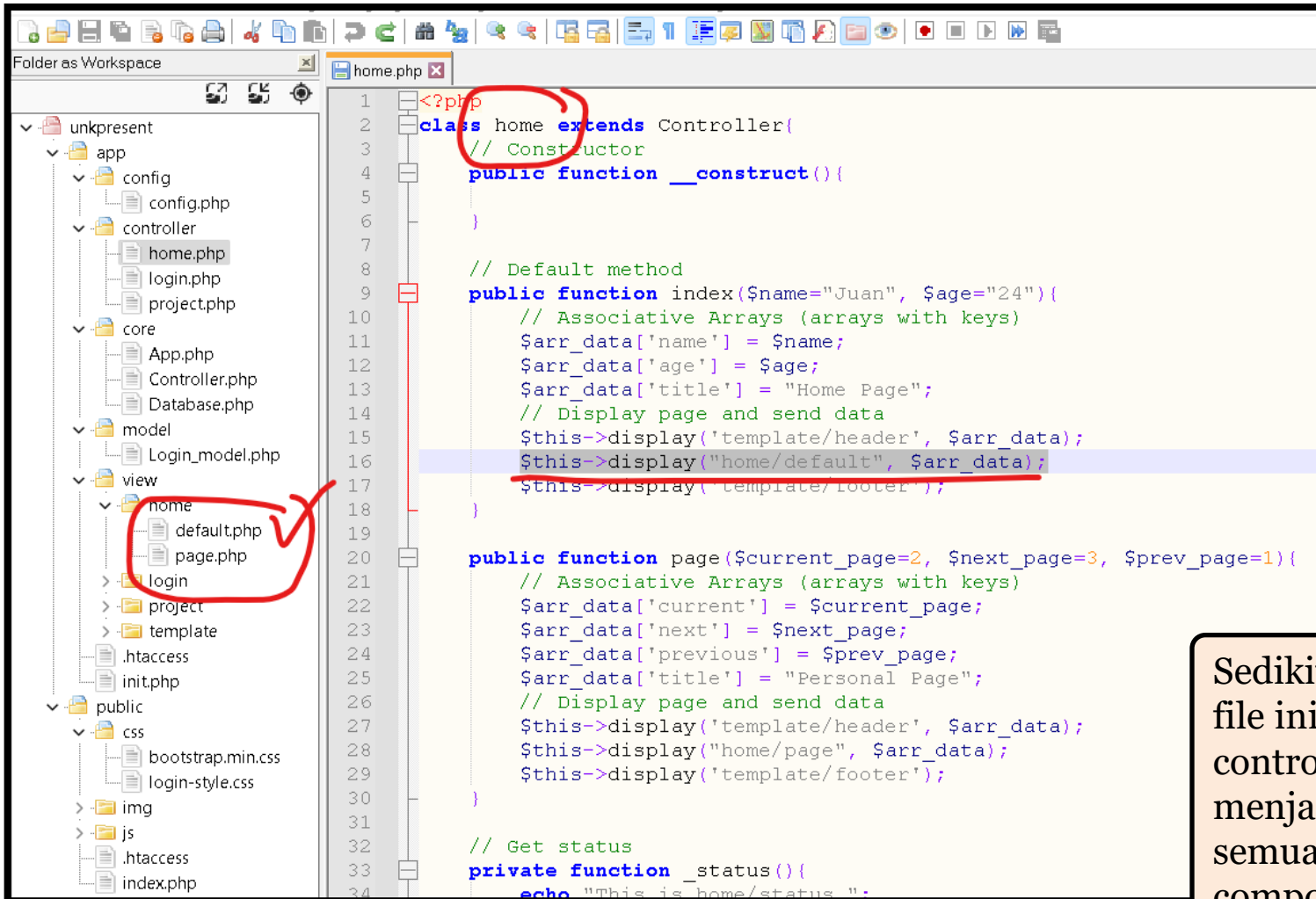
Comparison After Changes & Additions



Change “config.php” File



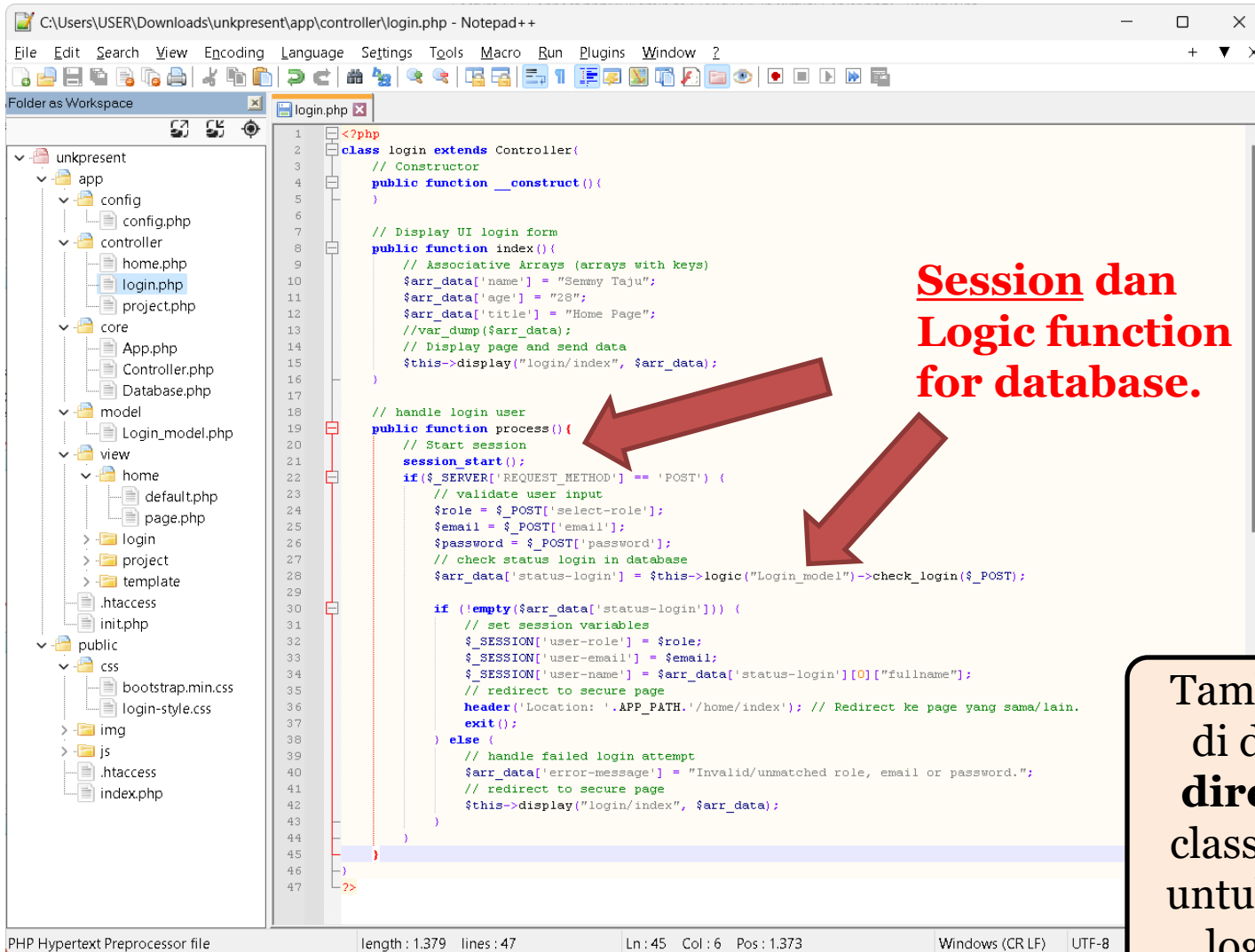
Modify “home.php” File



```
1 <?php
2 class home extends Controller{
3     // Constructor
4     public function __construct(){
5
6     }
7
8     // Default method
9     public function index($name="Juan", $age="24"){
10        // Associative Arrays (arrays with keys)
11        $sarr_data['name'] = $name;
12        $sarr_data['age'] = $age;
13        $sarr_data['title'] = "Home Page";
14        // Display page and send data
15        $this->display('template/header', $sarr_data);
16        $this->display('home/default', $sarr_data);
17        $this->display('template/footer');
18    }
19
20    public function page($current_page=2, $next_page=3, $prev_page=1){
21        // Associative Arrays (arrays with keys)
22        $sarr_data['current'] = $current_page;
23        $sarr_data['next'] = $next_page;
24        $sarr_data['previous'] = $prev_page;
25        $sarr_data['title'] = "Personal Page";
26        // Display page and send data
27        $this->display('template/header', $sarr_data);
28        $this->display('home/page', $sarr_data);
29        $this->display('template/footer');
30    }
31
32    // Get status
33    private function _status(){
34        echo "This is home/status ";
```

Sedikit perubahan pada file ini, seperti nama controller “home” sudah menjadi huruf kecil semua dan UI pada component view diganti dari *index.php* ke *default.php*.

Add Controller “login.php” File

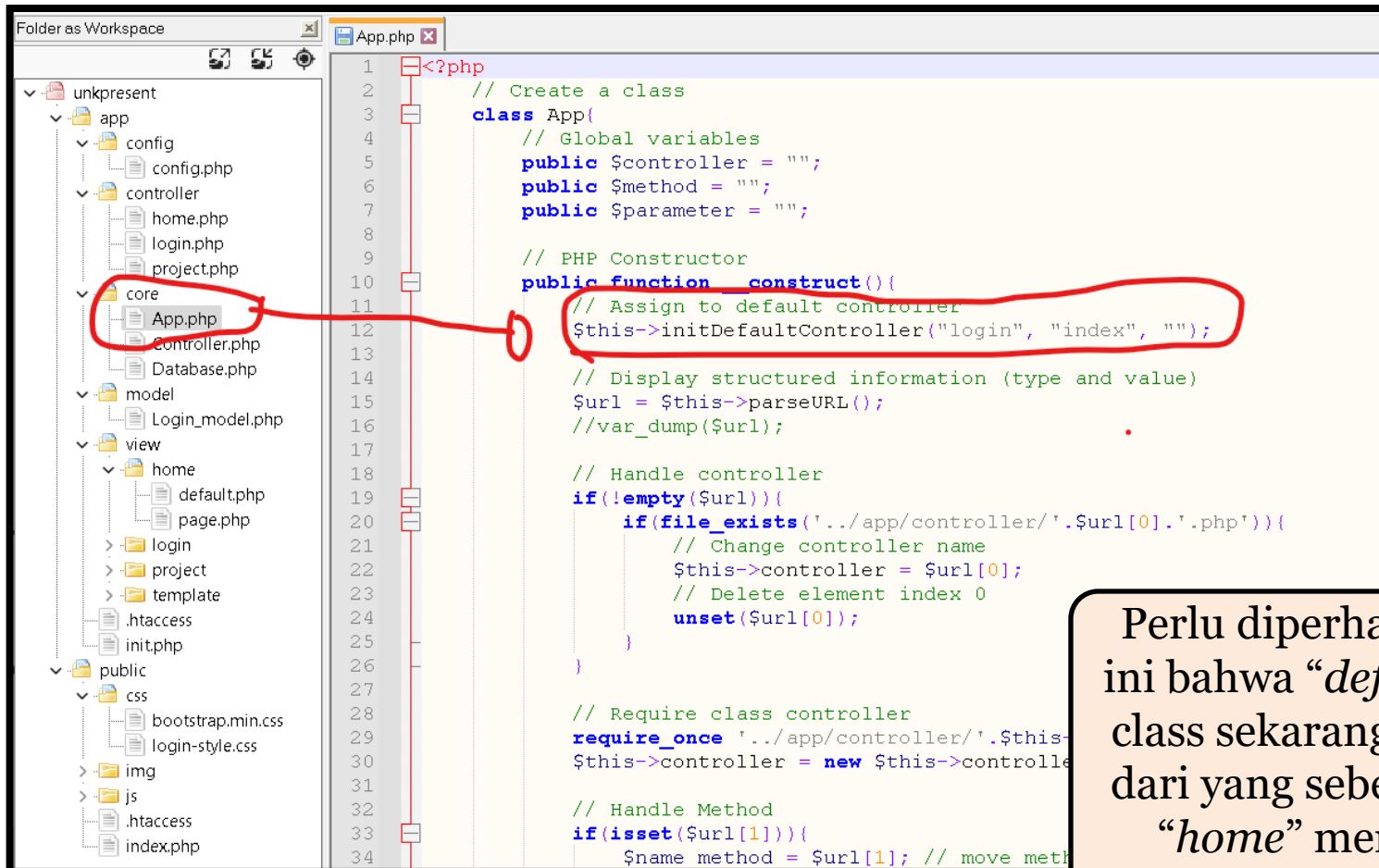


```
1 <?php
2 class login extends Controller{
3     // Constructor
4     public function __construct(){
5     }
6
7     // Display UI login form
8     public function index(){
9         // Associative Arrays (arrays with keys)
10        $arr_data['name'] = "Semmy Taju";
11        $arr_data['age'] = "28";
12        $arr_data['title'] = "Home Page";
13        //var_dump($arr_data);
14        // Display page and send data
15        $this->display("login/index", $arr_data);
16    }
17
18    // handle login user
19    public function process(){
20        // Start session
21        session_start();
22        if($_SERVER['REQUEST_METHOD'] == 'POST') {
23            // validate user input
24            $role = $_POST['select-role'];
25            $email = $_POST['email'];
26            $password = $_POST['password'];
27            // check status login in database
28            $arr_data['status-login'] = $this->logic("Login_model")->check_login($_POST);
29
30            if (!empty($arr_data['status-login'])) {
31                // set session variables
32                $_SESSION['user-role'] = $role;
33                $_SESSION['user-email'] = $email;
34                $_SESSION['user-name'] = $arr_data['status-login'][0]['fullname'];
35                // redirect to secure page
36                header('Location: ' . APP_PATH . '/home/index'); // Redirect ke page yang sama/lain.
37                exit();
38            } else {
39                // handle failed login attempt
40                $arr_data['error-message'] = "Invalid/unmatched role, email or password.";
41                // redirect to secure page
42                $this->display("login/index", $arr_data);
43            }
44        }
45    }
46 }
47 ?>
```

Session dan Logic function for database.

Tambahkan sebuah file di dalam **controller directory**. Controller class ini akan berfungsi untuk menghandle user login. Terdapat dua fungsi di dalam class ini.

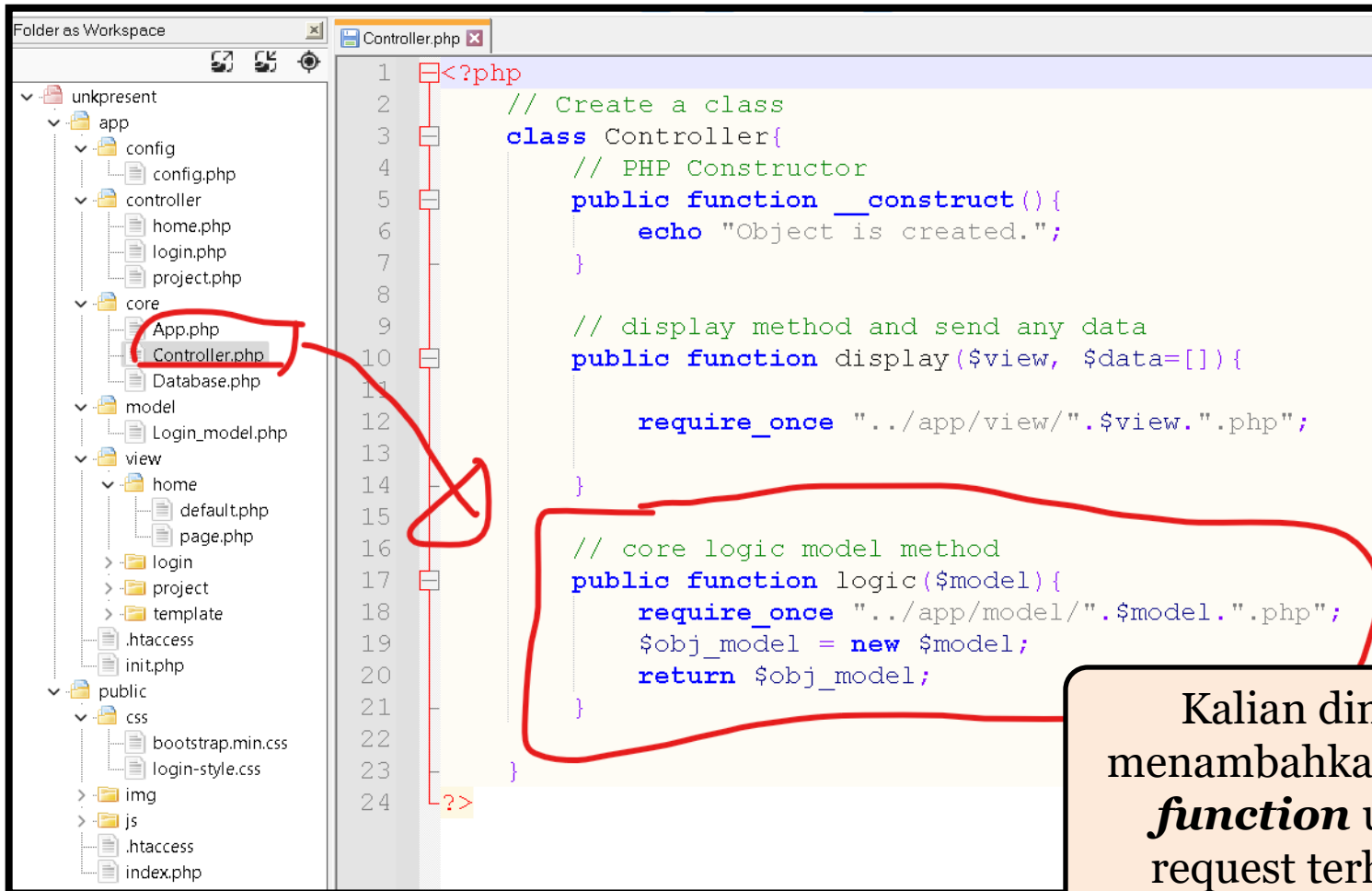
Modify Core “App.php” File



```
1 <?php
2 // Create a class
3 class App{
4     // Global variables
5     public $controller = "";
6     public $method = "";
7     public $parameter = "";
8
9     // PHP Constructor
10    public function construct(){
11        // Assign to default controller
12        $this->initDefaultController("login", "index", "");
13
14        // Display structured information (type and value)
15        $url = $this->parseURL();
16        //var_dump($url);
17
18        // Handle controller
19        if(!empty($url)){
20            if(file_exists('../app/controller/'.$url[0].'.php')){
21                // Change controller name
22                $this->controller = $url[0];
23                // Delete element index 0
24                unset($url[0]);
25            }
26        }
27
28        // Require class controller
29        require_once '../app/controller/'.$this->controller;
30        $this->controller = new $this->controller;
31
32        // Handle Method
33        if(isset($url[1])){
34            $name method = $url[1]; // move meth
```

Perlu diperhatikan ada code ini bahwa “*default controller*” class sekarang sudah di ganti dari yang sebelumnya adalah “*home*” menjadi “*login*”. Dengan harapan, pada saat user mengakses URL yang pertama kali terbuka adalah login form.

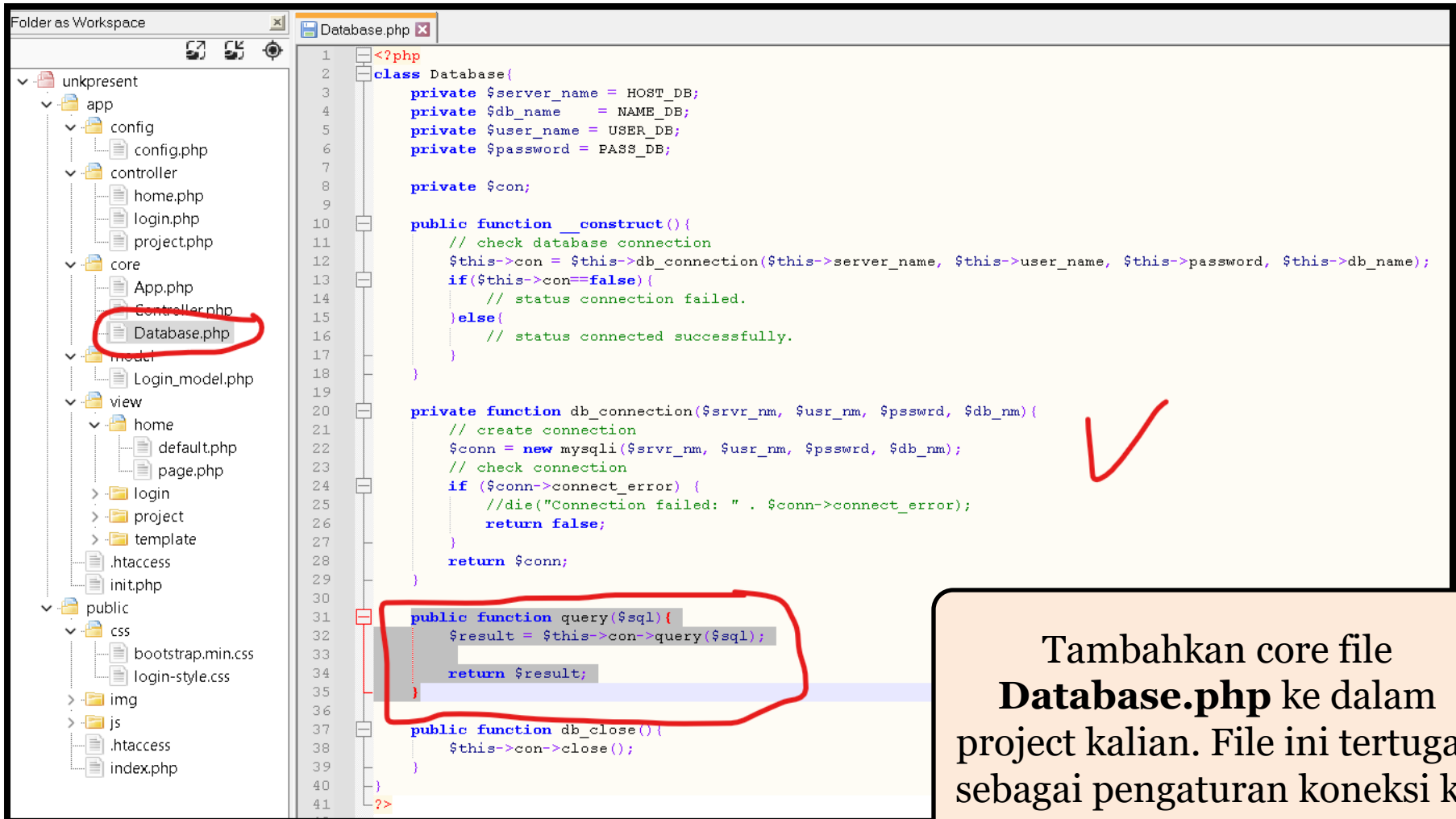
Modify Core “Controller.php” File



```
1 <?php
2 // Create a class
3 class Controller{
4     // PHP Constructor
5     public function __construct(){
6         echo "Object is created.";
7     }
8
9     // display method and send any data
10    public function display($view, $data=[]){
11
12        require_once "../app/view/".$view.".php";
13
14    }
15
16    // core logic model method
17    public function logic($model){
18        require_once "../app/model/".$model.".php";
19        $obj_model = new $model;
20        return $obj_model;
21    }
22 }
23
24 ?>
```

Kalian diminta untuk menambahkan sebuah **logic function** untuk handle request terhadap data di dalam database. Fungsi tersebut akan melakukan koneksi ke target model file.

Add Core “Database.php” File

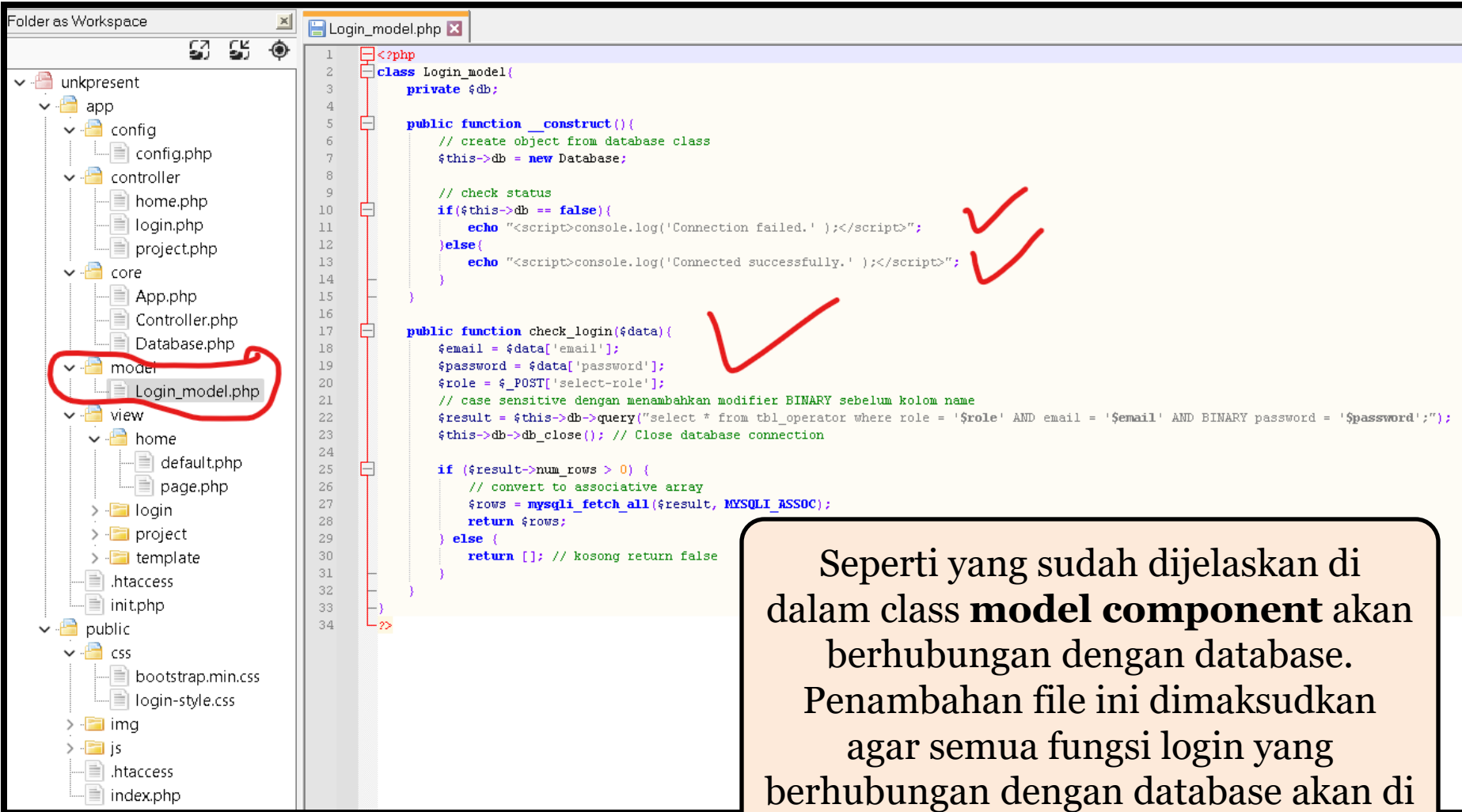


The screenshot shows a code editor with a file explorer on the left and a code editor on the right. The file explorer shows a project structure with a 'core' directory containing 'App.php', 'Controller.php', and 'Database.php'. The 'Database.php' file is highlighted with a red circle. The code editor shows the contents of 'Database.php', which defines a 'Database' class with private properties for database connection details and public methods for connection, query, and closing the connection. The 'query' method is highlighted with a red box. A red checkmark is visible next to the code.

```
1 <?php
2 class Database{
3     private $server_name = HOST_DB;
4     private $db_name     = NAME_DB;
5     private $user_name   = USER_DB;
6     private $password    = PASS_DB;
7
8     private $con;
9
10    public function __construct(){
11        // check database connection
12        $this->con = $this->db_connection($this->server_name, $this->user_name, $this->password, $this->db_name);
13        if($this->con==false){
14            // status connection failed.
15        }else{
16            // status connected successfully.
17        }
18    }
19
20    private function db_connection($server_nm, $usr_nm, $psswr, $db_nm){
21        // create connection
22        $conn = new mysqli($server_nm, $usr_nm, $psswr, $db_nm);
23        // check connection
24        if ($conn->connect_error) {
25            //die("Connection failed: " . $conn->connect_error);
26            return false;
27        }
28        return $conn;
29    }
30
31    public function query($sql){
32        $result = $this->con->query($sql);
33
34        return $result;
35    }
36
37    public function db_close(){
38        $this->con->close();
39    }
40
41    ?>
```

Tambahkan core file **Database.php** ke dalam project kalian. File ini bertugas sebagai pengaturan koneksi ke database dan handle query SQL.

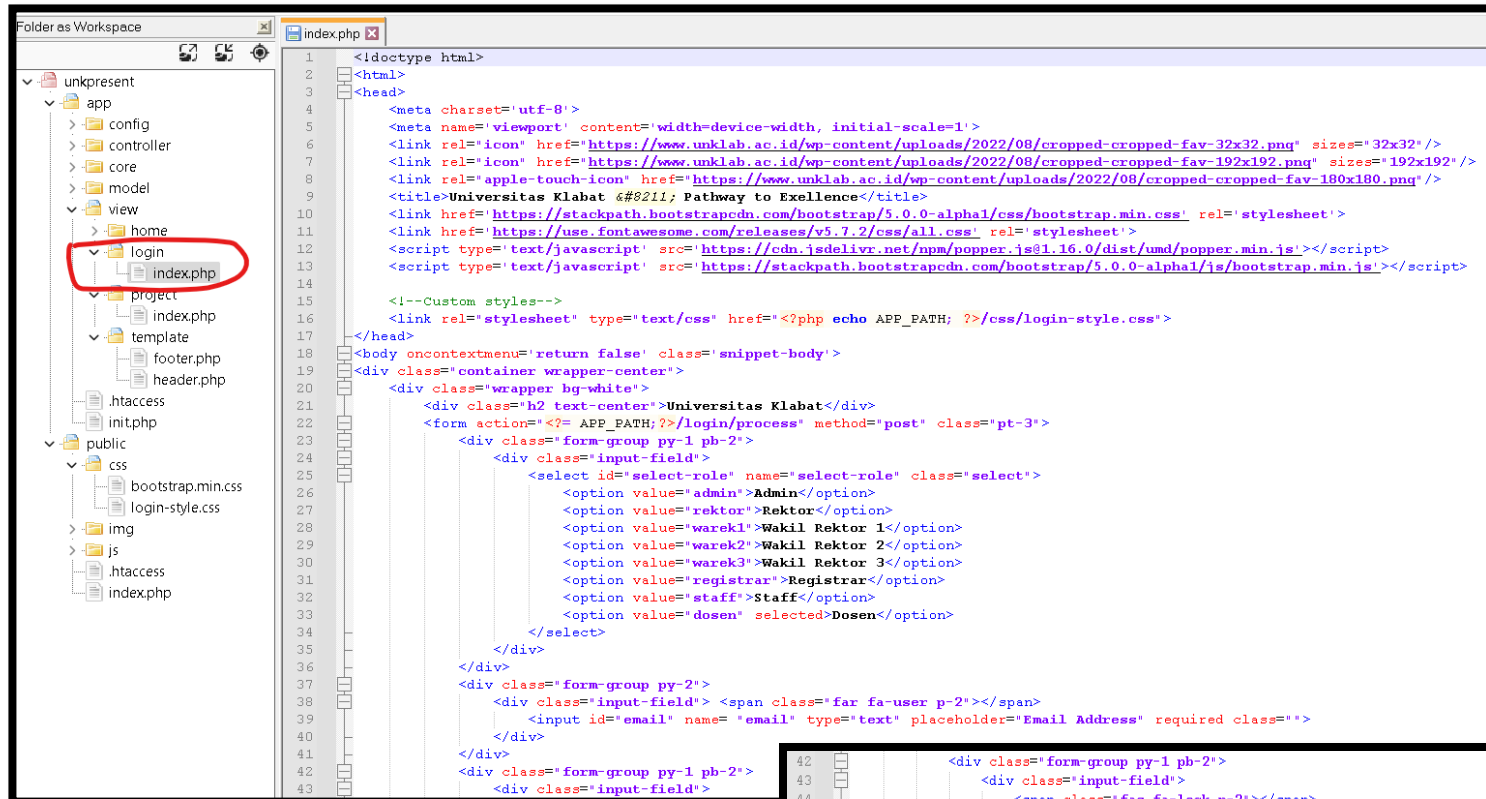
Add Model “Login_model.php” File



```
1 <?php
2 class Login_model{
3     private $db;
4
5     public function __construct(){
6         // create object from database class
7         $this->db = new Database;
8
9         // check status
10        if($this->db == false){
11            echo "<script>console.log('Connection failed.' );</script>";
12        }else{
13            echo "<script>console.log('Connected successfully.' );</script>";
14        }
15    }
16
17    public function check_login($data){
18        $email = $data['email'];
19        $password = $data['password'];
20        $role = $_POST['select-role'];
21        // case sensitive dengan menambahkan modifier BINARY sebelum kolom name
22        $result = $this->db->query("select * from tbl_operator where role = '$role' AND email = '$email' AND BINARY password = '$password'");
23        $this->db->db_close(); // Close database connection
24
25        if ($result->num_rows > 0) {
26            // convert to associative array
27            $rows = mysqli_fetch_all($result, MYSQLI_ASSOC);
28            return $rows;
29        } else {
30            return []; // kosong return false
31        }
32    }
33 }
34
```

Seperti yang sudah dijelaskan di dalam class **model component** akan berhubungan dengan database. Penambahan file ini dimaksudkan agar semua fungsi login yang berhubungan dengan database akan di handle pada file ini. Sebagai jembatan antara view dan controller.

Create Folder “login” & File “index.php”

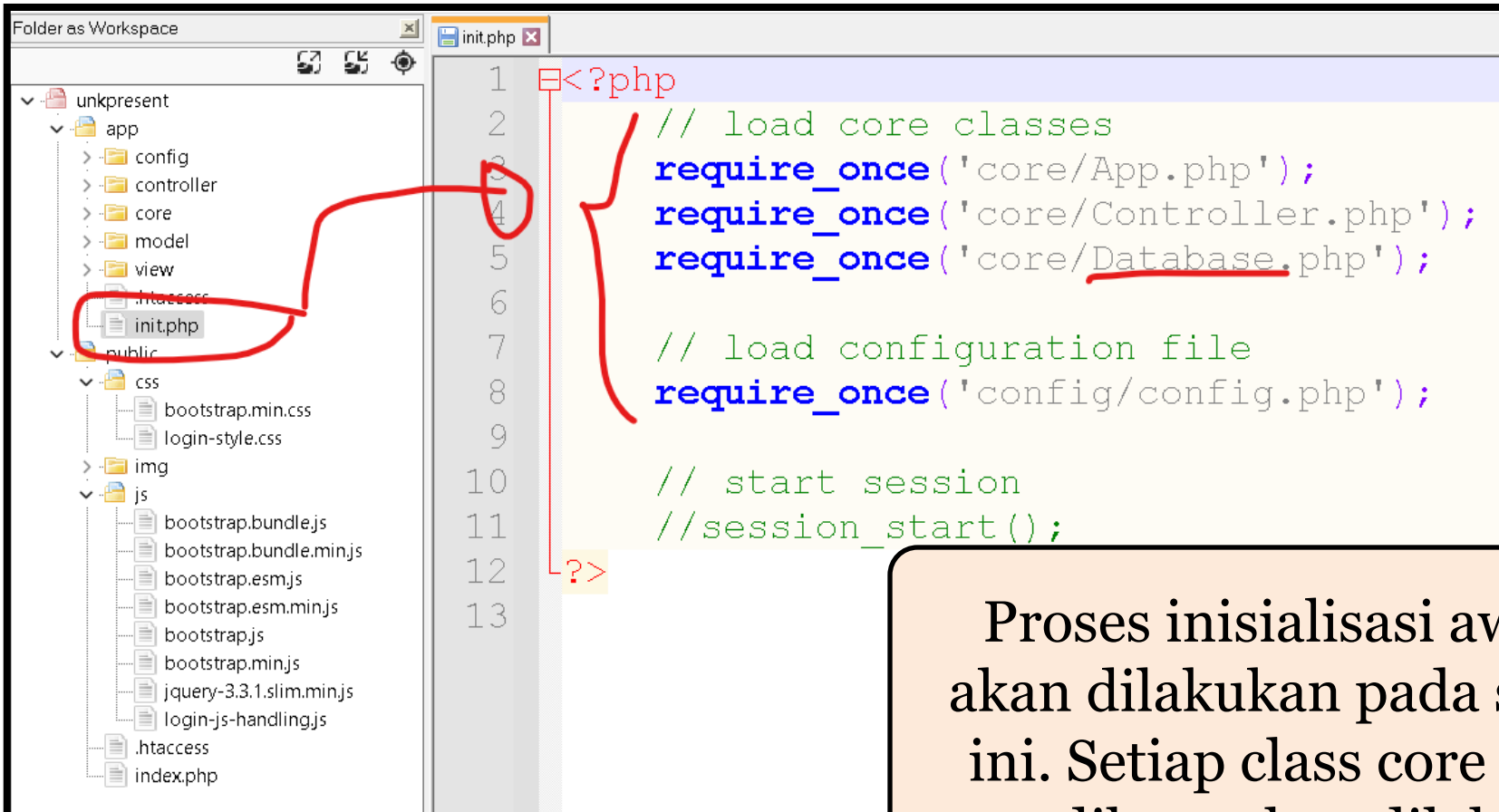


```
1 <!doctype html>
2 <html>
3 <head>
4 <meta charset='utf-8'>
5 <meta name='viewport' content='width=device-width, initial-scale=1'>
6 <link rel='icon' href='https://www.unklab.ac.id/wp-content/uploads/2022/08/cropped-cropped-fav-32x32.png' sizes='32x32' />
7 <link rel='icon' href='https://www.unklab.ac.id/wp-content/uploads/2022/08/cropped-cropped-fav-192x192.png' sizes='192x192' />
8 <link rel='apple-touch-icon' href='https://www.unklab.ac.id/wp-content/uploads/2022/08/cropped-cropped-fav-180x180.png' />
9 <title>Universitas Klabat &#8211; Pathway to Excellence</title>
10 <link href='https://stackpath.bootstrapcdn.com/bootstrap/5.0.0-alpha1/css/bootstrap.min.css' rel='stylesheet'>
11 <link href='https://use.fontawesome.com/releases/v5.7.2/css/all.css' rel='stylesheet'>
12 <script type='text/javascript' src='https://cdn.jsdelivr.net/npm/popper.js@1.16.0/dist/umd/popper.min.js'></script>
13 <script type='text/javascript' src='https://stackpath.bootstrapcdn.com/bootstrap/5.0.0-alpha1/js/bootstrap.min.js'></script>
14
15 <!--Custom styles-->
16 <link rel='stylesheet' type='text/css' href='<?php echo APP_PATH; ?>/css/login-style.css'>
17 </head>
18 <body oncontextmenu='return false' class='snippet-body'>
19 <div class='container wrapper-center'>
20 <div class='wrapper bg-white'>
21 <div class='h2 text-center'>Universitas Klabat</div>
22 <form action='<?php echo APP_PATH; ?>/login/process' method='post' class='pt-3'>
23 <div class='form-group py-1 pb-2'>
24 <div class='input-field'>
25 <select id='select-role' name='select-role' class='select'>
26 <option value='admin'>Admin</option>
27 <option value='rektor'>Rektor</option>
28 <option value='warek1'>Wakil Rektor 1</option>
29 <option value='warek2'>Wakil Rektor 2</option>
30 <option value='warek3'>Wakil Rektor 3</option>
31 <option value='registrar'>Registrar</option>
32 <option value='staff'>Staff</option>
33 <option value='dosen' selected>Dosen</option>
34 </select>
35 </div>
36 </div>
37 <div class='form-group py-2'>
38 <div class='input-field'> <span class='fas fa-user p-2'></span>
39 <input id='email' name='email' type='text' placeholder='Email Address' required class=''>
40 </div>
41 </div>
42 <div class='form-group py-1 pb-2'>
43 <div class='input-field'>
```

Perhatikan
nomor
baris.

```
42 <div class='form-group py-1 pb-2'>
43 <div class='input-field'>
44 <span class='fas fa-lock p-2'></span>
45 <input id='password' name='password' type='password' placeholder='Password' required class=''>
46 <button type='button' class='btn bg-white text-muted'>
47 <span class='fas fa-eye-slash'></span> </button>
48 </div>
49 </div>
50 <?php
51 if(isset($data['error-message'])) {
52     echo "<div id='error-mes-login' style='font-size: 12px;color:red'>". $data['error-message']. "</div>";
53 }
54 <?>
55 <button type='submit' class='btn btn-block text-center mt-3'>Log in</button>
56 </form>
57 </div>
58 </div>
59 <script src='<?php echo APP_PATH; ?>/js/jquery-3.3.1.slim.min.js'></script>
60 <script type='text/javascript' src='<?php echo APP_PATH; ?>/js/login-js-handling.js'></script>
61 </body>
62 </html>
```

Modify “*init.php*” File

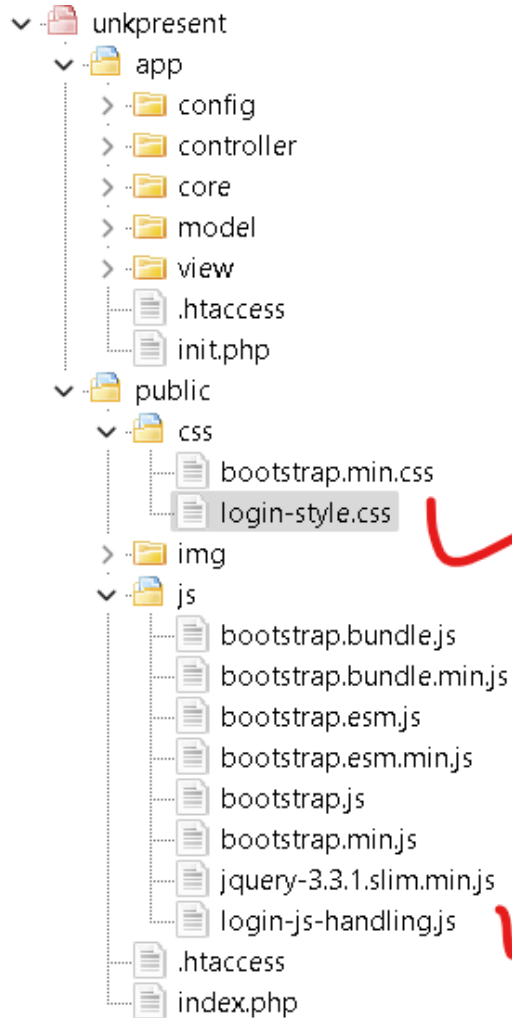


Proses inisialisasi awal akan dilakukan pada step ini. Setiap class core file yang dibuat akan dilakukan proses load pada setiap file dari core classes.

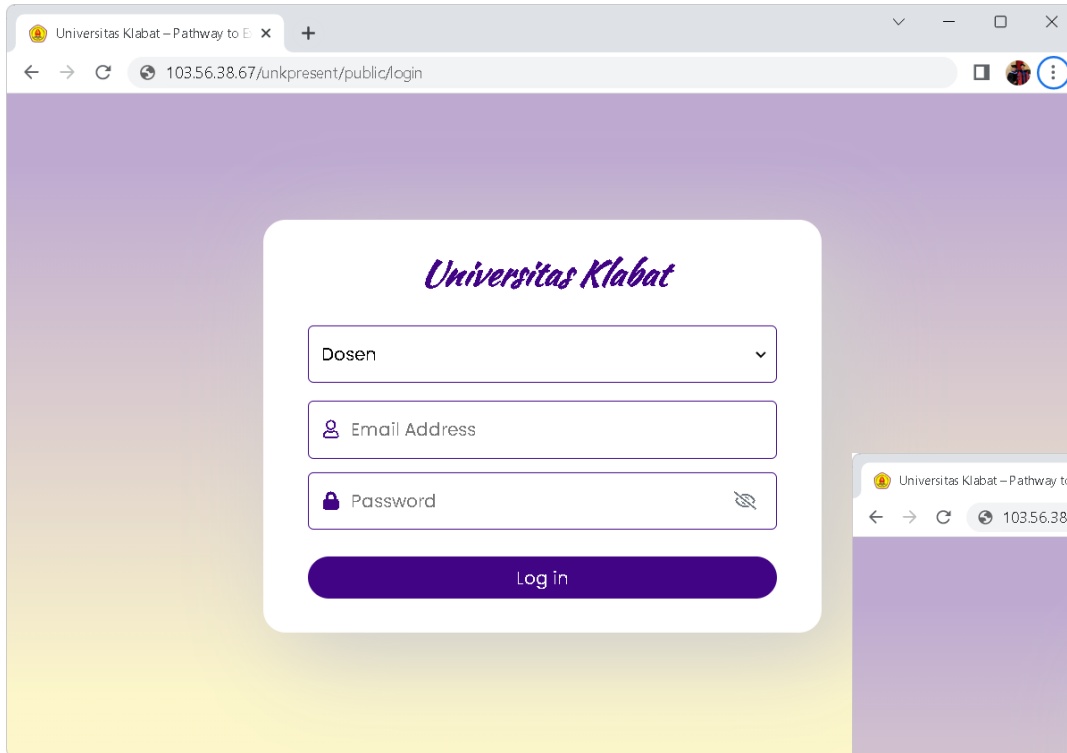
File Styles Login Page

Terdapat dua file yang akan dishare di Google Classroom:

- 1. login-style.css**
- 2. login-js-handling.js**

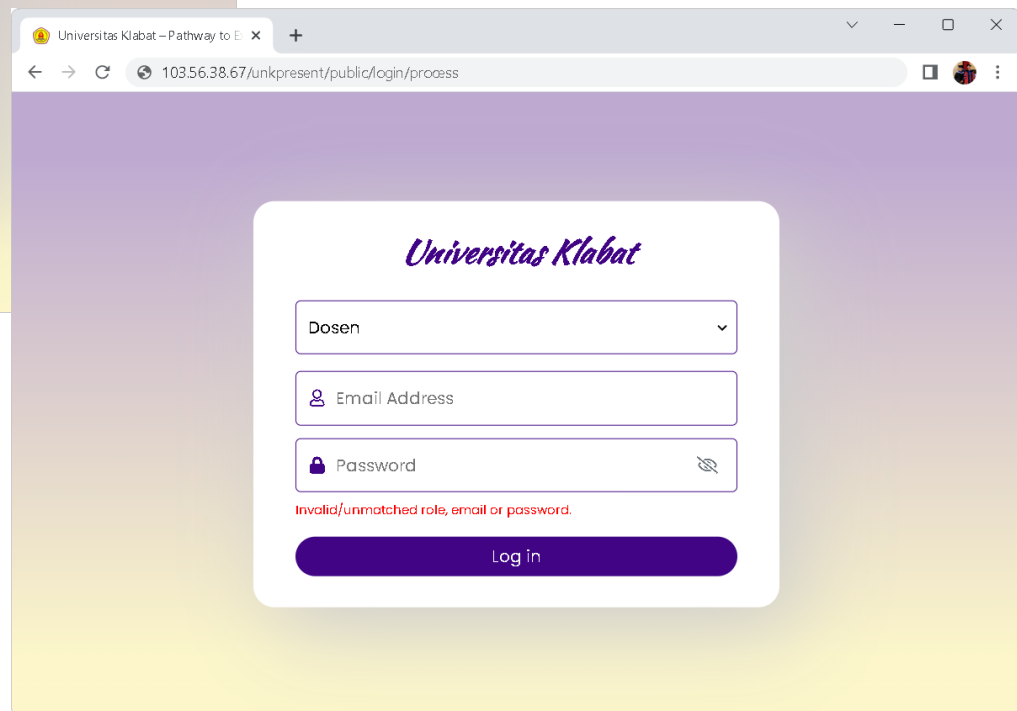


Expected Output



A screenshot of a web browser showing the login page for Universitas Klabat. The browser's address bar displays the URL `103.56.38.67/unkpresent/public/login`. The login form is centered on the page and contains the following elements:

- The text *Universitas Klabat* in a purple, italicized font.
- A dropdown menu with the text "Dosen" and a downward arrow.
- An input field with a person icon and the text "Email Address".
- An input field with a lock icon and the text "Password".
- A purple button with the text "Log in".

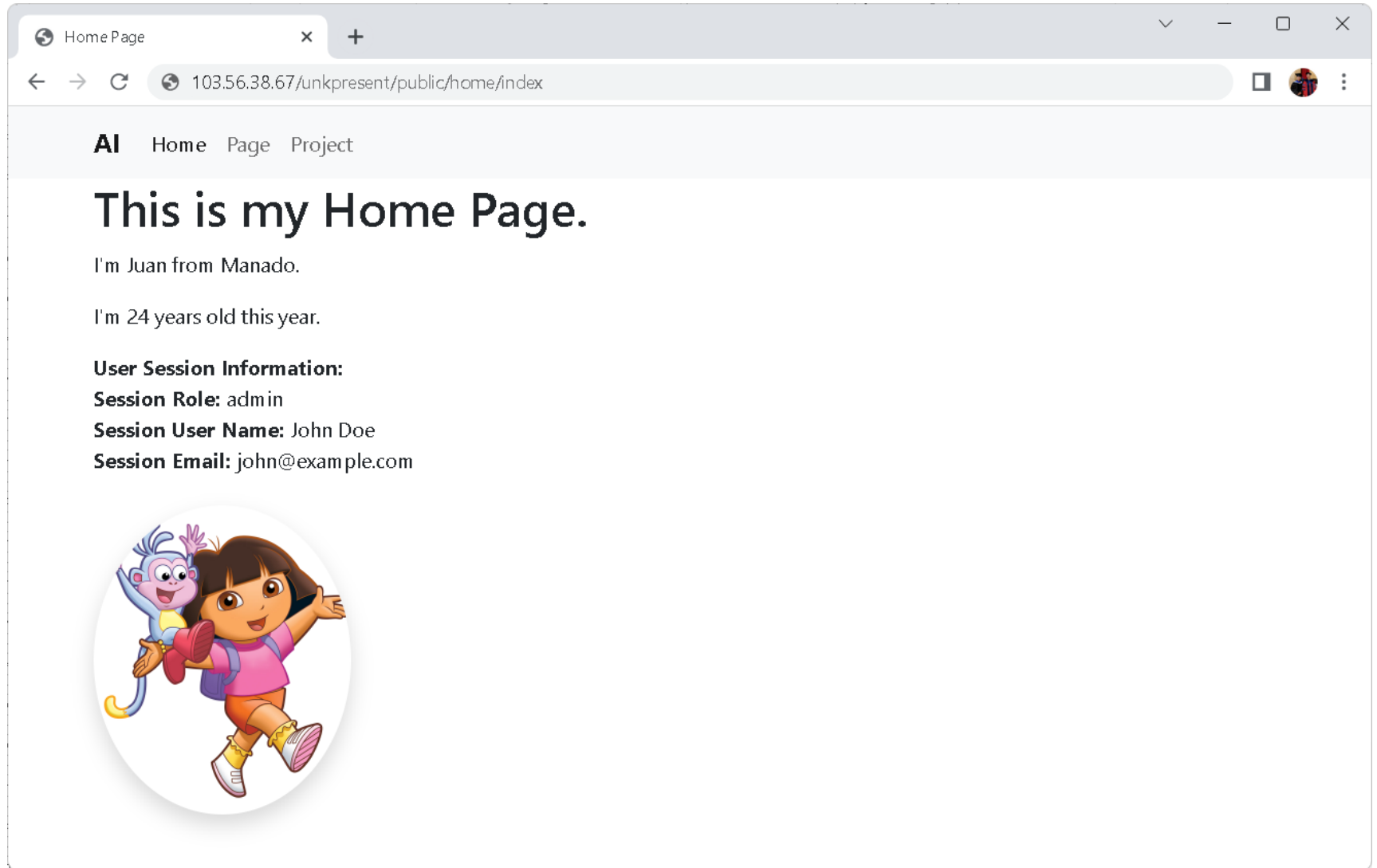


A screenshot of the same web browser showing the login page after an attempt. The address bar now displays `103.56.38.67/unkpresent/public/login/process`. The login form is identical to the previous one, but an error message is displayed below the password field:

Invalid/unmatched role, email or password.

The "Log in" button remains visible at the bottom of the form.

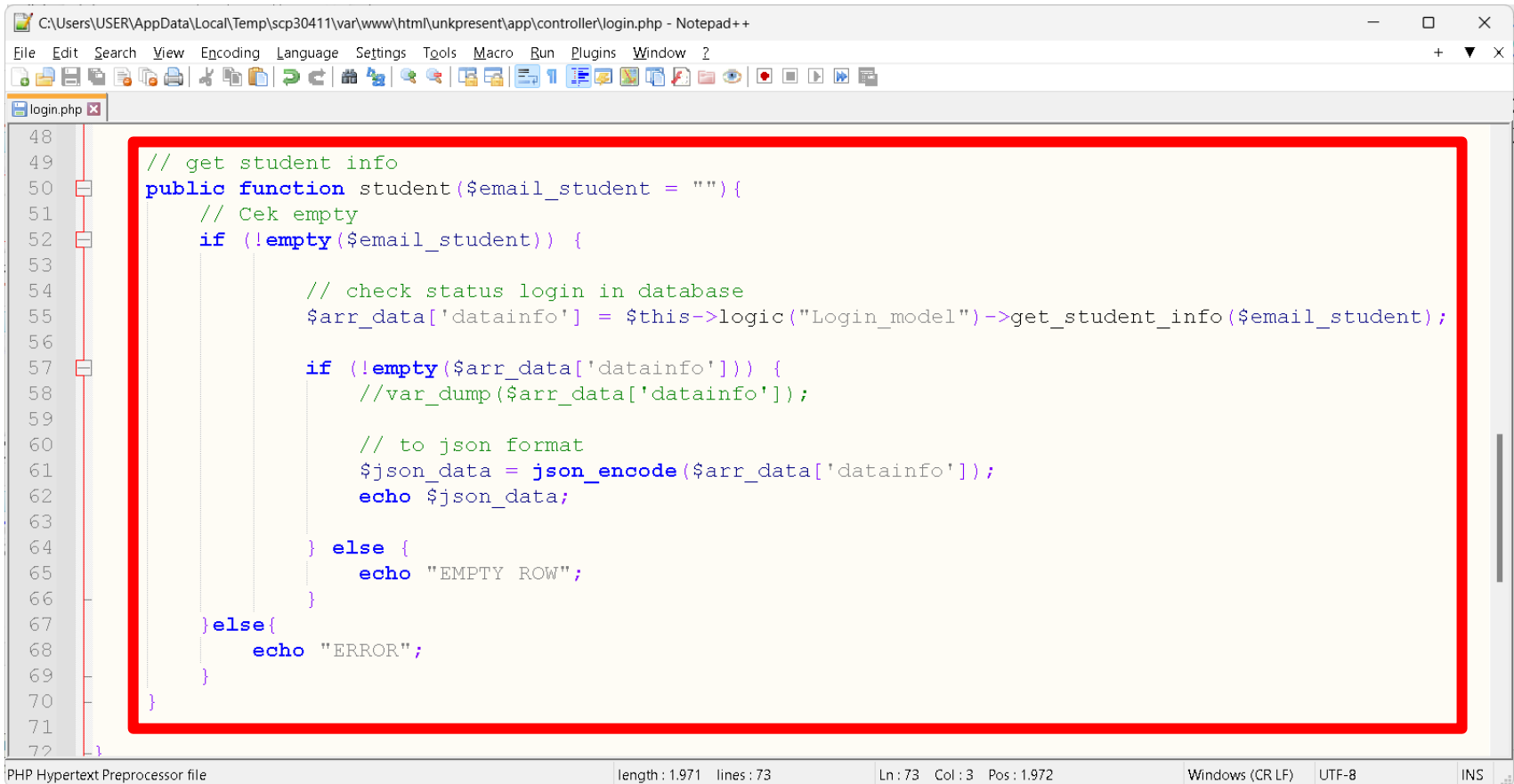
Expected Output



Exercise #3

1. Pelajari *framework* yang sudah digunakan untuk membuat login form.
2. Buat simple RESTful API untuk mengambil/menampilkan “*data mahasiswa*”.
3. Selanjutnya, di local PC kalian. Tangkap (gunakan **PHP** dan **jQuery Ajax**) dan tampilkan data dari RESTful API tersebut di dalam table HTML.

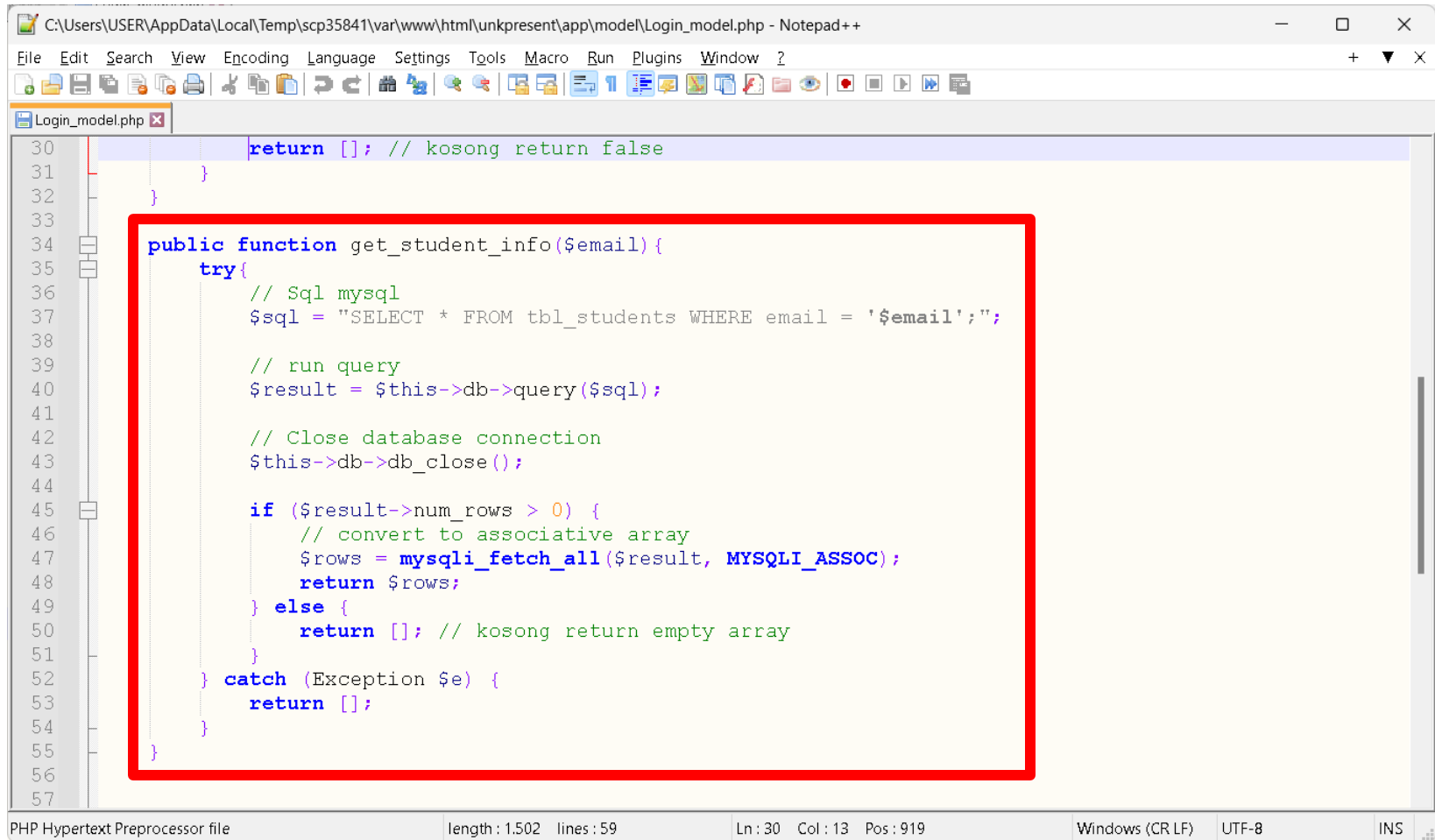
Add Function in “login.php”



```
48
49 // get student info
50 public function student($email_student = ""){
51     // Cek empty
52     if (!empty($email_student)) {
53
54         // check status login in database
55         $arr_data['datainfo'] = $this->logic("Login_model")->get_student_info($email_student);
56
57         if (!empty($arr_data['datainfo'])) {
58             //var_dump($arr_data['datainfo']);
59
60             // to json format
61             $json_data = json_encode($arr_data['datainfo']);
62             echo $json_data;
63
64         } else {
65             echo "EMPTY ROW";
66         }
67     }else{
68         echo "ERROR";
69     }
70 }
71
72
```

Tambahkah sebuah method “student” di *controller login* pada file “login.php”. *Function* ini yang akan memanggil **model function** untuk mengambil data di database berdasarkan email dari mahasiswa.

Add Function in “Login_model.php”



```
30         return []; // kosong return false
31     }
32 }
33
34 public function get_student_info($email){
35     try{
36         // Sql mysql
37         $sql = "SELECT * FROM tbl_students WHERE email = '$email'";
38
39         // run query
40         $result = $this->db->query($sql);
41
42         // Close database connection
43         $this->db->db_close();
44
45         if ($result->num_rows > 0) {
46             // convert to associative array
47             $rows = mysqli_fetch_all($result, MYSQLI_ASSOC);
48             return $rows;
49         } else {
50             return []; // kosong return empty array
51         }
52     } catch (Exception $e) {
53         return [];
54     }
55 }
56
57
```

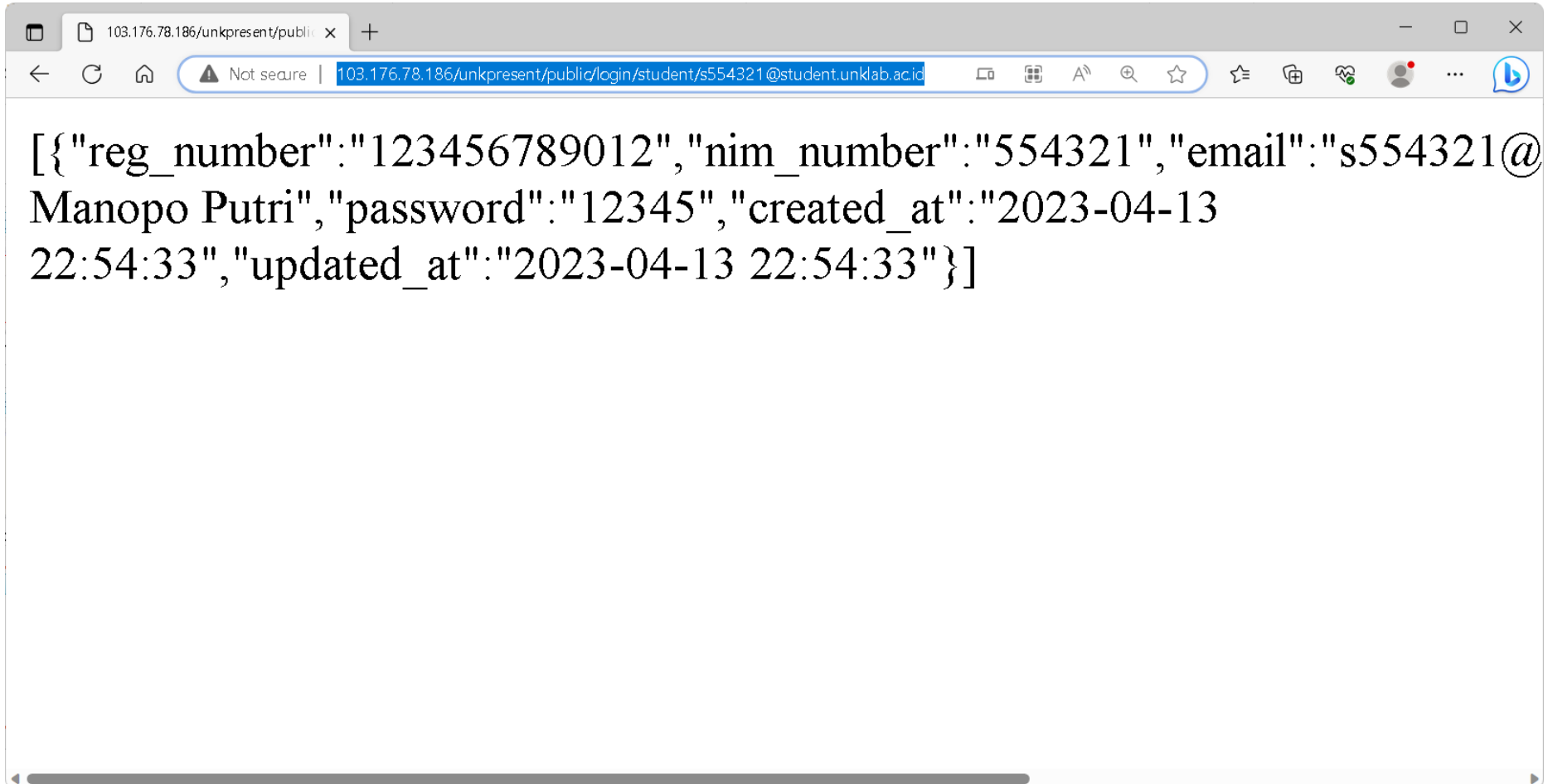
PHP Hypertext Preprocessor file | length: 1.502 | lines: 59 | Ln: 30 | Col: 13 | Pos: 919 | Windows (CR LF) | UTF-8 | INS

Tambahkan sebuah method “get_student_info” di *model Login* pada file “*Model_login.php*”. Function ini yang akan mengambil data secara langsung dari *database* berdasarkan email dari mahasiswa.

Output *from* RESTful API

Link API:

<http://103.176.78.186/unkpresent/public/login/student/s554321@student.unklab.ac.id>



Postman adalah platform kolaborasi untuk pengembangan API. Fitur Postman menyederhanakan setiap langkah pembuatan API dan menyederhanakan kolaborasi sehingga kita dapat membuat API yang lebih baik dan lebih cepat.

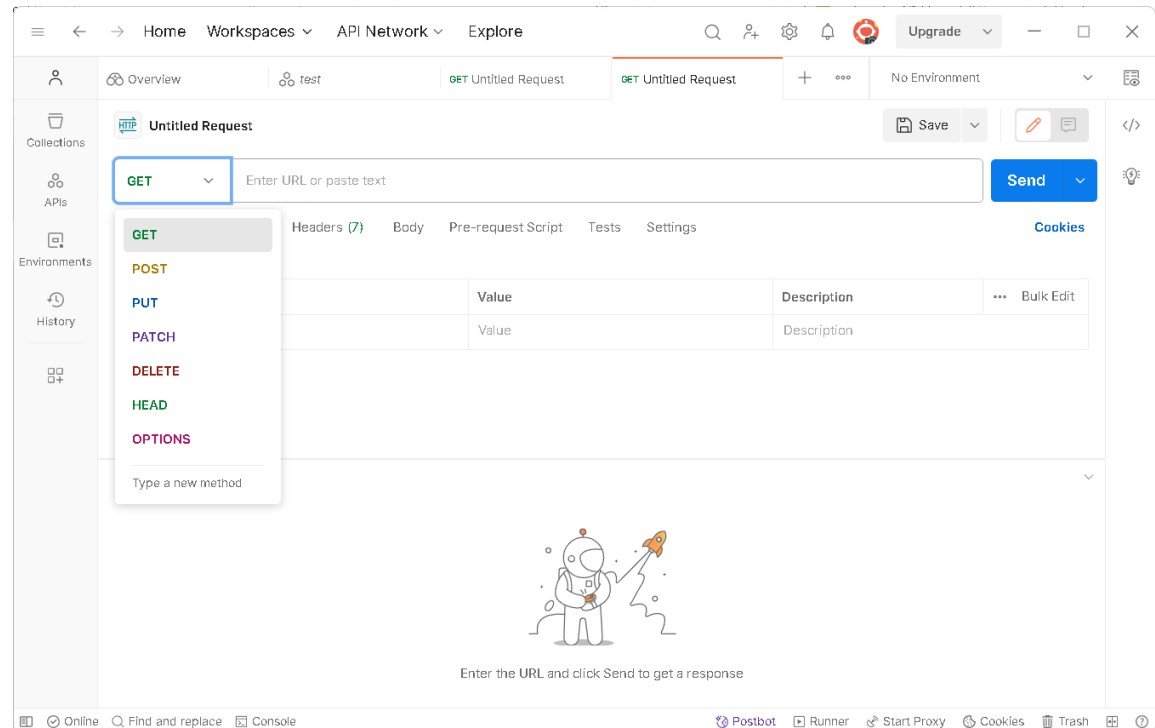
Gunakan URL API yang sudah dibuat (Contoh):

<http://103.176.78.186/unkpresent/public/login/student/s554321@student.unklab.ac.id>

Cara download **Postman**:

- 1) Buka official website postman di <https://www.postman.com/downloads/>
- 2) Pilih Download App
- 3) Selesai Download Double klik dan pilih install -> Tunggu sampai proses selesai.
- 4) Jika sudah selesai silahkan klik tombol Launch untuk menjalankan
- 5) Silahkan Registrasi menggunakan email aktif

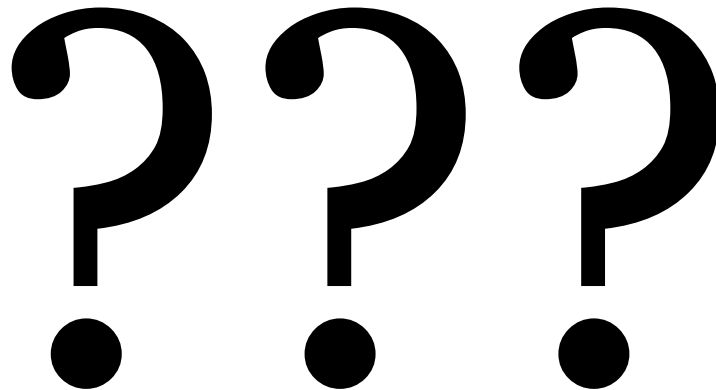
**Sign in dan
Silahkan
Demo cara
menggunakan
an Postman:**



Display Output *in* Table HTML

Link API:

<http://103.176.78.186/unkpresent/public/login/student/s554321@student.unklab.ac.id>



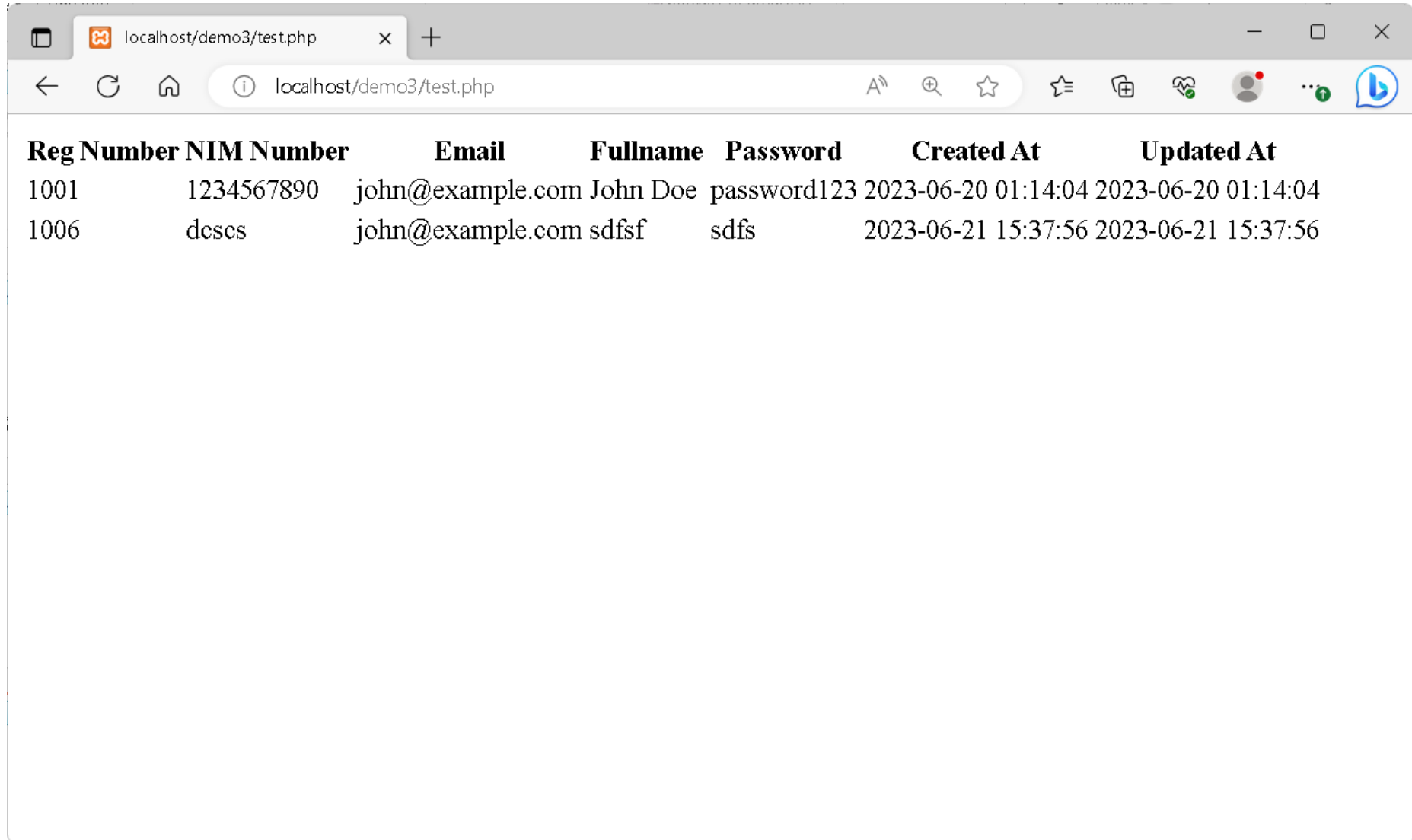
Bagaimana cara untuk tangkap (gunakan PHP dan jQuery Ajax) dan tampilkan output json data dari RESTful API tersebut di dalam *table* HTML dari local PC?

Display Output in Table HTML

```
C:\xampp\htdocs\demo3\test.php - Notepad++
File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window ? +
test.php x
1 <?php
2 // Create variable URL
3 $url = 'http://27.112.78.254/unkpresent/public/login/student/john@example.com';
4
5 // Mengambil konten dari URL
6 $data = file_get_contents($url);
7
8 // Replace format from database
9 $data = str_replace("<script>console.log('Connected successfully. ');</script>", "", $data);
10
11 // Menguraikan data JSON menjadi array PHP
12 $data = json_decode($data, true);
13
14 // Memeriksa apakah penguraian JSON berhasil
15 if ($data != null) {
16     // Membuat tabel HTML
17     echo '<table>';
18     echo '<tr>';
19     echo '<th>Reg Number</th>';
20     echo '<th>NIM Number</th>';
21     echo '<th>Email</th>';
22     echo '<th>Fullname</th>';
23     echo '<th>Password</th>';
24     echo '<th>Created At</th>';
25     echo '<th>Updated At</th>';
26     echo '</tr>';
27
28     // Menampilkan data dalam tabel
29     foreach ($data as $item) {
30         echo '<tr>';
31         echo "&<td>".$item['reg_number']. "</td>";
32         echo "<td>".$item['nim_number']. "</td>";
33         echo "<td>".$item['email']. "</td>";
34         echo "<td>".$item['fullname']. "</td>";
35         echo "<td>".$item['password']. "</td>";
36         echo "<td>".$item['created_at']. "</td>";
37         echo "<td>".$item['updated_at']. "</td>";
38         echo '</tr>';
39     }
40
41     echo '</table>';
42 } else {
43     // Kesalahan dalam penguraian JSON
44     echo "Gagal menguraikan data JSON";
45 }
46
```

length: 1.284 lines: 46 Ln: 20 Col: 36 Pos: 591 Windows (CR LF) UTF-8 INS

Expected Output



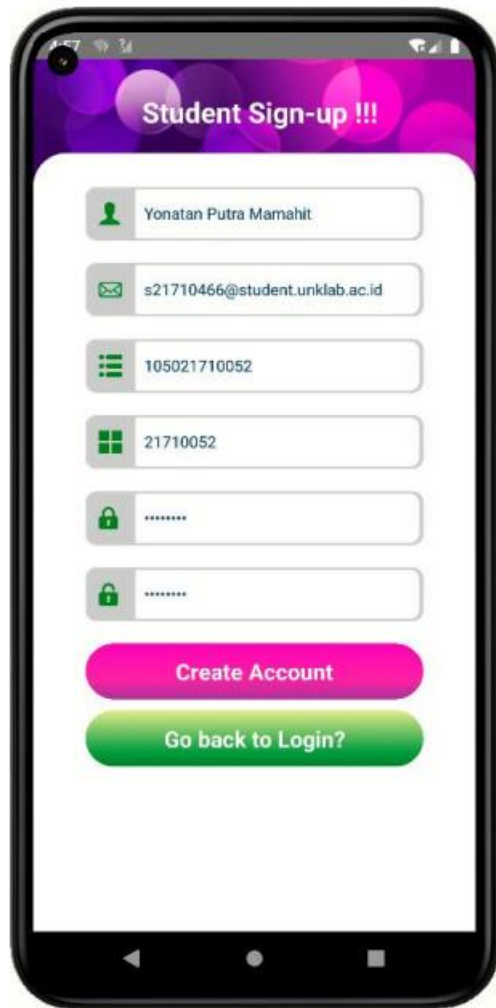
The image shows a web browser window with a single tab titled 'localhost/demo3/test.php'. The address bar also displays 'localhost/demo3/test.php'. The browser's toolbar includes navigation buttons (back, forward, home), search, and various extension icons. The main content area displays a table with the following data:

Reg Number	NIM Number	Email	Fullname	Password	Created At	Updated At
1001	1234567890	john@example.com	John Doe	password123	2023-06-20 01:14:04	2023-06-20 01:14:04
1006	dcscs	john@example.com	sdfs	sdfs	2023-06-21 15:37:56	2023-06-21 15:37:56

Exercise #4

1. Di local PC, buatlah sebuah **form PHP** sederhana untuk input data mahasiswa baru ke dalam *database* di *cloud server*.
2. Buat sebuah fungsi **simple RESTful API** untuk handle input data mahasiswa baru ke dalam table “*tbl_students*” di *cloud server*.
3. Tampilkan data mahasiswa dalam table HTML dari *framework* PHP yang telah dibuat pada saat operator/dosen login maka *table* akan terupdate dengan *current* data yang ada di *database cloud server*.

Students sign up dari platform yang berbeda ke cloud server.



Student Sign-up !!!

Yonatan Putra Mamahit

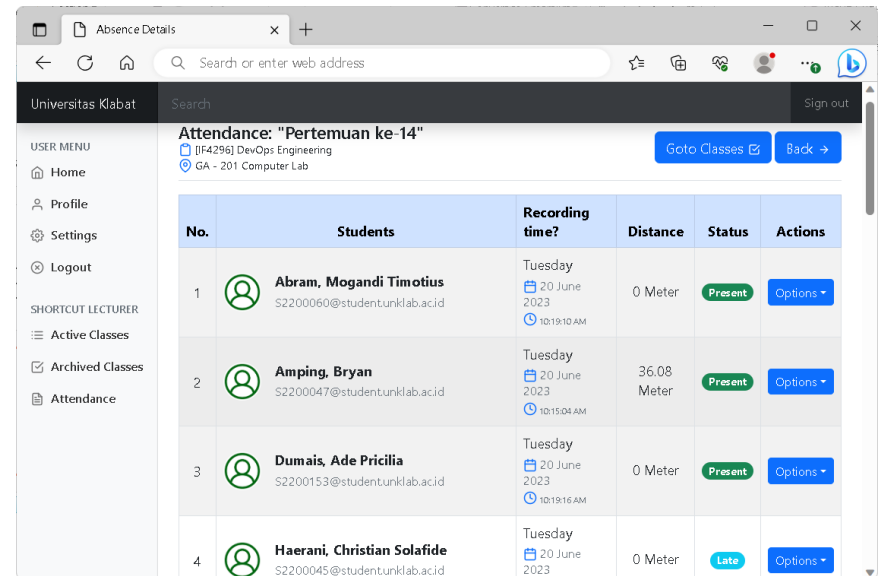
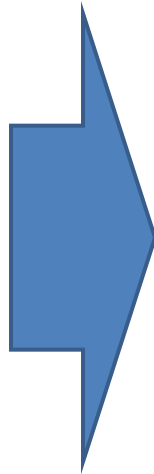
s21710466@student.unklab.ac.id

105021710052

21710052

Create Account

Go back to Login?



Universitas Klabat

Attendance: "Pertemuan ke-14"

[IF4296] DevOps Engineering

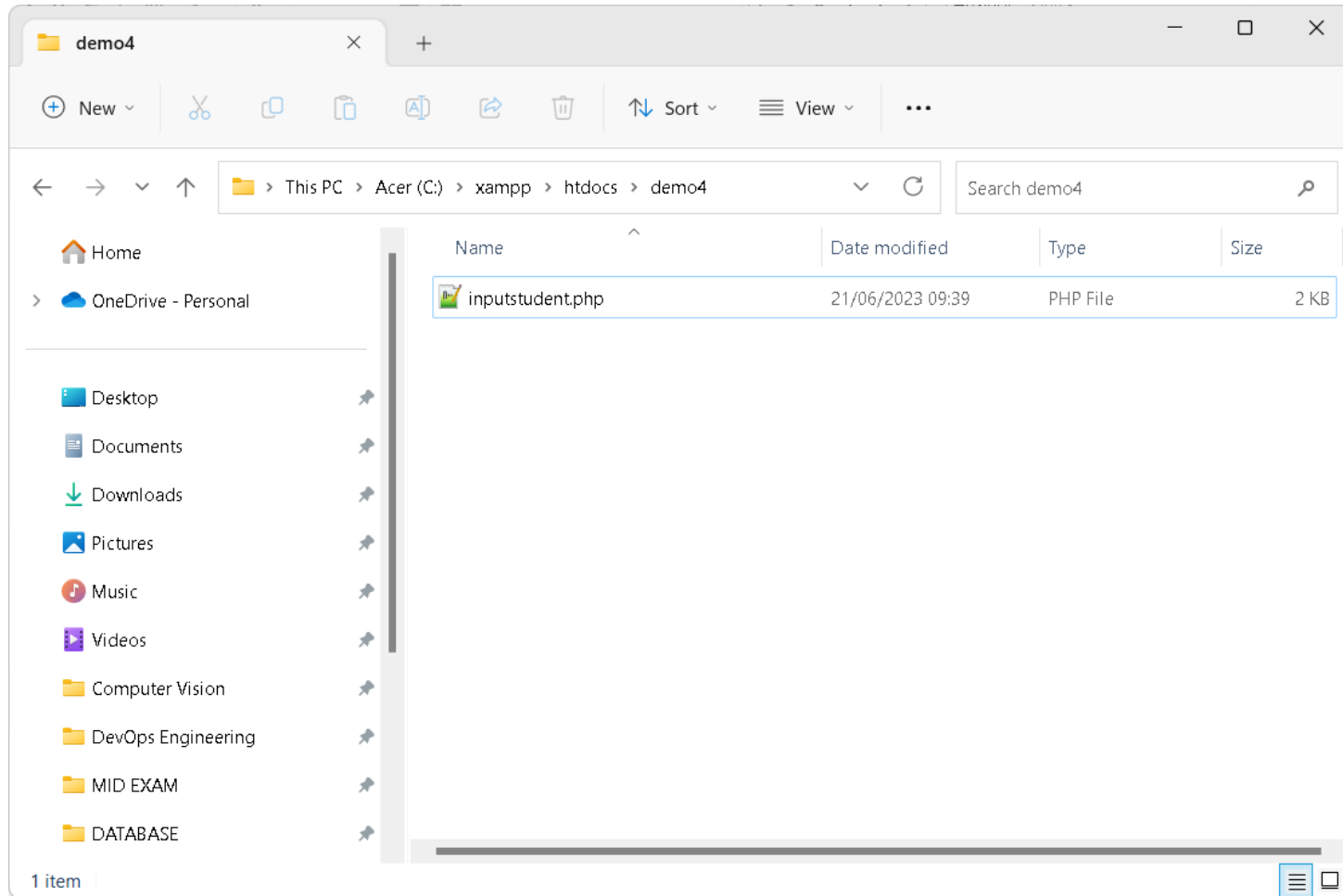
GA - 201 Computer Lab

Goto Classes Back

No.	Students	Recording time?	Distance	Status	Actions
1	Abram, Mogandi Timotius S2200060@student.unklab.ac.id	Tuesday 20 June 2023 10:19:10 AM	0 Meter	Present	Options
2	Amping, Bryan S2200047@student.unklab.ac.id	Tuesday 20 June 2023 10:15:04 AM	36.08 Meter	Present	Options
3	Dumais, Ade Pricilia S2200153@student.unklab.ac.id	Tuesday 20 June 2023 10:19:16 AM	0 Meter	Present	Options
4	Haerani, Christian Solafide S2200045@student.unklab.ac.id	Tuesday 20 June 2023	0 Meter	Late	Options

Localhost New Directory

Location: C:\xampp\htdocs\demo4

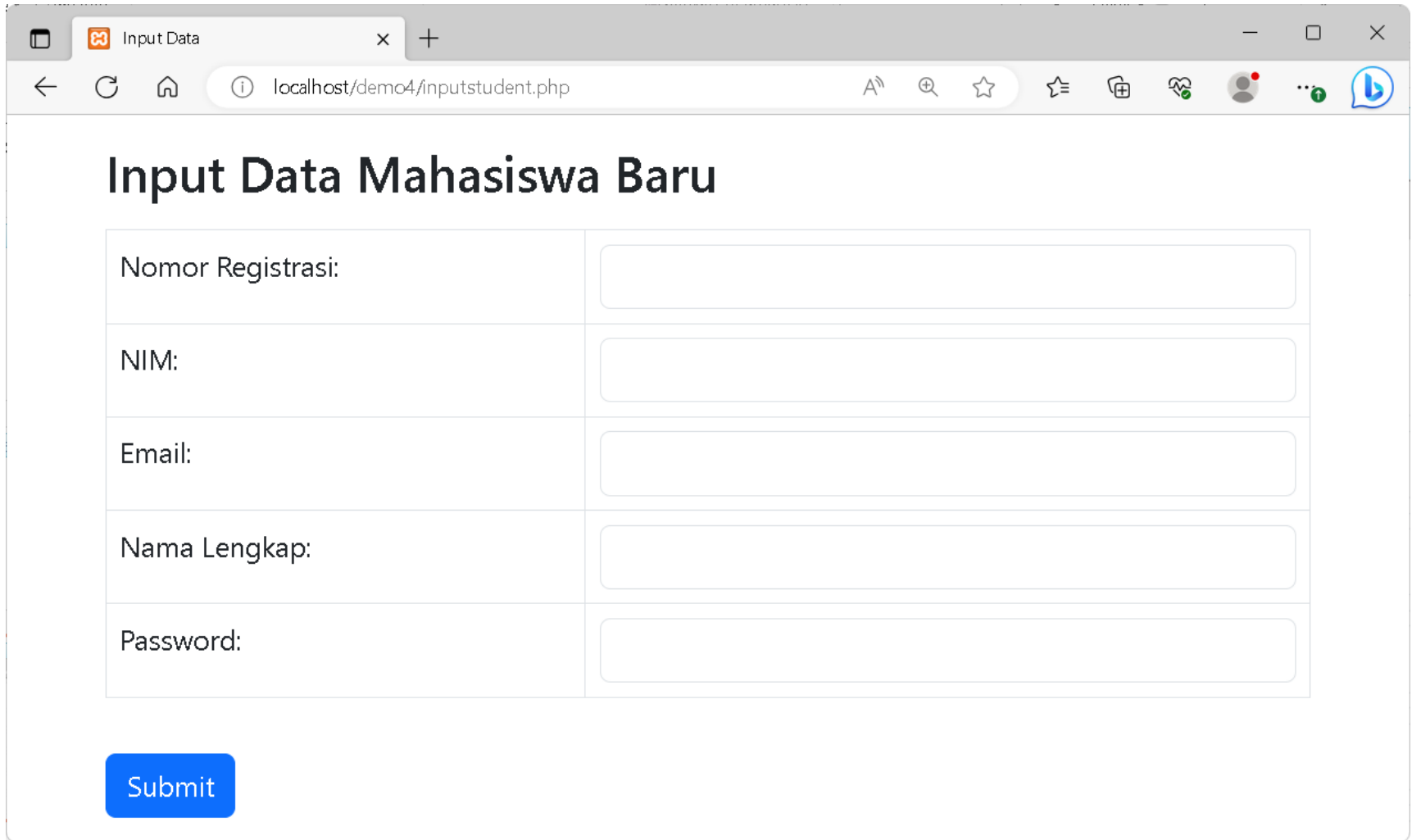


Create Input Form Bootstrap 5

```
C:\xampp\htdocs\demo4\inputstudent.php - Notepad++
File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window ?
inputstudent.php
1 <!DOCTYPE html>
2 <html>
3 <head>
4 <title>Input Data</title>
5 <!-- Tambahkan link CSS Bootstrap -->
6 <link rel="stylesheet" href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css">
7 </head>
8 <body>
9 <div class="container">
10 <h2 class="mt-3">Input Data Mahasiswa Baru</h2>
11 <form method="POST" action="http://103.176.78.186/unkpresnet/public/home/inputstudent">
12 <table class="table table-bordered mt-3">
13 <tbody>
14 <tr>
15 <td><label for="reg_number">Nomor Registrasi:</label></td>
16 <td><input type="text" id="reg_number" name="reg_number" class="form-control" required></td>
17 </tr>
18 <tr>
19 <td><label for="nim_number">NIM:</label></td>
20 <td><input type="text" id="nim_number" name="nim_number" class="form-control" required></td>
21 </tr>
22 <tr>
23 <td><label for="email">Email:</label></td>
24 <td><input type="email" id="email" name="email" class="form-control" required></td>
25 </tr>
26 <tr>
27 <td><label for="fullname">Nama Lengkap:</label></td>
28 <td><input type="text" id="fullname" name="fullname" class="form-control" required></td>
29 </tr>
30 <tr>
31 <td><label for="password">Password:</label></td>
32 <td><input type="password" id="password" name="password" class="form-control" required></td>
33 </tr>
34 </tbody>
35 </table>
36 <input type="submit" value="Submit" class="btn btn-primary mt-3">
37 </form>
38 </div>
39
40 <!-- Tambahkan skrip JavaScript Bootstrap -->
41 <script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/js/bootstrap.bundle.min.js"></script>
42 </body>
43 </html>
```

PHP Hypertext Preprocessor file length : 2,049 lines : 44 Ln : 11 Col : 96 Sel : 87 | 1 Windows (CR LF) UTF-8 INS

Preview in Browser

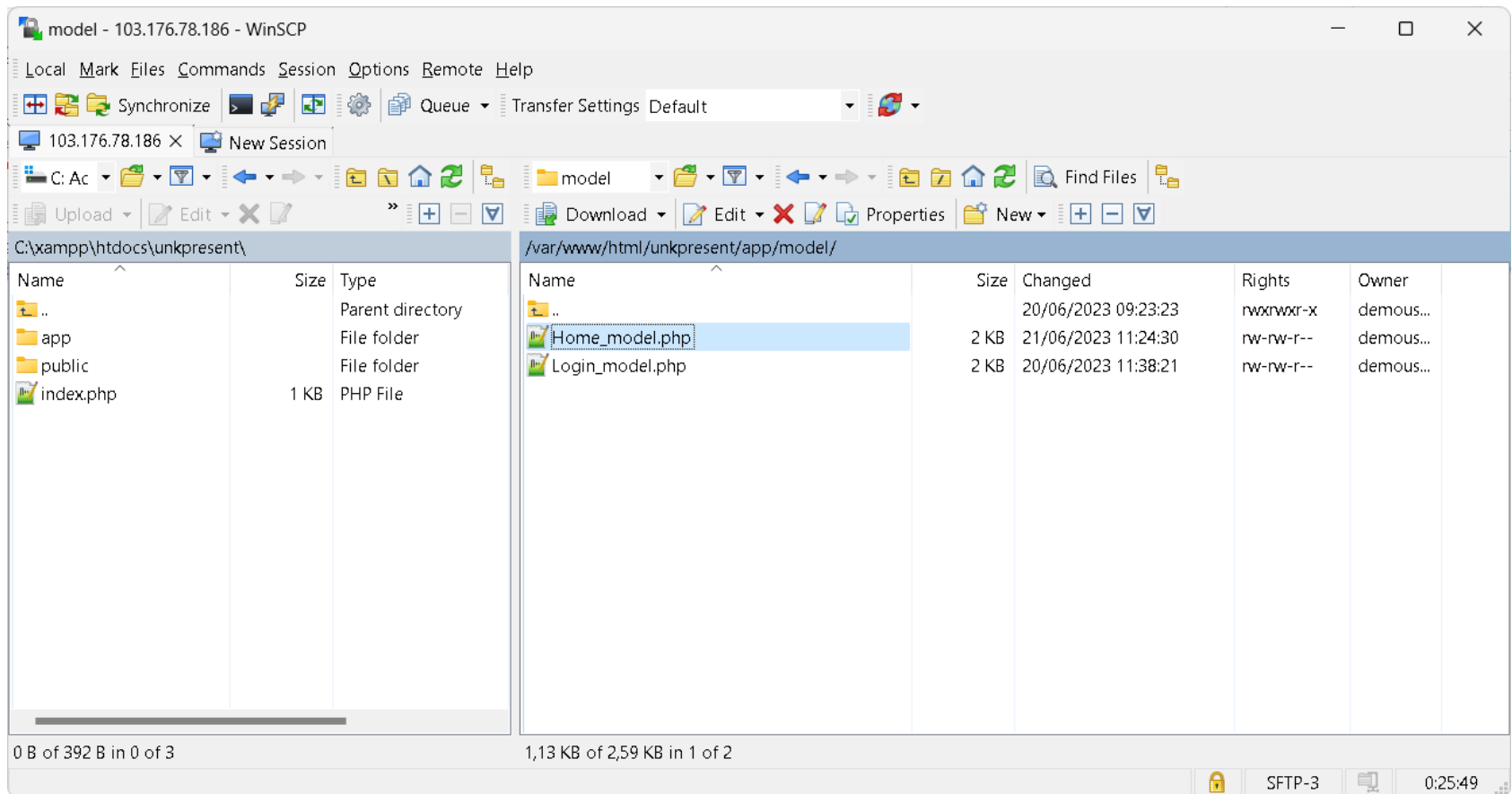


The screenshot shows a web browser window with the title 'Input Data' and the URL 'localhost/demo4/inputstudent.php'. The page content is a form titled 'Input Data Mahasiswa Baru'. The form consists of five rows, each with a label on the left and an input field on the right. The labels are 'Nomor Registrasi:', 'NIM:', 'Email:', 'Nama Lengkap:', and 'Password:'. Below the form is a blue 'Submit' button.

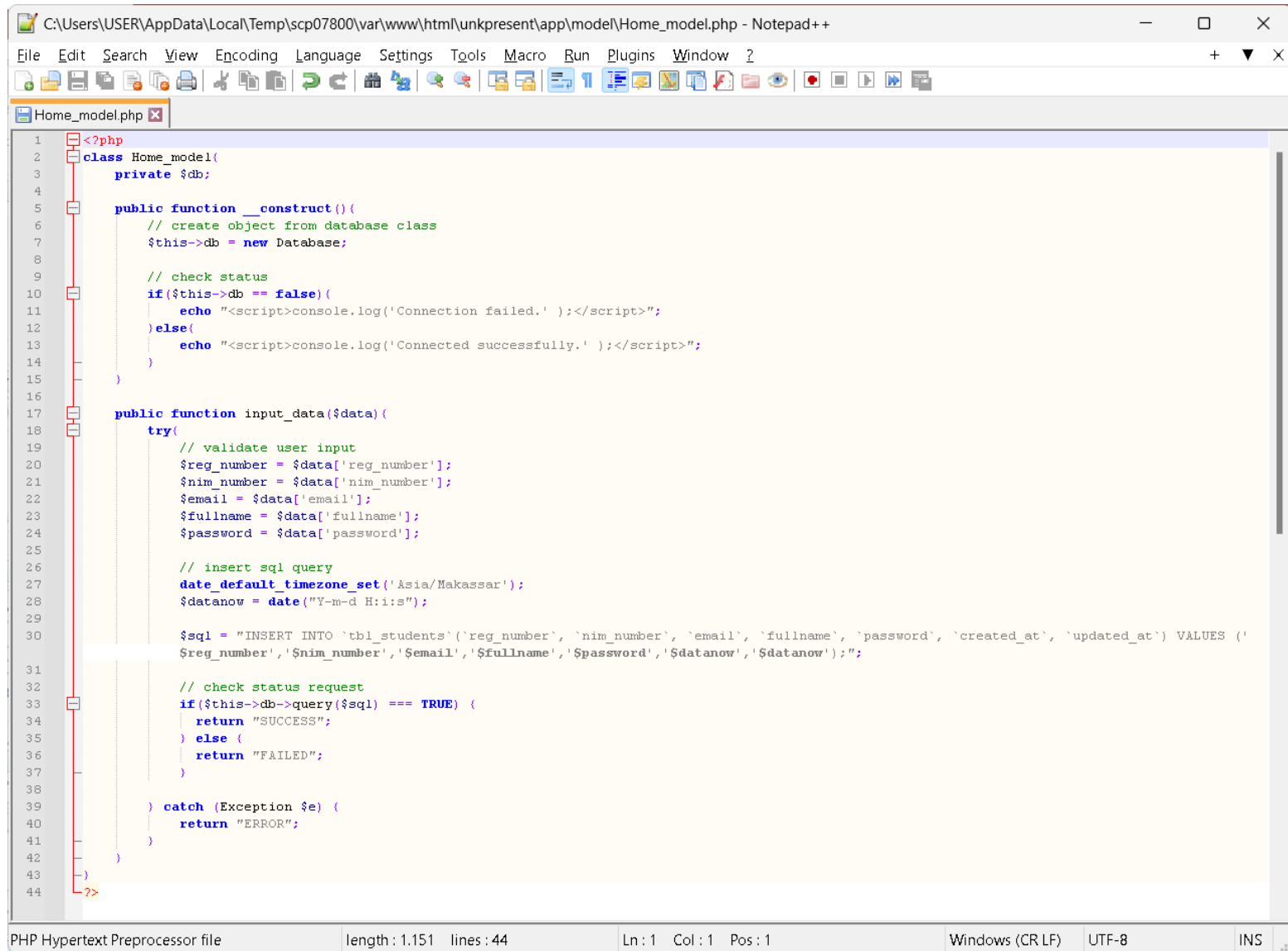
Nomor Registrasi:	
NIM:	
Email:	
Nama Lengkap:	
Password:	

Create New Model File

Buat sebuah model file di directory “*model*” dengan nama “*Home_model.php*”



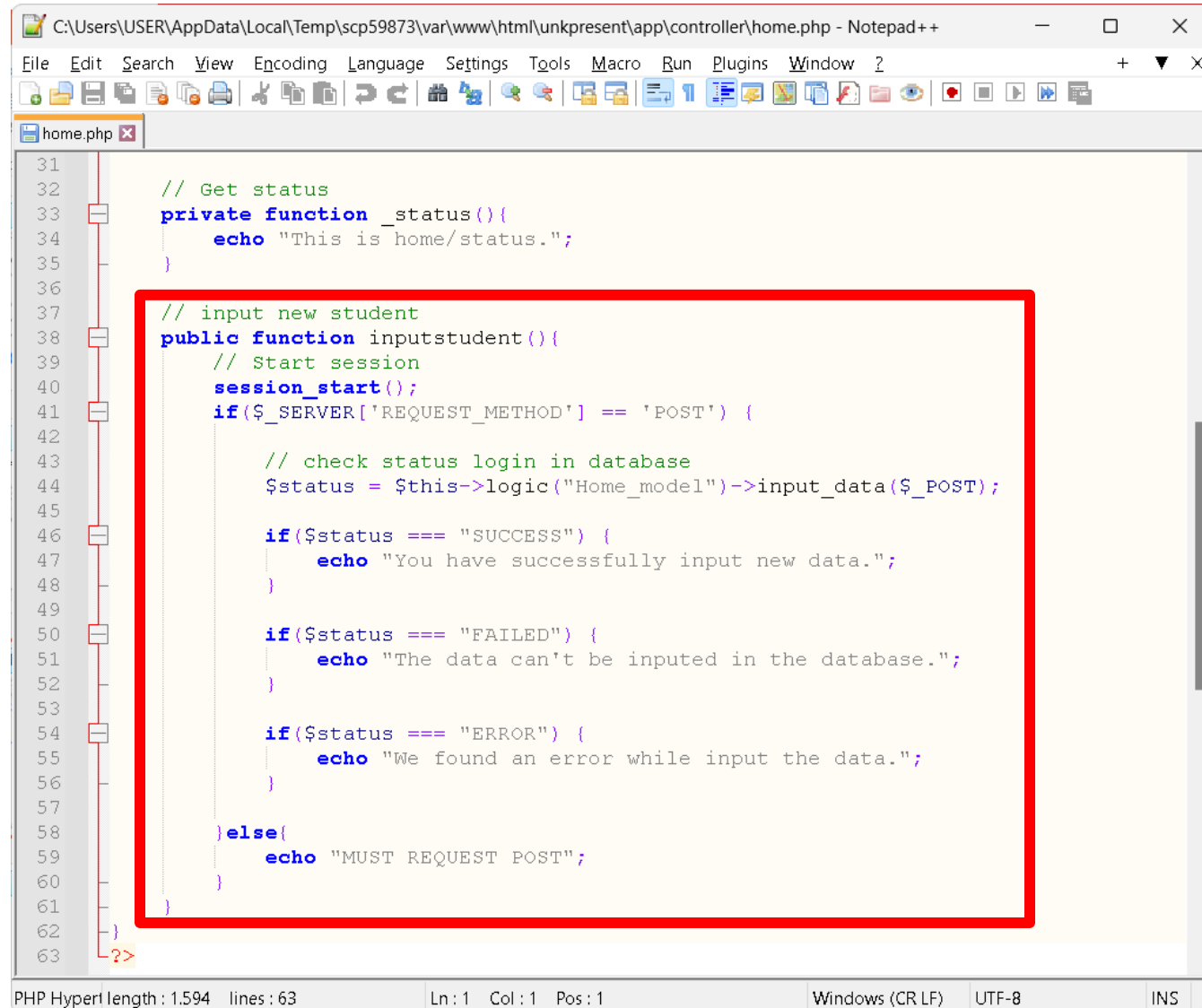
Add Script to “Home_model.php”



```
C:\Users\USER\AppData\Local\Temp\scp07800\var\www\html\unkpresent\app\model\Home_model.php - Notepad++
File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window ?
Home_model.php
1 <?php
2 class Home_model(
3     private $db;
4
5     public function __construct(){
6         // create object from database class
7         $this->db = new Database;
8
9         // check status
10        if($this->db == false){
11            echo "<script>console.log('Connection failed.' );</script>";
12        }else{
13            echo "<script>console.log('Connected successfully.' );</script>";
14        }
15    }
16
17    public function input_data($data){
18        try{
19            // validate user input
20            $reg_number = $data['reg_number'];
21            $nim_number = $data['nim_number'];
22            $email = $data['email'];
23            $fullname = $data['fullname'];
24            $password = $data['password'];
25
26            // insert sql query
27            date_default_timezone_set('Asia/Makassar');
28            $datanow = date("Y-m-d H:i:s");
29
30            $sql = "INSERT INTO `tbl_students`(`reg_number`, `nim_number`, `email`, `fullname`, `password`, `created_at`, `updated_at`) VALUES ('
31            $reg_number','$nim_number','$email','$fullname','$password','$datanow','$datanow')";
32
33            // check status request
34            if($this->db->query($sql) === TRUE) {
35                return "SUCCESS";
36            } else {
37                return "FAILED";
38            }
39        } catch (Exception $e) {
40            return "ERROR";
41        }
42    }
43 }
44 ?>
```

PHP Hypertext Preprocessor file length: 1.151 lines: 44 Ln: 1 Col: 1 Pos: 1 Windows (CR LF) UTF-8 INS

Add Function to Controller “Home.php”



```
C:\Users\USER\AppData\Local\Temp\scp59873\var\www\html\unkpresent\app\controller\home.php - Notepad++
File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window ?
home.php
31
32 // Get status
33 private function _status(){
34     echo "This is home/status.";
35 }
36
37 // input new student
38 public function inputstudent(){
39     // Start session
40     session_start();
41     if($_SERVER['REQUEST_METHOD'] == 'POST') {
42
43         // check status login in database
44         $status = $this->logic("Home_model")->input_data($_POST);
45
46         if($status === "SUCCESS") {
47             echo "You have successfully input new data.";
48         }
49
50         if($status === "FAILED") {
51             echo "The data can't be inputed in the database.";
52         }
53
54         if($status === "ERROR") {
55             echo "We found an error while input the data.";
56         }
57
58     }else{
59         echo "MUST REQUEST POST";
60     }
61 }
62
63
```

PHP Hyperl length : 1.594 lines : 63 Ln : 1 Col : 1 Pos : 1 Windows (CR LF) UTF-8 INS

Input New Student Data

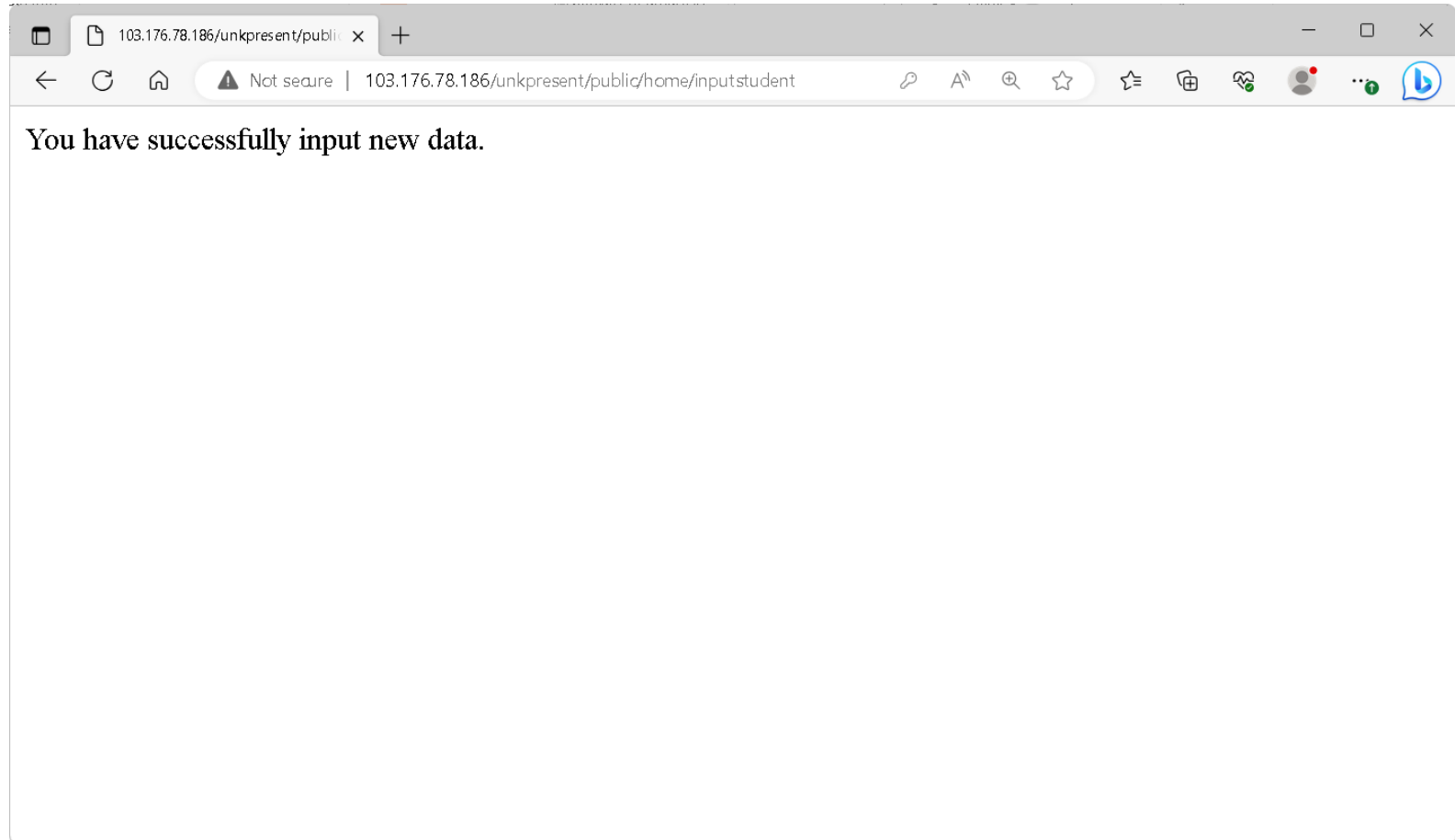
Input Data

localhost/demo4/inputstudent.php

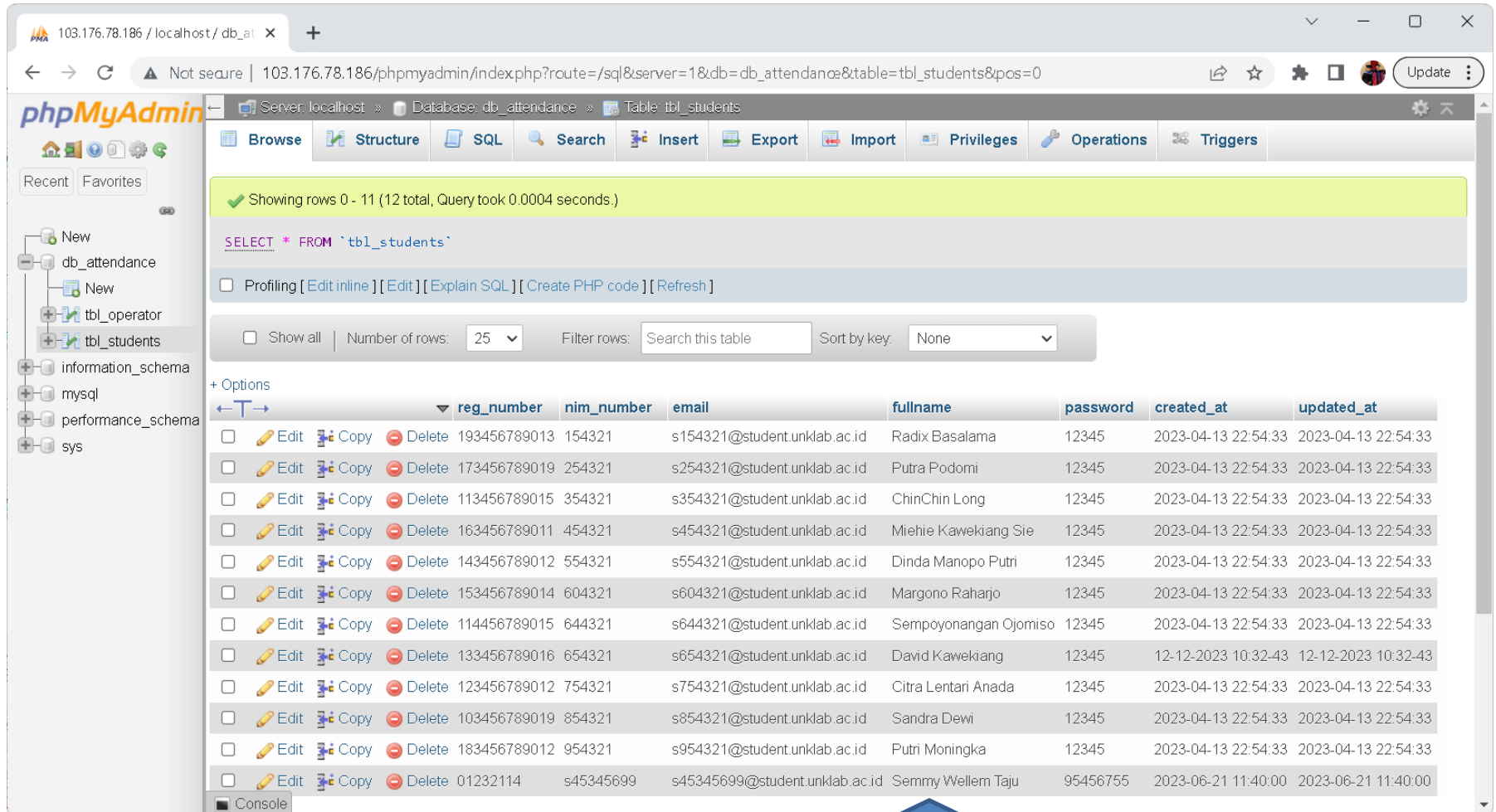
Input Data Mahasiswa Baru

Nomor Registrasi:	01232114
NIM:	s45345699
Email:	s45345699@student.unklab.ac.id
Nama Lengkap:	Semmy Wellem Taju
Password:	

Submit Response *from* Server



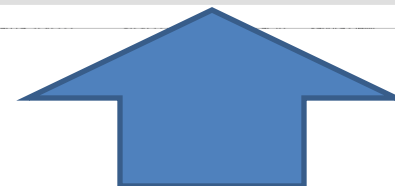
New Record Data Added



The screenshot shows the phpMyAdmin interface for a database named 'db_attendance' and a table named 'tbl_students'. The table contains 12 rows of student data. A new record has been added at the bottom, with the following details:

reg_number	nim_number	email	fullname	password	created_at	updated_at
193456789013	154321	s154321@student.unklab.ac.id	Radix Basalama	12345	2023-04-13 22:54:33	2023-04-13 22:54:33
173456789019	254321	s254321@student.unklab.ac.id	Putra Podomi	12345	2023-04-13 22:54:33	2023-04-13 22:54:33
113456789015	354321	s354321@student.unklab.ac.id	ChinChin Long	12345	2023-04-13 22:54:33	2023-04-13 22:54:33
163456789011	454321	s454321@student.unklab.ac.id	Miehie Kawekiang Sie	12345	2023-04-13 22:54:33	2023-04-13 22:54:33
143456789012	554321	s554321@student.unklab.ac.id	Dinda Manopo Putri	12345	2023-04-13 22:54:33	2023-04-13 22:54:33
153456789014	604321	s604321@student.unklab.ac.id	Margono Raharjo	12345	2023-04-13 22:54:33	2023-04-13 22:54:33
114456789015	644321	s644321@student.unklab.ac.id	Sempoyonangan Ojomiso	12345	2023-04-13 22:54:33	2023-04-13 22:54:33
133456789016	654321	s654321@student.unklab.ac.id	David Kawekiang	12345	12-12-2023 10:32:43	12-12-2023 10:32:43
123456789012	754321	s754321@student.unklab.ac.id	Citra Lentari Anada	12345	2023-04-13 22:54:33	2023-04-13 22:54:33
103456789019	854321	s854321@student.unklab.ac.id	Sandra Dewi	12345	2023-04-13 22:54:33	2023-04-13 22:54:33
183456789012	954321	s954321@student.unklab.ac.id	Putri Moningka	12345	2023-04-13 22:54:33	2023-04-13 22:54:33
01232114	s45345699	s45345699@student.unklab.ac.id	Semmy Wellem Taju	95456755	2023-06-21 11:40:00	2023-06-21 11:40:00

Diharapkan output yang kalian buat bisa ditampilkan via CMD atau putty.



**Create this lab
exercise in groups
(max 3 students)**

END PRESENTATION

Thank you for your attention

Instructor: S – W – T

THANK YOU

